

# Algebraicity of Kuga–Satake Correspondences for K3 Surfaces

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## Abstract

We prove that the Kuga–Satake correspondence is algebraic for every smooth projective complex K3 surface. More precisely, the prescribed embedding of its transcendental cohomology into the cohomology of its Kuga–Satake abelian variety is induced by a rational algebraic cycle, with the fixed normalization and full even-Clifford target. The result also holds for isogenous Kuga–Satake models with the transported embedding.

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# 1 Introduction

The Kuga–Satake construction places a polarized Hodge structure of K3 type inside the tensor square of a weight-one Hodge structure. For a projective K3 surface, this realizes its transcendental cohomology inside the cohomology of an abelian variety. The algebraicity question asks whether this Hodge-theoretic relation is induced by an algebraic cycle. It is a particular case of the Hodge conjecture, with a specified tensor to be realized.

## 1.1 The correspondence to be realized

Let  $S$  be a smooth projective complex K3 surface. Write

$$T(S) = \mathrm{NS}(S)_{\mathbb{Q}}^{\perp} \subset H^2(S, \mathbb{Q}),$$

where orthogonality is with respect to cup product. Equip this transcendental Hodge structure with the polarization used in the Kuga–Satake construction. Fix the standard construction data and let  $A_S = \mathrm{KS}(T(S))$  denote the full even-Clifford Kuga–Satake abelian variety, with its fixed rational polarization. If  $W_S = H^1(A_S, \mathbb{Q})$ , the construction gives an embedding

$$T(S)(1) \hookrightarrow \mathrm{End}(W_S).$$

The polarization identifies  $W_S^\vee$  with  $W_S(1)$ , so untwisting and using the Künneth inclusion gives a specified Hodge embedding

$$\kappa_S : T(S) \hookrightarrow W_S \otimes W_S \hookrightarrow H^2(A_S \times A_S, \mathbb{Q}). \quad (1.1)$$

The fixed data in this definition matter: the theorem concerns this map itself, including its normalization and its full Clifford target.

For smooth projective varieties  $S$  and  $Z$ , with  $S$  a surface, a class  $\Gamma \in \mathrm{CH}^2(S \times Z)_{\mathbb{Q}}$  acts on degree-two cohomology by

$$\Gamma_*(x) = (p_Z)_*(\mathrm{cl}(\Gamma) \smile p_S^*x).$$

Here  $\mathrm{CH}^2$  denotes codimension-two algebraic cycles modulo rational equivalence, with rational coefficients, and  $\mathrm{cl}$  is the Betti cycle class. Integration over  $S$  reduces degree by four, so this formula does take values in  $H^2(Z, \mathbb{Q})$ .

**Theorem 1.1.** *For every smooth projective complex K3 surface  $S$ , and for every fixed set of standard Kuga–Satake construction data as above, there is a cycle*

$$\Gamma_S \in \mathrm{CH}^2(S \times A_S \times A_S)_{\mathbb{Q}}$$

*such that  $\Gamma_{S*}|_{T(S)} = \kappa_S$ . The same assertion holds for an isogenous Kuga–Satake model when the prescribed map is transported by the corresponding rational Hodge isomorphism.*

## 1.2 Historical context and related work

Kuga and Satake introduced their construction in 1967 [KS67]. It associates an abelian variety to polarized K3 cohomology through the spin representation of its Clifford algebra. This association allows questions about weight-two cohomology to be studied through weight-one cohomology, where abelian varieties provide geometric tools. Deligne used the construction in his proof of the Weil conjecture for K3 surfaces [Del72]. His work also establishes the absolute-Hodge character of the Kuga–Satake correspondence; see [Del72, Del82]. André subsequently proved that this correspondence is motivated [And96, Section 7]. These results give the tensor strong functorial and arithmetic properties. Its realization by an algebraic cycle is the further assertion addressed here.

Geometric realizations in special cases have long played an important role. Morrison analyzed the Kuga–Satake variety attached to an abelian surface [Mor85]. Paranjape constructed the relevant algebraic correspondence for K3 surfaces obtained by resolving double covers of the plane branched along six lines in general position [Par88]. This gives a four-dimensional family whose general member has Picard number sixteen. Van Geemen developed the Clifford-algebra and Hodge-theoretic description in detail and explained formulations of the algebraicity problem using surface covers and products of holomorphic one-forms [van00, Sections 10–11]. His cyclic quartic examples provide another geometric realization [van00, Example 11.3]. These formulations motivate our search for a nonzero correspondence; recovery of the prescribed tensor is handled by the reduction stated below.

A more recent source of examples comes from hyperkähler varieties of generalized Kummer type. O’Grady related their Kuga–Satake varieties to intermediate Jacobians [O’G21]. Combining this description with Markman’s results on algebraic cycles and Abel–Jacobi maps [Mar23], Voisin proved algebraicity of the Kuga–Satake correspondence for projective varieties of generalized Kummer type of dimension at least four [Voi22, Theorem 0.5]. Floccari then used associated K3 surfaces to obtain algebraicity for further families of K3 surfaces [Flo24, Flo26]. In particular, [Flo26, Theorem 5.11] applies whenever  $T(S)$  admits an isometric embedding into  $U_{\mathbb{Q}}^{\oplus 3} \oplus \langle -m \rangle_{\mathbb{Q}}$  for a positive integer  $m$ , where  $U$  is the hyperbolic plane. That result also proves the Hodge conjecture for every power

of these K3 surfaces. The hyper-Kummer construction of Floccari and Fu proves that the Chow motives in certain families with generic Picard number sixteen are of abelian type [FF26].

Low Picard number also occurs in this special-family literature. Bolognesi and Laterveer study two nine-dimensional families of K3 surfaces with generic Picard number two, distinguished by a non-symplectic automorphism of order three fixing a curve. They state Kuga–Satake algebraicity results for these families [BL24, Theorem 4.4, Propositions 6.2 and 6.4]. Theorem 1.1 addresses this algebraicity problem for every projective K3 surface, in the fixed formulation stated above.

Two further strands of work enter the proof. Buskin proved that rational Hodge isometries between projective K3 surfaces are algebraic [Bus19]. We use this result to transfer algebraic correspondences between suitable K3 Hodge structures. For the deformation step, the relevant ancestry is the semiregularity theory of Buchweitz–Flenner and its extension to perfect complexes by Pridham [BF03, Pri24]. Markman’s secant-sheaf approach combines spinor geometry, Fourier–Mukai transforms, and semiregularity to construct algebraic Weil classes, including those on split abelian sixfolds [Mar25, Mar26]. Our proof likewise constructs an object that can deform while its characteristic classes remain of Hodge type. The specific task is to obtain such an object with enough deformation directions to cover polarized K3 moduli. The fixed mirror-symmetry theorems and the immersed-Lagrangian construction described below supply it.

### 1.3 The reduction to a nonzero correspondence

The proof uses the following reduction from the companion manuscript [Opeb]. A polarization below is ample and primitive in the integral K3 lattice. A property holds at a very general point if it holds outside a countable union of proper closed algebraic subsets of the relevant moduli space.

**Theorem 1.2** (Reduction to very general polarized K3 surfaces). *Assume that for every positive integer  $d$ , a very general primitively polarized complex K3 surface  $(S, h)$  with  $h^2 = 2d$  admits an abelian variety  $B_S$  and a cycle*

$$Z_S \in \mathrm{CH}^2(S \times B_S)_{\mathbb{Q}}$$

*whose action is nonzero on  $h^{\perp} \subset H^2(S, \mathbb{Q})$ . Then Theorem 1.1 holds.*

This is the combination of Theorems 1.1 and 1.2 and Corollary 1.3 of [Opeb]. Its content has two parts. At a very general polarized K3 surface, the full orthogonal Hodge group and the spin representation recover the prescribed full Kuga–Satake tensor from any nonzero algebraic correspondence of the displayed kind. Algebraic correspondences then specialize through the polarized moduli space. The reduction includes comparison with the stipulated construction data at special fibers. We shall use the reduction in this precise form; the main task of the present paper is to construct its nonzero algebraic input.

### 1.4 The proof mechanism

Fix a positive integer  $d$ . We begin with a K3 surface  $S$  of Picard number nineteen and an abelian variety  $B = E^g$ , where  $E$  is a non-CM elliptic curve and  $g = 512$ . These varieties are mirrors of a symplectic quartic  $P$  and a standard symplectic torus  $Y$ . Only mirror symmetry at these fixed varieties is used. The auxiliary dimension 512 permits a rational Clifford representation with eighteen specified symmetric generators; its size is not a dimension assertion about the Kuga–Satake variety in Theorem 1.1.

Choose an ample rational class  $h$  on the starting K3 with square  $2d$ . A rational class is sufficient at this stage; the relation between its primitive integral ray and degree  $2d$  will be handled at the

end. Put

$$M = P \times Y, \quad n = g + 2, \quad D_{\text{tot}} = 22 + \binom{2g}{2}.$$

The proof constructs a perfect complex  $\mathcal{E}$  on  $S \times B$  with two complementary properties.

First, topology gives the upper bound

$$\dim \text{Ext}^2(\mathcal{E}, \mathcal{E}) \leq D_{\text{tot}} - 19. \quad (1.2)$$

We construct a rational class  $\alpha \in H^n(M, \mathbb{Q})$  for which cup product  $H^2(M, \mathbb{Q}) \rightarrow H^{n+2}(M, \mathbb{Q})$  has a prescribed nineteen-dimensional kernel. A torus bundle and surgery realize a multiple of its Poincaré dual by an immersed Lagrangian  $L \rightarrow M$  with

$$b_2(L) = D_{\text{tot}} - 19, \quad H_{n-2}(L, \mathbb{Q}) \longrightarrow H_{n-2}(M, \mathbb{Q}) \text{ injective.}$$

The immersion has only odd-degree double-point sectors. The homological injection detects its Floer obstruction, and the absence of degree-two double-point generators bounds its second self-Floer group by  $b_2(L)$ . A representability and trace-pairing argument makes  $\mathcal{E}$  a direct summand of the resulting Floer object, proving (1.2).

Second, intersection numbers prescribe the component of the Mukai vector  $v(\mathcal{E}) = \text{ch}(\mathcal{E})\sqrt{\text{td}}$  with algebraic K3 coefficients. Eighteen anticommuting matrices control this component. The Hochschild cap action factors through

$$\text{HH}^2(S \times B) \longrightarrow \text{Ext}^2(\mathcal{E}, \mathcal{E}) \longrightarrow \text{HH}_{-2}(S \times B).$$

The domain has dimension  $D_{\text{tot}}$ , so (1.2) leaves a kernel of dimension at least nineteen. The explicit Mukai component forces this kernel to consist of ordinary polarized deformation directions and to project injectively to the nineteen-dimensional polarized K3 tangent space. Equality follows. It makes the degree-two semiregularity map injective and gives all polarized K3 directions.

These directions integrate to a Hodge-preserving period germ: the corresponding infinitesimal orthogonal and symplectic actions annihilate  $v(\mathcal{E})$ , so their exponentials fix it. Semiregularity then lifts the perfect complex formally. Existence for perfect complexes and finite-type spreading produce a family dominating polarized K3 moduli. The codimension-two Chern character is the required nonzero correspondence on a very general fiber. Finally, rational Hodge isometries transfer between the necessary polarization degrees, and Theorem 1.2 completes the proof.

## 1.5 Technical contributions and organization

The construction of algebraic classes through immersed Floer objects is related to the method in [Opea]. The additional requirement here is deformation across the full polarized K3 period space. Prescribing the second Betti number of the immersion and matching it with the Hochschild calculation provides that deformation mechanism.

Two analytic details are developed separately because they are needed to apply the fixed mirror-symmetry theorems to the constructed immersion. The first is homological detection of the immersed obstruction using an interior-marked disk identity. Finite constructions use both energy and arity bounds; a coarsened residue of an ultrapower turns successively accurate solutions into an uncurved object while retaining the trace pairings needed for representability. The second is a comparison between the clean immersed category and the ordinary categories in the mirror-symmetry theorems. We verify the parameterized disk constructions and recover full categories from ordered Hamiltonian copies. These statements concern finite clean collections and may be useful independently of the K3 application.

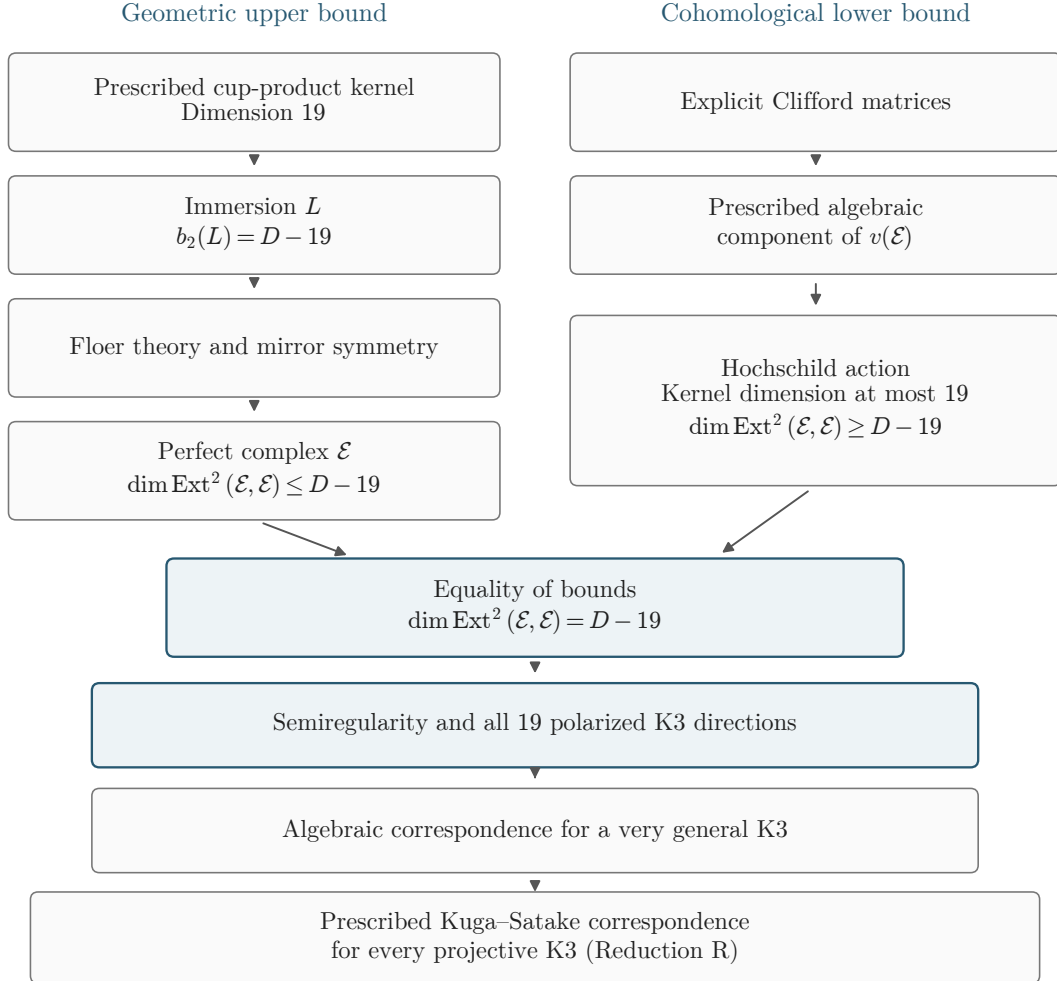


Figure 1: The two bounds that produce deformation. The topological construction bounds the second extension group from above; the prescribed Mukai component bounds the kernel of the Hochschild action from above. Together they force equality and semiregularity. Here  $D = D_{\text{tot}}$ .

Section 2 fixes the mirror varieties, the algebraic and symplectic cohomology dictionaries, and the generating labels. Section 3 constructs the matrices and the cohomology class  $\alpha$ . Section 4 realizes its prescribed kernel geometrically. Sections 5 and 6 supply the Floer construction and category comparisons just described. Section 7 constructs  $\mathcal{E}$  and computes the required part of its Mukai vector. Section 8 proves semiregularity, carries out the deformation and spreading argument, and establishes Theorem 1.1.

All cohomology is rational unless another field is indicated. Floer categories are integrally graded, spin, and use background class zero. Their perfect hulls include direct summands. We quote the foundational virtual-chain and homological mirror-symmetry theorems with their needed hypotheses at the points of application, and prove the extensions specific to the present construction there.

## 2 The starting mirrors and their numerical classes

The symplectic construction will take place on a product of a quartic surface and a torus. We first identify a finite collection of embedded product Lagrangians whose category is the category of perfect complexes on a projective variety. We establish the ordinary mirror categories and their numerical classes. The resulting assertion for clean product labels uses the comparison and Künneth propositions proved in Section 6. This is the only use of homological mirror symmetry. The numerical identifications in this section will later determine part of the Chern character of a complex from intersection numbers.

Let  $P$  be a smooth quartic in  $\mathbb{P}^3$ , equipped with the restricted Fubini–Study form normalized to have integral cohomology class  $a$ . Write

$$a^2 = q \text{ pt}_P, \quad q = 4. \quad (2.1)$$

Let  $g = 512$  and let

$$Y = (\mathbb{R}/\mathbb{Z})_x^g \times (\mathbb{R}/\mathbb{Z})_y^g, \quad \omega_Y = \sum_{i=1}^g X_i Y_i, \quad X_i = dx_i, \quad Y_i = dy_i.$$

Products of differential forms denote exterior products. Thus  $M = P \times Y$  has complex dimension  $n = g + 2 = 514$  and integral symplectic class  $a + \sum_i X_i Y_i$ . Choose a nonzero holomorphic two-form on the initial complex quartic and the flat holomorphic volume form on the torus. Their product trivializes the complex determinant of  $TM$ . Transport this trivialization to other compatible almost complex structures through the contractible space of compatible structures. Its homotopy class is the volume grading used on  $M$  and on all its branes.

All Floer categories in this section use the Novikov field  $\Lambda$  with complex coefficients and real exponents, supported on sets locally finite to the left. The area coefficient  $T^\lambda$  has degree zero. The branes are integrally graded and spin, with zero background class. We use the finite models and coefficient extensions described in Sections 5 and 6. An algebraically closed extension of  $\Lambda$  of cardinality at most continuum is abstractly isomorphic to  $\mathbb{C}$ ; the precise field argument is recorded in Remark 5.10. Néron–Severi groups and the numerical lattices below do not change under algebraically closed extensions of characteristic zero. Consequently we may make these choices before making a final identification of the coefficient field with  $\mathbb{C}$ . To interpret the Betti and Hodge notation in this section, choose a temporary abstract identification of its current algebraically closed coefficient field with  $\mathbb{C}$  and transport the varieties to that field. This identification need not fix the original coefficient copy of  $\mathbb{C}$ . The K3 numerical space is intrinsically  $\mathbb{Q} \oplus \text{NS}(S)_{\mathbb{Q}} \oplus \mathbb{Q}$ , with its unit and point summands; the torus classes used below are rational combinations of divisor classes

specified by integral matrices. These are the data retained under subsequent coefficient extensions. The Hodge assertions used here hold for every such temporary realization, because the elliptic curve remains non-CM.

## 2.1 The quartic factor

We use Seidel's undeformed quartic theorem [Sei15, Theorem 1.3]. Its coefficient field is the algebraic closure of a Laurent-series field, and its mirror is the minimal resolution of the quotient of the Dwork quartic by the diagonal group of order 16. Write  $u$  for Seidel's series variable. His prequantum normalization in Section 11a uses  $\mathcal{O}(4)$ , so his symplectic form is four times the hyperplane-normalized form used here. Accordingly we extend his symplectic coefficients by  $f(u) = T^{1/4}$ : a disk of area  $A$  in our normalization has his area  $4A$ , and  $u^{4A}$  becomes  $T^A$ . His theorem also twists the mirror scalar action by an invertible formal reparametrization  $\psi$ . We absorb that twist in the mirror base map; with his convention that a scalar  $\lambda$  acts as  $\psi(\lambda)$ , the untwisted mirror base map is  $f \circ \psi^{-1}$ . The theorem is obtained by identifying the endomorphism  $A_\infty$  algebras on a collection of 64 Lagrangian spheres; these spheres split-generate. We therefore use its enhanced conclusion,

$$\mathrm{Perf}\langle \text{the chosen spheres in } P \rangle \simeq \mathrm{Perf}(S), \quad (2.2)$$

where  $S$  is the mirror K3 surface over the enlarged coefficient field. Here and below  $\mathrm{Perf}$  of an  $A_\infty$  category means its ordinary split-closed pretriangulated hull.

After this rescaling, the symplectic deformation identification between smooth quartics permits the use of our specified  $P$ . We enlarge the finite sphere collection to include vanishing spheres spanning  $a^\perp \subset H^2(P, \mathbb{Q})$ . Indeed, the vanishing cohomology of a Lefschetz pencil of quartics in  $\mathbb{P}^3$  is the orthogonal complement of the hyperplane class. Its vanishing cycles are Lagrangian spheres for the restricted Kähler form and admit gradings. They are objects of Seidel's category: the flat prequantum multisection required there exists on every sphere. His theorem identifies the full category, so adding these objects preserves (2.2). Scalar extension preserves this enhanced equivalence and its split-generation assertion, as is seen by extending a finite perfect resolution of the diagonal.

The mirror family is non-isotrivial, and its geometric generic member has Picard number at most 19. One may use the lattice-polarized mirror-quartic description for this assertion [Dol96]. Alternatively, the Dwork period varies: at the Fermat member its infinitesimal variation is represented by the product of the four coordinates, which is nonzero in the Jacobian ring. The diagonal group is symplectic, and its quotient and crepant resolution carry the same varying period line. These correspondences exist in families after restricting the parameter base. A non-isotrivial one-parameter K3 family cannot have geometric generic Picard number 20: a generic divisor class spreads after a finite base change, whereas the complex K3 surfaces of Picard number 20 have only countably many period points. Seidel's nonconstant formal parameter, including its invertible change of coordinate, selects this geometric generic fiber.

Put

$$\Lambda_S^{\mathrm{alg}} = H^0(S, \mathbb{Q}) \oplus \mathrm{NS}(S)_{\mathbb{Q}} \oplus H^4(S, \mathbb{Q}), \quad m((r, x, s), (r', x', s')) = xx' - rs' - sr'.$$

The Euler characteristic of Floer cohomology of two graded Lagrangian surfaces is minus their oriented intersection number. The Euler characteristic on  $S$  is minus the Mukai pairing. Consequently the assignments of our sphere classes to the Mukai vectors of their mirrors induce a rational isometry

$$(a^\perp, \cup) \xrightarrow{\sim} (\Lambda_S^{\mathrm{alg}}, m). \quad (2.3)$$

For completeness, choose 21 sphere classes forming a basis of  $a^\perp$ . Their intersection matrix is nonsingular, so the matrix of their mirror Mukai pairings has rank 21. Thus  $\dim \Lambda_S^{\text{alg}} \geq 21$ . The preceding Picard bound gives the reverse inequality. The resulting isometry also takes every additional sphere class to its mirror Mukai vector: pair its difference with the chosen basis and use nondegeneracy. In particular  $\rho(S) = 19$ .

We specify the Euler sign used here. In complex dimension  $2l$ , the sign of a transverse intersection generator of degree  $j$  satisfies

$$(-1)^j = (-1)^l \text{sign}_{\text{or}}(x). \quad (2.4)$$

This follows by rotating the graded tangent planes to coordinate planes; each of the  $2l$  rotation directions contributes its Maslov crossing sign, and comparison with the oriented direct-sum convention gives  $(-1)^l$ . We always transport orientations with these grading conventions.

## 2.2 An ample class in every rational squareclass

Fix a positive integer  $d$  and put  $r = 2d$ . The rational K3 lattice and Witt cancellation in (2.3) give

$$a^\perp \simeq U^2 \oplus E_8(-1)^2 \oplus \langle -4 \rangle, \quad \text{NS}(S)_\mathbb{Q} \simeq U \oplus E_8(-1)^2 \oplus \langle -4 \rangle.$$

Here  $U$  is the hyperbolic plane; it is also the rational Mukai lattice of  $H^0 \oplus H^4$ . The quadric  $x^2 = r$  in  $\text{NS}(S)_\mathbb{R}$  has a rational point, because the rational form contains  $U$ . Projection from a rational point parametrizes a Zariski-open subset of this quadric rationally. It follows, using rational points close to the parameters of any real smooth point, that rational points are dense on each relevant real sheet. The ample cone is open and meets the positive sheet. We can therefore choose

$$h \in \text{NS}(S)_\mathbb{Q} \quad \text{ample}, \quad h^2 = r.$$

Clearing denominators and taking the primitive generator of the ray gives an integral primitive ample class. Its degree has squareclass  $r$ ; the normalization  $h^2 = r$  will be retained in the calculations.

Decompose the Mukai space orthogonally as

$$\Lambda_S^{\text{alg}} = \Delta \perp \mathcal{N}, \quad \Delta = H^0(S, \mathbb{Q}) \oplus \mathbb{Q}h \oplus H^4(S, \mathbb{Q}), \quad \mathcal{N} = h^\perp \cap \text{NS}(S)_\mathbb{Q}. \quad (2.5)$$

Choose any rational basis  $d_1, d_2, d_3$  of  $\Delta$ . There is a basis  $n_1, \dots, n_{18}$  of  $\mathcal{N}$  satisfying

$$m(n_i, n_j) = -\delta_{ij} \lambda_i, \quad (\lambda_1, \dots, \lambda_{18}) = (1, \dots, 1, r). \quad (2.6)$$

To verify this, Witt's theorem moves  $h$  into  $U$ , where its orthogonal line has squareclass  $-r$ . Each rational  $E_8$  form is the standard sum of eight squares: use its usual full-rank lattice in  $\mathbb{Q}^8$  with Euclidean form. The remaining line  $\langle -4 \rangle$  is rationally  $\langle -1 \rangle$ . This gives seventeen negative unit squares and one negative square of size  $r$ . Through (2.3), the same symbols  $d_i, n_i$  will denote degree-two classes on  $P$ .

## 2.3 The torus factor and graph objects

The mirror of  $Y$  is

$$B = E^g,$$

where  $E$  is the Tate elliptic curve with parameter  $T$ , the area-one Novikov parameter. After any final identification of the coefficient field with  $\mathbb{C}$ ,  $E$  is non-CM. Indeed its  $j$ -series is transcendental over the original complex coefficient field, hence also over  $\mathbb{Q}$ ; field identification preserves the latter

property, whereas a CM  $j$ -invariant is algebraic. Thus the Hodge group of  $B$  is the diagonal  $SL_2$  acting on the  $g$  copies of  $H^1(E)$ .

Choose forms  $z \in H^{1,0}(E)$  and  $w \in H^{0,1}(E)$  once on  $E$ , normalized so that  $zw$  is its principal polarization class, and let  $z_i, w_i$  be their pullbacks from the  $i$ th factor of  $B$ . Thus the normalization is coherent across all factors. Define

$$B(A) = z^t A w, \quad \theta = B(I). \quad (2.7)$$

If  $A$  is symmetric integral,  $B(A)$  is the divisor class described by  $A$  under the principal polarization of  $E^g$ . The rational Hodge algebra of  $B$  is generated by these classes. Indeed the invariants of the standard  $SL_2$  tensor representation are generated by the alternating pairings of two factors. Passing from tensor powers to exterior powers shows that all invariants in  $H^*(B)$  are generated by the corresponding degree-two classes; on  $E^g$  these are exactly the symmetric matrix divisors. In particular the classes  $e^{B(A)}$ , as  $A$  ranges over symmetric integral matrices, span the entire rational Hodge-class space. This last assertion follows as well by polarizing the finite exponential polynomial: its coefficients are all products of divisor classes.

For a symmetric integral matrix  $A$ , let  $D_A \subset Y$  be the graph  $y = Ax$  modulo  $\mathbb{Z}^g$ , with trivial local system and spin structure induced by the base framing. Small exact periodic perturbations replace it by  $y = Ax + \nabla f(x)$ . These remain Hamiltonian isotopic graphs. The inclusion of any such section induces an injection on fundamental groups, so  $\pi_2(Y, D_A) = 0$ . The sections and the torus fibers are therefore tautologically unobstructed. The base has  $\pi_2 = 0$ , and its framing and the zero section trivialize the twists in the family Floer construction.

Abouzaid's finite-list  $A_\infty$  functor [Abo17, Theorem 2.10 and the chain-level construction] is cohomologically faithful on these objects. His fully faithful theorem [Abo21, Theorem 1.1 and Corollary 1.2] also applies. These are undeformed theorems for our fixed torus fibration. Its rigid analytic mirror has the independent multiplicative periods  $T$  in each coordinate, and is the analytification of  $B$ . Projective rigid GAGA identifies the resulting perfect complexes with algebraic ones. Opposite boundary-order conventions are matched by taking the  $A_\infty$  opposite and perfect duality.

Each  $D_A$  meets each fiber once and transversely. Its local family Floer complex is consequently a line in one degree. Its image is a shifted line bundle, with numerical Chern class

$$c_1(\mathcal{L}_A) = \epsilon B(A), \quad \epsilon \in \{1, -1\} \quad (2.8)$$

for a single sign determined by conventions. We give the numerical calculation, following [Opea, Section 9.1], to fix what information is being used. On the real universal cover of the base, lift the section to  $Ax + \nabla f(x)$  and use the paths from the zero section given by this lift. On a contractible overlap with the same lift, the winding number in the continuation coefficient is zero: after closing with these paths it is the boundary winding of a disk over the contractible base cover. The scalar continuation factors can be trivialized there. Under the deck translation  $b \in \mathbb{Z}^g$ , the change of lift gives the monomial multiplier  $Z^{-\epsilon Ab}$ . The standard theta line has precisely this character. One can check all symmetric matrices by tensoring pullbacks for rank-one symmetric matrices, which span the integral symmetric matrices. Equal monomial characters differ by a scalar character, hence by an element of  $\text{Pic}^0(B)$ . This proves (2.8); no identification of that  $\text{Pic}^0$  element is needed.

The shifts can be chosen common to our list. Continue the graph gradings from the flat volume grading through the Lagrangian graph planes  $\text{graph}(tA)$  in the fixed tangent space. These planes remain transverse to the vertical plane, so their intersection index is constant. This is a path of tangent planes; it does not assert an isotopy between closed graph tori of different integral slopes. Exact perturbations have tangent planes in the same contractible space of symmetric graphs. There

is also a check using morphisms. Choose all successive slope differences positive definite and include three ordered scalar slopes. Opposite Floer and sheaf degree concentrations would require the same nonzero shift difference between each pair of three objects, which is impossible. With matching concentration, all shift differences are zero. Remove the common shift.

On such a list faithfulness already proves fullness without using the stronger theorem. Between different slopes all Floer intersections lie in one degree and their number is  $|\det(A' - A)|$ . Riemann–Roch and ample or anti-ample cohomology vanishing give the identical Ext dimensions for the mirror lines, independently of their  $\text{Pic}^0$  twists. Self Floer and self Ext have the cohomology ranks of a  $g$ -torus. Hence the faithful maps are isomorphisms on all graded Homs.

We require a finite list for two reasons: its images must recover  $\text{Perf}(B)$ , and its numerical classes must give enough Euler-pairing tests to recover a Hodge-class component. The next lemma arranges both properties on the same list.

**Lemma 2.1.** *There is a finite set of symmetric integral slopes, with all successive differences positive definite, whose graph images split-generate  $\text{Perf}(B)$  and whose classes  $e^{B(A)}$  span the rational Hodge classes of  $B$ .*

*Proof.* For generation, take a vector bundle  $G$  which is a split-generator of  $\text{Perf}(B)$  [TV07, Section 3.5], and cover it by sums of sufficiently negative powers of the principal polarization, tensored with arbitrary elements of  $\text{Pic}^0(B)$ . Serre generation holds uniformly in these twists: apply relative Serre vanishing and generation to the Poincaré bundle over the proper parameter space  $\text{Pic}^0(B)$ . Here is the uniform construction, with all twists left arbitrary. Put  $\mathcal{P} = \text{Pic}^0(B)$ , choose the normalized Poincaré line bundle, and put  $K_0 = G$ . Suppose that  $K_j$  has been constructed as a vector bundle on  $B \times \mathcal{P}^j$ . Pull it back to  $B \times \mathcal{P}^{j+1}$ , and denote the new Poincaré line by  $P_{\text{new}}$ . If  $H$  represents the principal polarization and  $p$  is projection to  $\mathcal{P}^{j+1}$ , relative Serre vanishing and generation permit a sufficiently large  $m_j$  such that

$$F_j = p_*(K_j \otimes H^{m_j} \otimes P_{\text{new}}^{-1})$$

is locally free, commutes with base change, and has surjective evaluation. Tensoring that evaluation gives a surjection

$$V_j = p^* F_j \otimes H^{-m_j} \otimes P_{\text{new}} \longrightarrow K_j.$$

Its kernel  $K_{j+1}$  is a vector bundle. This construction is uniform over the proper parameter space  $\mathcal{P}^{j+1}$ , and  $m_j$  may be increased arbitrarily. After  $g+1$  steps, every tuple of twists gives the required sequence of surjections; the fiber of  $V_j$  is a sum of the prescribed line  $H^{-m_j} \otimes P_j$ . Thus all powers have been chosen before the unknown graph-image twists are specified.

The resulting exact sequence has the form

$$0 \longrightarrow K \longrightarrow V_g \longrightarrow \cdots \longrightarrow V_0 \longrightarrow G \longrightarrow 0,$$

where each  $V_j$  is a sum of one of the chosen lines. Its class in  $\text{Ext}^{g+1}(G, K)$  vanishes because  $G$  is locally free and  $\dim B = g$ . The corresponding truncated complex of the  $V_j$  therefore has  $G$  as a direct summand. These lines split-generate. Positive powers work by duality. This accommodates either sign in (2.8) and allows the scalar slopes to be as far apart as desired.

For spanning, the exponential polynomial and divisor generation above show that all integral symmetric slopes span the Hodge-class space. The integral points in any translate of the positive-definite cone are Zariski dense in the vector space of symmetric matrices. For example, dilates of a small rational box inside that cone provide arbitrarily large Cartesian grids, which cannot lie in the zero set of a fixed nonzero polynomial. Inductively choose the next slope outside the annihilating

polynomial conditions for the span already obtained and with positive-definite difference from every preceding slope. Finitely many choices suffice. Append them to the scalar slopes used for generation.  $\square$

## 2.4 The cohomology dictionary and rational areas

For computations on the symplectic torus, regard  $z_i, w_i$  as formal exterior generators and define the rational vector-space isomorphism

$$F(\gamma) = \gamma(\iota_{\partial/\partial Y_i}, X_i \wedge)(Y_1 \cdots Y_g). \quad (2.9)$$

Here a monomial is evaluated in its displayed exterior order. Multiplication by  $Y_i$  after applying  $F$  corresponds to  $\partial_{z_i}$  before applying  $F$ , and multiplication by  $X_i$  corresponds to left multiplication by  $w_i$ . Direct expansion gives

$$F(e^{B(A)}) = \prod_{i=1}^g \left( Y_i - \sum_j A_{ij} X_j \right). \quad (2.10)$$

This is the Poincaré dual of  $D_A$ , up to one orientation convention independent of  $A$ . In particular graph intersections and Euler pairings of the mirror lines are both the determinant pairing in the slope matrices, with the common sign already specified.

The numerical dictionary is now fixed. Before collecting the mirror input, we arrange the discrete polygon-area groups used by the finite category comparisons in Section 6.

**Lemma 2.2.** *The finite sphere and graph collections above can be put in transverse position, without triple intersections, so that polygon areas on each finite collection lie in a fixed discrete rational subgroup of  $\mathbb{R}$ . Their product collection is clean and has the same property. The construction can be made successively for countably many Hamiltonian copies, with a denominator depending on the finite subcollection.*

*Proof.* Use integral prequantum bundles. A sphere has trivial flat holonomy. On an undisplaced integral symmetric graph, the quadratic primitive on the real cover gives rational holonomy (the offsets are half-integral). Thus a finite prequantum power has a parallel section on each label. Hamiltonian perturbation preserves this property. First arrange transverse position and no triples. At each of the finitely many intersection points, compare the values of the chosen parallel sections. Small exact graph perturbations in Weinstein neighborhoods of the labels make all these phase differences rational: take functions constant near each intersection preimage, with separately prescribed constants and disjoint supports there. Their differentials vanish near the intersections, so the intersection positions are unchanged, while the continued prequantum sections change phase by the corresponding endpoint primitive differences. There are no triples, hence the phase conditions at distinct pair intersections are independent. Choose rational target phases sufficiently close to the original ones to keep all transversality conditions. For torus labels these are exact periodic perturbations.

After passing to a common power, all phase ratios are roots of unity. Parallel transport around a polygon boundary therefore gives its symplectic area in  $D^{-1}\mathbb{Z}$  for a fixed  $D > 0$ . This argument allows switches between labels at corners. The tensor product prequantum sections give the assertion for product labels: their intersections are products of clean components, and the phase ratios are constant roots of unity on each component. At each new layer of Hamiltonian copies only finitely many old intersections impose conditions; performing the construction successively proves the last assertion.  $\square$

**Proposition 2.3** (Starting mirror category). *For every integer  $d > 0$  there exist finite collections of graded spin spheres in  $P$  and graded spin graph tori in  $Y$ , an algebraically closed extension  $k \supset \Lambda$  of cardinality at most continuum, a projective K3 surface  $S/k$  of Picard number 19, and  $B = E^g/k$  as above, with the following properties.*

- (i) *The product labels, with their induced zero-curvature bounding cochains, form a full subcategory of the clean Fukaya category of  $P \times Y$ . Its ordinary perfect hull is equivalent to  $\text{Perf}(S \times B)$ , with all Homs and products respected.*
- (ii) *The sphere classes and mirror Mukai vectors give (2.3). The ample rational class  $h$ , decomposition (2.5), and bases satisfying (2.6) can be chosen with  $h^2 = 2d$ .*
- (iii) *The graph images are line bundles with numerical classes  $e^{\epsilon B(A)}$ , for one common  $\epsilon = \pm 1$ . They split-generate  $\text{Perf}(B)$ , and these classes span all rational Hodge classes of  $B$ . Their symplectic classes are given by (2.10).*
- (iv) *Each individual product label has self Floer cohomology with the full graded ranks of its ordinary cohomology. The labels obey the rational-area convention of Lemma 2.2.*

*Proof.* The undeformed sphere and graph calculations above first give the ordinary categories on each factor. Proposition 6.1 identifies these with their finite clean models, preserving the labeled objects, after a coefficient extension as stated. Apply Proposition 6.11 to the two clean collections. The exterior products of the two sets of mirror generators split-generate  $\text{Perf}(S \times B)$ ; this follows, for example, by taking exterior products of perfect diagonal resolutions. The full Hom comparison in that proposition proves (i). Parts (ii) and (iii) were proved above. For (iv), choose a common compatible almost complex structure on  $P$  having neither spheres nor individual-label disks, as in [Sei15, Lemma 8.4]. On  $Y$  the relative homotopy argument excludes disks for every compatible structure. Each factor label has zero bounding cochain and self Floer ranks equal to its ordinary cohomology ranks. Künneth gives the product ranks. Finally apply Lemma 2.2. All coefficient extensions can be included in one algebraically closed field of cardinality at most continuum.  $\square$

### 3 A middle-dimensional class with a prescribed cup-product kernel

We construct a rational class on  $M = P \times Y$  whose annihilator in  $H^2(M, \mathbb{Q})$  has dimension 19 and contains the symplectic class. The construction also makes cup product injective on the tensor product of this annihilator with  $H^1(M, \mathbb{Q})$ . These two properties will control the second cohomology of the manifold constructed in Section 4.

We retain the cohomology dictionary of Section 2. Thus  $g = 512$ ,  $n = g + 2$ ,  $q = 4$ , and  $a \in H^2(P, \mathbb{Q})$  satisfies  $a^2 = q \text{pt}_P$ . Fix a positive integer  $d$  and put

$$r = 2d, \quad c = qr, \quad (\lambda_1, \dots, \lambda_{18}) = (1, \dots, 1, r).$$

Under the isometry (2.3), regard the basis  $d_1, d_2, d_3$  of  $\Delta$  and the basis  $n_1, \dots, n_{18}$  of  $\mathcal{N}$  as classes in  $a^\perp \subset H^2(P, \mathbb{Q})$ . In particular,

$$ad_l = an_i = d_l n_i = 0, \quad n_i n_j = -\delta_{ij} \lambda_i \text{pt}_P. \quad (3.1)$$

The products  $d_l d_k$  are given by the nondegenerate restriction of the cup-product form to  $\Delta$ ; their individual values will not be needed. All cohomology classes in this section have rational coefficients unless a field extension is specified.

We will arrange that the annihilator projects isomorphically onto  $\mathcal{N} \oplus \mathbb{Q}a \subset H^2(P, \mathbb{Q})$ . Thus its torus component must be determined uniquely by its nineteen coefficients on  $P$ . The following matrices provide the linear relation between these two components.

### 3.1 Rational symmetric Clifford matrices

The required matrix sizes and square factors matter: the negative directions  $n_i$  have different lengths, and the matrices must compensate for them. We use the normed-algebra construction of Clifford matrices (see, for example, [Bae02, Sections 2.2–2.3]), followed by rational changes of basis. We give the construction explicitly.

**Proposition 3.1.** *There are symmetric matrices  $G_1, \dots, G_{18} \in M_g(\mathbb{Q})$  such that*

$$G_i^2 = \frac{r}{\lambda_i} I_g, \quad G_i G_j = -G_j G_i \quad (i \neq j). \quad (3.2)$$

*They generate  $M_g(\mathbb{Q})$ . Consequently their common commutant over any field of characteristic zero is the scalar matrices. The ordered product*

$$K = \frac{2}{r^8} G_1 \cdots G_{18} \quad (3.3)$$

*satisfies*

$$K^t = -K, \quad K^2 = -cI_g, \quad KG_i = -G_i K. \quad (3.4)$$

*Proof.* First construct eight symmetric anticommuting involutions on  $\mathbb{Q}^{16}$ . Let  $\mathbb{H}_{\mathbb{Q}}$  be the quaternion algebra with basis  $1, i, j, k$ , relations  $i^2 = j^2 = k^2 = ijk = -1$ , and norm  $|a|^2 = a\bar{a}$ . On  $\mathbb{O}_{\mathbb{Q}} = \mathbb{H}_{\mathbb{Q}} \oplus \mathbb{H}_{\mathbb{Q}}$  use the product

$$(a, b)(u, v) = (au - \bar{v}b, va + b\bar{u})$$

and the norm  $|(a, b)|^2 = |a|^2 + |b|^2$ . This norm is multiplicative. Indeed, on expanding the two squared norms, their mixed terms are  $-2\operatorname{Re}(aub\bar{v})$  and  $2\operatorname{Re}(vau\bar{b})$ , which cancel because the real part of a product of quaternions is invariant under cyclic permutation. The remaining four terms give  $(|a|^2 + |b|^2)(|u|^2 + |v|^2)$ .

Use the standard orthonormal rational basis  $e_1, \dots, e_8$  of  $\mathbb{O}_{\mathbb{Q}}$ , and let  $L_x$  be the  $8 \times 8$  matrix of left multiplication by  $x$ . Multiplicativity gives  $L_x^t L_x = |x|^2 I_8$ . For  $x \neq 0$  it also gives  $L_x L_x^t = |x|^2 I_8$ , by invertibility; the latter equality is automatic at  $x = 0$ . Polarizing both identities shows that

$$L_x^t L_y + L_y^t L_x = 2\langle x, y \rangle I_8, \quad L_x L_y^t + L_y L_x^t = 2\langle x, y \rangle I_8.$$

It follows that the rational symmetric matrices

$$A_j = \begin{pmatrix} 0 & L_{e_j}^t \\ L_{e_j} & 0 \end{pmatrix} \quad (1 \leq j \leq 8)$$

satisfy  $A_i A_j + A_j A_i = 2\delta_{ij} I_{16}$ . Their product  $J_8 = A_1 \cdots A_8$  is symmetric, has square  $I_{16}$ , and anticommutes with every  $A_j$ : reversing eight factors gives the sign  $(-1)^{28} = 1$ , and commuting a chosen factor through the other seven gives the sign  $-1$ .

On  $\mathbb{Q}^{16} \otimes \mathbb{Q}^{16}$  define sixteen matrices

$$T_j = A_j \otimes I_{16}, \quad T_{8+j} = J_8 \otimes A_j \quad (1 \leq j \leq 8).$$

They are symmetric anticommuting involutions. Their product  $J_{16} = T_1 \cdots T_{16}$  is a symmetric involution anticommuting with each  $T_j$ , since  $(-1)^{120} = 1$  and  $(-1)^{15} = -1$ .

By Lagrange's four-square theorem, choose integers  $b_0, b_1, b_2, b_3$  with  $r = b_0^2 + b_1^2 + b_2^2 + b_3^2$ . The matrix of multiplication by the corresponding quaternion is

$$Q_r = \begin{pmatrix} b_0 & -b_1 & -b_2 & -b_3 \\ b_1 & b_0 & -b_3 & b_2 \\ b_2 & b_3 & b_0 & -b_1 \\ b_3 & -b_2 & b_1 & b_0 \end{pmatrix}, \quad Q_r Q_r^t = Q_r^t Q_r = r I_4.$$

For  $0 \leq a \leq 3$  and  $1 \leq i \leq 4$ , replace each block of four generators by

$$U_{4a+i} = \sum_{j=1}^4 (Q_r)_{ij} T_{4a+j}.$$

The orthogonality of the rows of  $Q_r$  gives  $U_i U_j + U_j U_i = 2r \delta_{ij} I_{256}$ . The  $U_i$  remain symmetric, and each anticommutes with  $J_{16}$ .

Equip  $\mathbb{Q}^{256} \otimes \mathbb{Q}^2$  with the symmetric form of matrix  $H = I_{256} \otimes \text{diag}(1, r)$ . The eighteen matrices

$$\tilde{G}_i = U_i \otimes I_2 \quad (1 \leq i \leq 16), \quad \tilde{G}_{17} = J_{16} \otimes \begin{pmatrix} 0 & r \\ 1 & 0 \end{pmatrix}, \quad \tilde{G}_{18} = J_{16} \otimes \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

are self-adjoint for  $H$ . They anticommute in pairs, the first seventeen have square  $r I_g$ , and the last has square  $I_g$ . The form  $H$  is rationally isometric to  $I_g$ : reorder its coordinates as  $I_{256} \oplus r I_{256}$  and use  $Q_r/r$  on each four-dimensional block of the second summand. Indeed  $(Q_r/r)^t (r I_4) (Q_r/r) = I_4$ . Thus there is an explicit rational matrix  $V$  with  $V^t H V = I_g$ . Set  $G_i = V^{-1} \tilde{G}_i V$ . Self-adjointness for  $H$  now becomes ordinary symmetry, and (3.2) follows.

To prove generation without using a classification of Clifford algebras, write  $G_I = G_{i_1} \cdots G_{i_k}$  for  $I = \{i_1 < \cdots < i_k\} \subset \{1, \dots, 18\}$ , with  $G_\emptyset = I_g$ . Every nonempty  $G_I$  has trace zero. If  $|I|$  is even, conjugation by any  $G_j$  with  $j \in I$  changes its sign. If  $|I|$  is odd, choose  $j \notin I$ ; such a  $j$  exists because 18 is even, and conjugation again changes its sign. Each  $G_j$  is invertible, so trace invariance proves the assertion. If  $I \neq J$ , the product  $G_I G_J$  is a nonzero rational multiple of  $G_{I \Delta J}$  and hence has trace zero. In contrast,  $G_I^2$  is a nonzero rational scalar matrix. The  $2^{18} = g^2$  matrices  $G_I$  are therefore linearly independent, as one sees by multiplying a linear relation by  $G_J$  and taking its trace. They form a basis of  $M_g(\mathbb{Q})$ . Generation and the commutant assertion follow, also after extension of scalars.

Finally, reversing the eighteen factors introduces  $(-1)^{153} = -1$ , and commuting a generator through the other seventeen introduces  $-1$ . Since  $\prod_i (r/\lambda_i) = r^{17}$ , we obtain

$$K^t = -K, \quad K^2 = \frac{4}{r^{16}} (-1)^{153} r^{17} I_g = -4r I_g = -c I_g.$$

This also proves the anticommutation in (3.4). □

### 3.2 Separating the exterior-algebra operators

Recall the formal exterior generators  $z_1, \dots, z_g, w_1, \dots, w_g$ , the forms  $B(A) = z^t A w$  and  $\theta = \sum_i z_i w_i$ , and the complement transform  $F$  of (2.9). For a skew matrix  $C$  put

$$W(C) = \frac{1}{2} \sum_{i,j} C_{ij} w_i w_j, \quad Z(C) = \frac{1}{2} \sum_{i,j} C_{ij} z_i z_j.$$

Both assignments are injective on skew matrices. For any matrix  $D$  and any skew matrix  $R$ , define operators on the exterior algebra by

$$e_D = \sum_{i,j} D_{ij} w_i \partial_{z_j}, \quad A_2(C, R, D) = W(C) \wedge (-) + \frac{1}{2} \sum_{i,j} R_{ij} \partial_{z_i} \partial_{z_j} + e_D. \quad (3.5)$$

Here  $\partial_{z_i}$  is the left exterior derivative:  $\partial_{z_i}(uv) = (\partial_{z_i} u)v + (-1)^{\deg u} u \partial_{z_i} v$ . Products of operators act on the rightmost input first.

These are precisely the operators obtained by transporting multiplication by a degree-two torus class through  $F$ . In fact the anticommutation relations for exterior multiplication and contraction give

$$X_i F(\gamma) = F(w_i \gamma), \quad Y_i F(\gamma) = F(\partial_{z_i} \gamma).$$

For the second identity, commute  $Y_i \wedge (-)$  through the substituted operators in the definition of  $F$ ; its anticommutator with  $\iota_{\partial/\partial Y_j}$  is  $\delta_{ij}$ , and it kills the vacuum  $Y_1 \cdots Y_g$ . Thus multiplication by

$$\frac{1}{2} X^t C X + X^t D Y + \frac{1}{2} Y^t R Y \quad (3.6)$$

corresponds to  $A_2(C, R, D)$ , with the displayed signs.

Choose pairwise disjoint sets  $J_1, J_2, J_3$ , each consisting of three distinct positive integers, and put

$$\beta_l = \sum_{j \in J_l} e^{j\theta} \quad (1 \leq l \leq 3). \quad (3.7)$$

The exponential is the finite exterior-algebra sum  $e^{j\theta} = \sum_{k=0}^g j^k \theta^k / k!$ . Three distinct coefficients in each sum will eliminate the three matrix parameters  $C$ ,  $R$ , and  $D - D^t$ .

**Lemma 3.2.** *Let  $k$  be a field of characteristic zero, let  $g \geq 10$ , and let  $J_1, J_2, J_3 \subset k^\times$  be pairwise disjoint sets of three distinct elements. Define  $\beta_l$  by (3.7) in  $\bigwedge_k \langle z_1, \dots, z_g, w_1, \dots, w_g \rangle$ . For skew matrices  $C, R$ , an arbitrary matrix  $D$ , scalars  $s_l$ , and an arbitrary exterior form  $\rho$ , the equations*

$$A_2(C, R, D) \beta_l + s_l \rho = 0 \quad (1 \leq l \leq 3) \quad (3.8)$$

imply

$$C = R = 0, \quad D = D^t, \quad s_l \rho = 0 \quad (1 \leq l \leq 3).$$

Moreover, the three  $\beta_l$  are linearly independent modulo the subspace of exterior forms of degrees at most two.

*Proof.* Write  $U = \bigwedge^2 \langle w_1, \dots, w_g \rangle$ . We first show that the nine subspaces  $e^{j\theta} U$ , for  $j \in J_1 \cup J_2 \cup J_3$ , have direct sum. The elementary Lefschetz fact needed here is that

$$U \longrightarrow \bigwedge^{2k+2} \langle z, w \rangle, \quad u \longmapsto \theta^k u \quad \text{is injective for } 0 \leq k \leq g-2. \quad (3.9)$$

To see this directly, write  $u = \sum_{a < b} u_{ab} w_a w_b$ . For each pair  $a < b$ , choose  $k$  indices distinct from  $a, b$ . The coefficient in  $\theta^k u$  with exactly these  $z$ -indices and with these  $w$ -indices together with  $a, b$  is  $\pm k! u_{ab}$ . No other coefficient of  $u$  contributes. This proves injectivity. Now suppose  $\sum_j e^{j\theta} u_j = 0$  with  $u_j \in U$ . Taking total exterior degrees  $2, 4, \dots, 18$  and applying (3.9) gives

$$\sum_j j^k u_j = 0 \quad (0 \leq k \leq 8).$$

The Vandermonde matrix is invertible because the nine  $j$ 's are distinct, so every  $u_j$  is zero.

Our sign convention for exterior derivatives gives

$$\partial_{z_i} e^{j\theta} = j w_i e^{j\theta}, \quad \partial_{z_i} \partial_{z_h} e^{j\theta} = j^2 w_i w_h e^{j\theta}.$$

The minus sign from differentiating past  $w_h$  in the second formula is canceled by interchanging  $w_h$  and  $w_i$ . It follows that

$$A_2(C, R, D) e^{j\theta} = e^{j\theta} W(C + j(D - D^t) + j^2 R). \quad (3.10)$$

In particular  $A_2 \beta_l$  lies in  $V_l = \bigoplus_{j \in J_l} e^{j\theta} U$ , and the three  $V_l$  have direct sum. If  $\rho = 0$ , each  $A_2 \beta_l$  vanishes. If  $\rho \neq 0$ , at most one of the  $s_l$  can be nonzero: two nonzero ones would place  $\rho$  in the intersection of two distinct  $V_l$ . Thus in either case at least one  $A_2 \beta_l$  vanishes. By the direct sum just proved and (3.10), the matrix polynomial  $C + j(D - D^t) + j^2 R$  vanishes at the three elements of that  $J_l$ . Each entry has degree at most two, so all three coefficient matrices are zero. Equation (3.10) then makes every  $A_2 \beta_l$  zero, and (3.8) gives  $s_l \rho = 0$ .

For the final assertion, suppose  $\sum_l a_l \beta_l$  has degree at most two, and set  $b_j = a_l$  for  $j \in J_l$ . The components of degrees  $2k$ ,  $2 \leq k \leq 10$ , give  $\sum_j b_j j^k = 0$ , since  $\theta^k \neq 0$  for  $k \leq g$ . This coefficient matrix is a Vandermonde matrix multiplied by the invertible diagonal matrix with entries  $j^2$ . Hence  $b_j = 0$  for every  $j$ , and every  $a_l$  is zero.  $\square$

### 3.3 The class and its annihilator

Fix the matrices of Proposition 3.1. Set

$$\chi = \frac{2}{q} W(K), \quad \eta = -\frac{1}{q} B(K), \quad \nu = Z(K), \quad (3.11)$$

and define

$$\alpha = \sum_{l=1}^3 d_l F(\beta_l) + \sum_{i=1}^{18} n_i F(B(G_i)) + F(\chi) + a F(\eta) + \text{pt}_P F(\nu). \quad (3.12)$$

Products in this formula mean exterior products from the two factors of  $M$ . Although  $\beta_l$  has components in several exterior degrees,  $\alpha$  is homogeneous of cohomological degree  $n$ . Indeed, a monomial with  $p$   $z$ -factors and  $s$   $w$ -factors has image under  $F$  in degree  $g - p + s$ . The first two groups in (3.12) therefore have torus degree  $g$ ; the last three have torus degrees  $g + 2, g, g - 2$ , respectively. The accompanying  $P$ -degrees are  $2, 2, 0, 2, 4$ , so every term has degree  $g + 2 = n$ . All coefficients are rational.

The terms of (3.12) serve three purposes in the kernel calculation. The  $\Delta$ -terms, together with  $\chi \neq 0$ , allow Lemma 3.2 to remove the  $\Delta$ -coordinates, the  $XX$ - and  $YY$ -components, and the skew part of the torus matrix. The  $\mathcal{N}$ -terms then turn the coefficients of the  $n_i$  into commutator equations involving  $G_i$ . Finally,  $F(\chi) + a F(\eta) + \text{pt}_P F(\nu)$  couples these equations so that the surviving torus matrix is uniquely determined by arbitrary coefficients in  $\mathcal{N} \oplus \mathbb{Q}a$ . The proof below verifies each of these claims.

**Proposition 3.3.** *Let  $\alpha \in H^n(M, \mathbb{Q})$  be the class (3.12), where  $M = P \times Y$ ,  $g = 512$ , and  $n = g + 2$ . Put*

$$D_{\text{tot}} = \dim H^2(M, \mathbb{Q}) = 22 + \binom{2g}{2}.$$

The kernel of  $\alpha \cup (-) : H^2(M, \mathbb{Q}) \longrightarrow H^{n+2}(M, \mathbb{Q})$  is

$$\Omega = \left\{ \sum_{i=1}^{18} t_i n_i + ua + X^t DY : D = uI_g + K^{-1} \sum_{i=1}^{18} \lambda_i t_i G_i, \quad t_i, u \in \mathbb{Q} \right\}. \quad (3.13)$$

Consequently  $\dim \Omega = 19$ , and  $\Omega$  contains the symplectic class  $a + \sum_i X_i Y_i$ . Furthermore cup product induces an injection

$$\Omega \otimes H^1(M, \mathbb{Q}) \longrightarrow H^3(M, \mathbb{Q}). \quad (3.14)$$

*Proof.* Since  $H^1(P, \mathbb{Q}) = 0$ , every degree-two class on  $M$  has a unique expression

$$x = \sum_{l=1}^3 s_l d_l + \sum_{i=1}^{18} t_i n_i + ua + \frac{1}{2} X^t C X + X^t D Y + \frac{1}{2} Y^t R Y,$$

where  $C, R$  are skew and  $D$  is arbitrary. Let  $A_2 = A_2(C, R, D)$ . Project  $x\alpha$  to the  $\Delta$ -summand of its degree-two  $P$ -component, and apply  $F^{-1}$  to the torus factor. The resulting equations are

$$s_l \chi + A_2 \beta_l = 0 \quad (1 \leq l \leq 3).$$

Here  $s_l \chi$  comes from multiplying the  $P$ -component of  $x$  by  $F(\chi)$ , and the other term comes from the torus component of  $x$ . Since  $K$  is invertible and skew,  $W(K) \neq 0$ . Lemma 3.2 therefore gives

$$s_1 = s_2 = s_3 = 0, \quad C = R = 0, \quad D = D^t. \quad (3.15)$$

We now compute the remaining components, keeping the exterior signs explicit. For symmetric  $D$  and  $G$ , differentiation and reordering give

$$\begin{aligned} e_D B(G) &= \sum_{i,j,b} D_{ij} G_{jb} w_i w_b = W(DG - GD), \\ e_D B(K) &= \sum_{i,j,b} D_{ij} K_{jb} w_i w_b = W(DK + KD), \\ e_D Z(K) &= \sum_{i,j,b} D_{ij} K_{jb} w_i z_b = B(KD), \\ e_D W(K) &= 0. \end{aligned}$$

For the third equality,  $\partial_{z_j} Z(K) = \sum_b K_{jb} z_b$ ; moving  $w_i$  past  $z_b$  gives the coefficient matrix  $-(DK)^t = KD$ . In the first two equalities we used  $\sum_{i,b} A_{ib} w_i w_b = W(A - A^t)$ .

Using (3.1), the degree-zero, degree-two, and degree-four  $P$ -components of  $x\alpha = 0$ , after (3.15), are exactly

| $P$ -component | equation before applying $F$                           |
|----------------|--------------------------------------------------------|
| 1              | $e_D \chi = 0,$                                        |
| $n_i$          | $t_i \chi + e_D B(G_i) = 0,$                           |
| $a$            | $u \chi + e_D \eta = 0,$                               |
| $\text{pt}_P$  | $-\sum_i \lambda_i t_i B(G_i) + qu\eta + e_D \nu = 0.$ |

The first equation is automatic. The remaining equations are equivalent, by injectivity of  $W$  on skew matrices and of  $B$  on all matrices, to

$$\begin{aligned} [D, G_i] &= -\frac{2}{q} t_i K \quad (1 \leq i \leq 18), \\ DK + KD &= 2uK, \\ KD &= uK + \sum_i \lambda_i t_i G_i. \end{aligned} \quad (3.16)$$

The last equation determines  $D$  uniquely as in (3.13). It remains to check that it imposes no further conditions on  $u, t_i$ .

Write  $A = \sum_i \lambda_i t_i G_i$  and  $D = uI_g + K^{-1}A$ . Both  $A$  and the  $G_i$  are symmetric, and  $AK = -KA$ . Since  $(K^{-1})^t = -K^{-1}$ , we have

$$(K^{-1}A)^t = -AK^{-1} = K^{-1}A,$$

so  $D$  is symmetric. Anticommutation also gives  $K^{-1}AK = -A$ , whence  $DK + KD = 2uK$ . For the first equation of (3.16), use  $G_i K^{-1} = -K^{-1}G_i$  and (3.2) to obtain

$$\begin{aligned} [K^{-1}A, G_i] &= K^{-1}(AG_i + G_iA) = 2\lambda_i t_i \frac{r}{\lambda_i} K^{-1} \\ &= 2rt_i K^{-1} = -\frac{2}{q} t_i K, \end{aligned}$$

where  $K^{-1} = -K/c$  and  $c = qr$ . Finally  $KD = uK + A$  by definition. Thus every choice of  $u, t_i$  satisfies all the equations, proving (3.13).

Projection of  $\Omega$  to  $H^2(P, \mathbb{Q})$  identifies it with  $\mathcal{N} \oplus \mathbb{Q}a$ , so its dimension is 19. Taking  $t_i = 0$  and  $u = 1$  yields the symplectic class. For the last assertion, use  $H^1(M, \mathbb{Q}) = H^1(Y, \mathbb{Q})$  and project the proposed cup product to  $H^2(P, \mathbb{Q}) \otimes H^1(Y, \mathbb{Q}) \subset H^3(M, \mathbb{Q})$ . On  $\Omega \otimes H^1(Y, \mathbb{Q})$  this projection is the tensor product of the injective projection  $\Omega \rightarrow H^2(P, \mathbb{Q})$  with the identity. It is injective, proving (3.14).  $\square$

The construction supplies a rational 19-dimensional subspace  $\Omega$  with  $\Omega\alpha = 0$  and no relations after multiplying by  $H^1(M, \mathbb{Q})$ . These are the precise cohomological properties used to realize a multiple of  $\alpha$  by an immersed Lagrangian in the next section. The matrices and Lemma 3.2 will also be used in the deformation calculation.

## 4 A graded immersed representative with controlled second cohomology

We now realize the middle-dimensional class constructed in Section 3 by a Lagrangian immersion. Two features of the representative will be used in the Floer argument: its second Betti number is prescribed, and its double points contribute no even-degree generators. The latter condition will be obtained without performing surgery on the Lagrangian double points, so that the former condition is retained.

Throughout this section homology and cohomology have rational coefficients unless indicated otherwise. Put

$$M = P \times Y, \quad g = 512, \quad n = g + 2 = 514, \quad \omega = \omega_P + \omega_Y.$$

The form is normalized so that  $[\omega]$  is integral. We use the determinant trivialization and grading convention fixed in Section 2. Let  $\alpha \in H^n(M; \mathbb{Q})$  and  $\Omega \subset H^2(M; \mathbb{Q})$  be the class and subspace in (3.12) and (3.13). The properties from Proposition 3.3 needed here are

$$\begin{aligned} \ker(\alpha \cup - : H^2(M) \longrightarrow H^{n+2}(M)) &= \Omega, \quad \dim \Omega = 19, \quad [\omega] \in \Omega, \\ \Omega \otimes H^1(M) \longrightarrow H^3(M), \quad \zeta \otimes v &\longmapsto \zeta \cup v, \quad \text{is injective.} \end{aligned} \tag{4.1}$$

Write  $D_{\text{tot}} = 22 + \binom{2g}{2} = b_2(M)$ . Fix the finite list  $\mathcal{D}$  of embedded product labels from Section 2, with the rational prequantization data of Lemma 2.2.

**Proposition 4.1.** *There are a closed connected oriented spin  $n$ -manifold  $L$ , a positive integer  $\mu$ , and an integrally graded Lagrangian immersion  $i : L \rightarrow M$  with the following properties.*

- (i) *The immersion is self-transverse, has only double points, and*

$$\text{PD}(i_*[L]) = \mu\alpha.$$

- (ii) *Pullback induces an isomorphism*

$$H^2(M)/\Omega \xrightarrow{\sim} H^2(L).$$

*In particular,  $b_2(L) = D_{\text{tot}} - 19$ . Pushforward  $i_* : H_{n-2}(L) \rightarrow H_{n-2}(M)$  is injective.*

- (iii) *The two ordered intersection degrees at every double point are odd.*

- (iv) *The immersion is transverse to each member of  $\mathcal{D}$ . Its double values avoid those labels, and its intersections with them avoid intersections between distinct labels. There is an integer  $N_0 > 0$  such that the symplectic area of every piecewise smooth disk whose boundary lifts, between successive corners, to members of  $\mathcal{D} \cup \{i\}$  belongs to  $N_0^{-1}\mathbb{Z}$ .*

*Here an immersion boundary segment is specified by a lift to its source; corners may switch between the two preimages of a double point.*

The topology and local handle construction adapt [Opea, Lemmas 3.3–3.6]. The adaptations requiring proof are that  $M$  has nontrivial tangent bundle and nonzero second homotopy group. We first construct the source with its prescribed stable tangent bundle, then arrange zero-area filling classes explicitly, and finally carry out the local replacement inside a contractible domain.

#### 4.1 The torus bundle and a tangentially prescribed representative

Choose linearly independent integral classes  $e_1, \dots, e_{19}$  spanning  $\Omega$  over  $\mathbb{Q}$ . Let

$$\pi : \widehat{M} \longrightarrow M$$

be the principal  $(S^1)^{19}$ -bundle with these Chern classes. The purpose of this bundle is to kill exactly the unwanted degree-two classes while retaining a lift of the required fundamental class.

**Lemma 4.2.** *Pullback identifies  $H^2(\widehat{M})$  with  $H^2(M)/\Omega$ . Moreover,  $\text{PD}(\alpha) \in H_n(M)$  belongs to the image of  $\pi_* : H_n(\widehat{M}) \rightarrow H_n(M)$ .*

*Proof.* The manifold  $M$  admits a compact Kähler structure, independently of the almost complex structures later used for holomorphic curves. Its de Rham algebra is therefore formal [DGMS75, Section 6, Main Theorem]. The connection model for a principal torus bundle, combined with formality, gives the finite-dimensional real differential graded algebra

$$\mathcal{A} = H^*(M; \mathbb{R}) \otimes \bigwedge(t_1, \dots, t_{19}), \quad |t_j| = 1, \quad dt_j = e_j, \quad (4.2)$$

as a model for  $\widehat{M}$ , compatible with pullback from  $M$ ; see [DGMS75, Hirsch Lemma (3.1) and the principal-fibration discussion]. To specify this compatibility, a connection on the bundle gives the model  $\Omega^*(M) \otimes \bigwedge(t_j)$  with  $dt_j$  the curvature representatives. Along a formality quasi-isomorphism, replacing a curvature representative by one differing by  $d\beta_j$  is absorbed by replacing  $t_j$  by  $t_j + \beta_j$ . Thus the usual Hirsch extension construction supplies a zigzag respecting the base inclusion.

The degree-two part of (4.2) is the sum of  $H^2(M)$ ,  $H^1(M) \otimes \text{span}(t_j)$ , and  $\bigwedge^2 \text{span}(t_j)$ . The differential on the second summand is injective by (4.1). Its value lies in  $H^3(M)$ . The differential on the third summand lies in  $H^2(M) \otimes \text{span}(t_j)$  and is injective: the coefficient of each  $t_j$  is a linear combination of the independent  $e_i$ , and its vanishing makes all coefficients of the original exterior two-form zero. These two images lie in different exterior degrees and cannot cancel. The boundaries in  $H^2(M)$  are exactly the span of the  $e_j$ . This proves the first assertion over  $\mathbb{R}$ .

For the second assertion, define a functional on  $\mathcal{A}^n$  by

$$c(\gamma) = \int_M \alpha \cup \gamma \quad (\gamma \in H^n(M; \mathbb{R})), \quad c(H^*(M) \otimes \bigwedge^{>0} \text{span}(t_j)) = 0.$$

The only boundaries on which this functional could be nonzero are the terms  $e_j \cup \beta$  with  $\beta \in H^{n-2}(M; \mathbb{R})$ . They are annihilated because  $\alpha \cup e_j = 0$ . Hence  $c$  is a cycle in the dual complex, and its image under the dual of the base inclusion is  $\text{PD}(\alpha)$ . This proves the homology lifting assertion over  $\mathbb{R}$ . Both the pullback and pushforward maps in question are defined over  $\mathbb{Q}$ . Their kernels and images commute with scalar extension; a rational vector in the real extension of a rational image already belongs to that image. Both assertions therefore hold over  $\mathbb{Q}$ .  $\square$

**Lemma 4.3.** *There is a stable oriented spin real bundle  $\xi$  on  $M$  whose complexification is stably isomorphic to  $TM$  as a complex bundle with the specified determinant trivialization.*

*Proof.* It suffices to construct the bundle on  $P$ . On a four-dimensional CW complex stable spin bundles are classified by  $p_1/2 \in H^4(-; \mathbb{Z})$ : the space  $B\text{Spin}$  is three-connected, its fourth homotopy group is  $\mathbb{Z}$ , and  $p_1/2$  represents a generator in that degree. Choose a stable spin bundle with

$$\int_P p_1(\xi)/2 = -12.$$

For a real oriented bundle,  $c_1(\xi \otimes_{\mathbb{R}} \mathbb{C}) = 0$  and  $c_2(\xi \otimes_{\mathbb{R}} \mathbb{C}) = -p_1(\xi)$ . Thus its complexification has second Chern number 24, the second Chern number of the quartic K3 surface. Bundles with structure group  $SU$  over a four-complex are classified stably by  $c_2$ , by the same obstruction-theoretic argument using  $BSU$ . This identifies the complexification with  $TP$  together with its determinant trivialization. Pull back to  $M$  and add the trivial real summands whose complexifications give the tangent bundle of  $Y$ .  $\square$

We use a stable tangent identification, not just the vanishing of a characteristic class. Its existence will provide both the spin structure needed for orientations and the formal graded Lagrangian monomorphism.

**Lemma 4.4.** *There are a closed connected oriented spin  $n$ -manifold  $L_0$ , a map  $f_0 : L_0 \rightarrow M$ , and a positive integer  $\mu$  such that*

$$\begin{aligned} \text{PD}(f_{0*}[L_0]) &= \mu\alpha, & TL_0 &\simeq_{\text{st}} f_0^*\xi, \\ f_0^* : H^2(M) &\rightarrow H^2(L_0), & \ker f_0^* &= \Omega, & f_{0*} : H_{n-2}(L_0) &\hookrightarrow H_{n-2}(M). \end{aligned} \tag{4.3}$$

*The map on fundamental groups is surjective. The source has an additional free circle generator mapping trivially to  $M$ , and its Euler characteristic may be made arbitrarily large without changing the other assertions.*

*Proof.* Let  $\tilde{\alpha} \in H_n(\widehat{M}; \mathbb{Q})$  be the lift supplied by Lemma 4.2. Apply bordism with the stable tangent bundle prescribed by  $\pi^*\xi$ . More explicitly, normalize the bundle to virtual rank zero and denote it by  $\tilde{\xi}$ . Pontryagin–Thom identifies this bordism group with the degree- $n$  stable homotopy of the

Thom spectrum of  $-\pi^*\bar{\xi}$ ; see [RF24, Theorems 2.5–2.6] for this tangential formulation. Rational stable Hurewicz is an isomorphism, because the rational sphere spectrum is the Eilenberg–Mac Lane spectrum  $H\mathbb{Q}$ . The ordinary Thom isomorphism, applicable because the bundle is oriented, identifies the rational homology in question with  $H_n(\widehat{M}; \mathbb{Q})$ . Under these identifications the map sends a manifold to its pushed-forward fundamental class. Clearing denominators therefore gives a representative of  $\mu\bar{\alpha}$ , for some positive integer  $\mu$ , with a stable identification of its tangent bundle with the pullback of  $\pi^*\xi$ .

We modify this representative by surgeries over  $\widehat{M}$ , retaining the stable identification on the surgery cobordisms. Set  $\Gamma = \pi_1(\widehat{M})$ . The homotopy sequence of the torus bundle exhibits  $\Gamma$  as an extension of  $\pi_1(M) = \mathbb{Z}^{2g}$  by a quotient of  $\mathbb{Z}^{19}$ . The latter subgroup is central: the structure group of the principal bundle acts trivially on the fundamental group of its fiber. Consequently  $\Gamma$  is finitely generated nilpotent, hence polycyclic, and  $\mathbb{Z}\Gamma$  is Noetherian [Hal54].

First join components by one-handles mapped along paths in  $\widehat{M}$ . Add circle handles mapped to a finite generating set of  $\Gamma$ . This makes the fundamental-group map surjective. Its kernel is normally finitely generated, since the source and target groups are finitely presented. Represent normal generators by disjoint embedded circles and use their target nullhomotopies to perform surgeries over  $\widehat{M}$ . The normal bundles of these circles are trivial, and a normal framing can be twisted by  $\pi_1(SO(n-1))$  to make the stable tangent identification extend over each surgery trace. The stabilization map on this  $\pi_1$  is surjective. The reverse handles have index  $n-1 > 2$ , so the resulting map induces an isomorphism on  $\pi_1$ .

The second homotopy groups of both the new source and  $\widehat{M}$  are finitely generated  $\mathbb{Z}\Gamma$ -modules. Indeed each space has a finite CW structure, its simply connected universal cover has a cellular chain complex of finite free  $\mathbb{Z}\Gamma$ -modules in every degree, and Hurewicz identifies  $\pi_2$  with the second homology of that complex. Noetherianity makes the relevant kernels and quotients finitely generated. Choose module generators of the target  $\pi_2$ . For each, take a connected sum with  $S^2 \times S^{n-2}$  mapped through its first factor to a sphere representing that generator. The pullback of  $\pi^*\xi$  to  $S^2$  is stably trivial because it is spin. Thus the stable tangent data on this added summand may be chosen to extend over  $S^2 \times D^{n-1}$ , with the map extending through the first factor. The operation is a bordism over  $\widehat{M}$  and makes the map on  $\pi_2$  surjective without changing  $\pi_1$ .

Its kernel is again a finitely generated  $\mathbb{Z}\Gamma$ -module. Represent generators by disjoint embedded two-spheres. General position permits this because  $n = 514$ . Each normal bundle is stably trivial: along a kernel sphere the target map is nullhomotopic, and the prescribed stable tangent bundle is therefore trivial. Its rank is  $n-2$ , so stable triviality implies actual triviality over  $S^2$ . Use a nullhomotopy in  $\widehat{M}$  for the map and a normal framing for surgery. The only remaining stable framing extension obstruction lies in  $\pi_2(SO)$ , which is zero. The surgery trace attaches three-cells along the chosen module generators, killing exactly their generated submodule of  $\pi_2$ . Viewed from its other boundary, it has handles of index  $n-2 > 3$ . The final map thus induces isomorphisms on both  $\pi_1$  and  $\pi_2$ .

A map inducing these low homotopy isomorphisms induces an isomorphism on  $H_2$ , for example by the low-degree universal-cover spectral sequence. Hence its pullback on  $H^2$  is an isomorphism. Compose with  $\pi$  and apply Lemma 4.2. The composite already has the asserted degree-two surjectivity and kernel, and all the surgeries were bordisms over the target, so its fundamental class still maps to  $\mu \text{PD}(\alpha)$ .

Finally take a connected sum with a constantly mapped  $S^1 \times S^{n-1}$ , and with as many constantly mapped copies of  $S^4 \times S^{n-4}$  as desired. These products are stably parallelizable; use stable tangent data bounding their evident product bordisms. They preserve the pushed-forward fundamental class and  $H^2$ . The first supplies a free null-image circle generator. Since  $n$  is even, each of the latter summands increases the Euler characteristic by  $4 - 2 = 2$ . Denote the resulting map by  $f_0$ .

It remains to verify the asserted homology detection. For  $u \in H_{n-2}(L_0)$ , Poincaré duality and the degree-two surjectivity give  $u = f_0^* \gamma \cap [L_0]$  for some  $\gamma \in H^2(M)$ . The projection formula gives

$$f_{0*} u = \mu(\gamma \cup \alpha) \cap [M].$$

If this vanishes, (4.1) implies  $\gamma \in \Omega$ ; hence  $f_0^* \gamma = 0$  and  $u = 0$ . This also shows explicitly why the precise kernel computation, rather than only its dimension, is needed.  $\square$

## 4.2 A graded immersion and zero-area paths between branches

The source of Lemma 4.4 already has the required homology. We next realize its formal tangent data and adjust the actions of its double points. These adjustments change neither the source nor its homotopy class as a map to  $M$ .

**Lemma 4.5.** *The map  $f_0$  is homotopic to an oriented spin Lagrangian immersion with zero Maslov class. It can be made self-transverse with only double points. Its signed number  $I$  of unordered double points satisfies*

$$2I = \mu^2 \int_M \alpha^2 - (-1)^{n/2} \chi(L_0). \quad (4.4)$$

We may choose  $L_0$  so that  $I$  is nonzero and has sign  $(-1)^{n/2-1}$ .

*Proof.* Complexify the stable identification in Lemma 4.4 and use Lemma 4.3. It gives a stable complex isomorphism  $TL_0 \otimes \mathbb{C} \simeq f_0^* TM$ , respecting the determinant homotopy coming from the orientation and the ambient trivialization. Complex rank  $n$  lies in the stable range over an  $n$ -dimensional CW complex, so this stable isomorphism destabilizes in its homotopy class. Taking its unitary part sends the real tangent bundle to Lagrangian planes. The determinant phase has a nullhomotopy, so this formal monomorphism has zero Maslov class. Also  $f_0^*[\omega] = 0$ , by (4.1) and (4.3).

The Gromov–Lees theorem [Gro71, Lee76] says precisely that a Lagrangian bundle monomorphism  $TL \rightarrow TM$  covering  $f$  is homotopic through such monomorphisms to the derivative of a Lagrangian immersion when  $f^*[\omega] = 0$ ; we use the formulation in [EM02, Theorem 16.3.2], also recalled in [RF24, Theorem 2.3]. Apply it to the preceding data. The formal homotopy preserves the phase nullhomotopy and therefore supplies an integral grading. Choose its parity to give the specified orientation. A small Lagrangian perturbation makes the immersion self-transverse. Triple points have negative expected dimension and can be avoided.

For an oriented immersion into an oriented  $2n$ -manifold, the self-intersection formula is

$$[f_0(L_0)]^2 = \langle e(\nu), [L_0] \rangle + 2I.$$

The isomorphism  $J : TL_0 \rightarrow \nu$  changes the induced orientation by  $(-1)^{n(n-1)/2} = (-1)^{n/2}$ , since  $n$  is even. Consequently the normal Euler number is  $(-1)^{n/2} \chi(L_0)$ , which proves (4.4). Increasing  $\chi(L_0)$  as permitted by Lemma 4.4 makes the sign of  $I$  equal to  $(-1)^{n/2-1}$ . Here  $n/2 = 257$ , so explicitly  $I = (\mu^2 \int_M \alpha^2 + \chi(L_0))/2 > 0$ .  $\square$

For an immersion  $f$  and a path in its source between the preimages of a double point, the image is a loop. When that loop is nullhomotopic, its filling area need not be independent of the filling:  $P$  has nonzero  $\pi_2$ . The next argument therefore chooses a filling class, rather than assigning a real action without qualification.

**Lemma 4.6.** *The immersion in Lemma 4.5 may be perturbed so that each pair of double-point preimages is joined by an embedded arc whose image bounds a disk of symplectic area exactly zero. The arcs may be taken pairwise disjoint and disjoint from all other double-point preimages.*

*Proof.* The map  $\pi_1(L_0) \rightarrow \pi_1(M)$  is surjective. Starting with any path between the two preimages, insert a source loop whose image cancels the class of its image loop. The new image loop is nullhomotopic. Its filling area is well defined modulo spherical periods.

The spherical periods of  $M$  are exactly  $\mathbb{Z}$  with the chosen normalization. Indeed  $\pi_2(M) = \pi_2(P)$ , the Hurewicz map  $\pi_2(P) \rightarrow H_2(P; \mathbb{Z})$  is an isomorphism, and the primitive hyperplane class  $a$  has pairing image  $\mathbb{Z}$  on the unimodular K3 lattice. In particular every integer change of filling area is available by adding a sphere.

The extra free circle of Lemma 4.4 maps trivially to  $M$ . Its filling area modulo  $\mathbb{Z}$  can be varied independently by a closed one-form graph perturbation in an immersed Weinstein neighborhood. In more detail, the circle gives a direct summand of  $H_1(L_0; \mathbb{Z})$ ; choose a closed one-form with any sufficiently small specified period there and zero periods on a complementary basis. The graph of this form is Lagrangian, and the swept annulus changes the action by that period. Choose the new action  $\rho$  irrational modulo  $\mathbb{Z}$ . The subgroup  $\mathbb{Z} + \rho\mathbb{Z}$  is dense in  $\mathbb{R}$ . Restore transverse double points by a further arbitrarily small perturbation if necessary, retaining the irrationality of this period by the same adjustable parameter.

For every double-point path insert multiples of the null-image circle. Combining this operation with changes of filling by spheres makes the filling areas as small in absolute value as desired. All these changes are finite, since there are finitely many double points. Choose disjoint small source neighborhoods of the double preimages. A function  $F$  which is constant near one preimage and zero near its partner and outside its neighborhood defines a small exact graph perturbation  $dF$ . It fixes the two sheets near their intersection, while changing the path action by the difference of the endpoint values of  $F$  (with a fixed sign determined by the graph convention). Choose these values to cancel the residual filling areas. Disjoint supports make the corrections independent. They may be made  $C^\infty$ -small, so no new intersections appear and transversality persists. The homotopy annuli transport the chosen fillings and prove that their corrected areas are exactly zero.

Finally represent the resulting path classes by mutually disjoint embedded arcs with the specified endpoints. General position in the  $n$ -manifold gives this and avoids the other double preimages. Homotopies of these paths within the Lagrangian source sweep zero symplectic area. They therefore preserve the chosen zero-area filling property.  $\square$

### 4.3 The exact domain containing the double points

The cancellation theorem to be used below requires a simply connected Liouville target. Rather than imposing a condition on  $M$ , we construct such a target locally around the part of the immersion to be replaced.

**Lemma 4.7.** *For the immersion  $f$  of Lemma 4.6, there is an embedded compact contractible Weinstein domain  $(U, \lambda) \subset M$  with  $d\lambda = \omega|_U$  such that  $V = f^{-1}(U)$  is a disk containing all double-point preimages. On this disk  $f^*\lambda = 0$ . The immersion has a positive cylindrical collar at  $\partial V$ , and at one interior point it is a single linear Lagrangian sheet in a radial Darboux ball.*

*Proof.* Complete the disjoint arcs to an embedded finite tree  $\Theta \subset L_0$ , using connector edges avoiding all double preimages. Include an ordinary vertex with only one preimage. The graph  $G = f(\Theta)$  has exactly the identifications of the paired double preimages; its fundamental group is freely generated by the cycles coming from their arcs. These cycles are nullhomotopic in  $M$  and have the specified zero-area fillings. Conjugating a cycle by a connector changes neither property.

Thickening the source tree gives a disk, whereas thickening its image produces one target loop for each double-point identification. We first realize both thickenings by an exact zero- and one-handlebody. We then kill the target loops by two-handles whose cores lie outside that handlebody

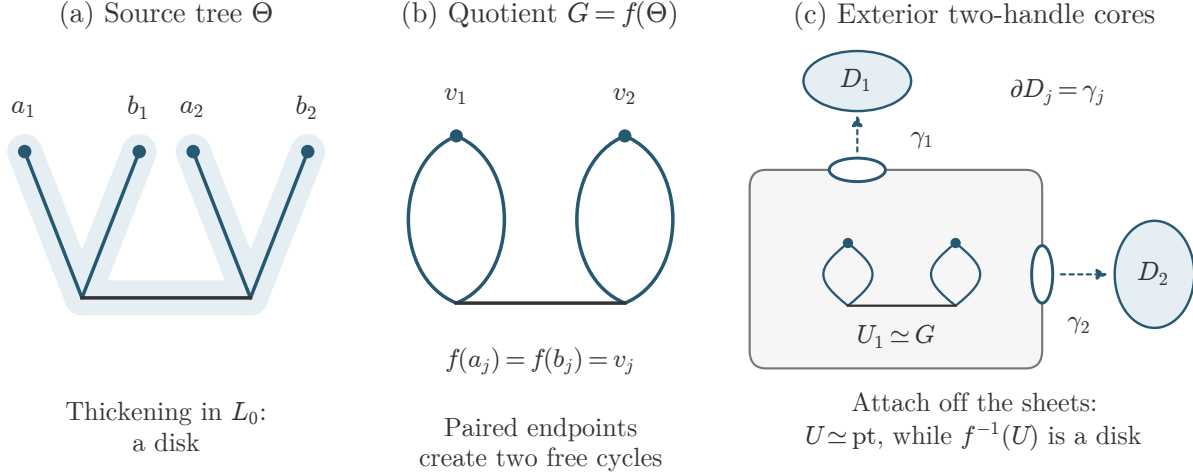


Figure 2: Source and target topology for two double points. The source tree  $\Theta$  joins the disjoint paths between paired preimages  $a_j, b_j$ ; its thickening is a disk. Identifying each pair produces  $G$ , the spine of the zero- and one-handlebody  $U_1$ . Boundary circles  $\gamma_j$  represent its free generators, and exterior isotropic disks  $D_j$  provide the cores of two-handles that make  $U$  contractible without changing its source preimage. Panel (c) records attachment data, not a planar embedding of disks and sheets: the dashed arrows indicate attachments, the immersed sheets are omitted, and the actual disk interiors lie outside  $U_1$  and avoid  $f(L_0)$  in the 1028-dimensional ambient manifold.

and avoid the immersed source. Thus the target becomes contractible while its source preimage remains the original disk; Figure 2 summarizes these different roles of the source and target topology.

*Zero- and one-handles.* Choose small disjoint radial Darboux balls at the vertices. The single sheet, or the pair of transverse Lagrangian sheets at a double point, can be made linear in these coordinates. For a pair, straighten the first sheet, write the other as a gradient graph over its transverse linear plane, and use the corresponding symplectic shear to straighten it while fixing the first. Arrange the incident tree edges as distinct radial rays. Join the balls by sufficiently thin Weinstein one-handles along the remaining edges, carrying strips of the corresponding sheets.

Here are explicit handle and attachment data, also fixing what will be needed for two-handles. In coordinates  $(u, v, z) \in \mathbb{R}^k \times \mathbb{R}^k \times \mathbb{R}^{2(n-k)}$ , with form  $\sum du_j \wedge dv_j + \omega_z$ , use the Liouville field

$$Z_k = - \sum u_j \partial_{u_j} + 2 \sum v_j \partial_{v_j} + \frac{1}{2} \sum z_j \partial_{z_j}, \quad k = 1, 2. \quad (4.5)$$

It has an index- $k$  quadratic Lyapunov function. A thin handle box has core  $v = z = 0$ , negative face near  $|u| = 1$ , and the remaining faces positive after rounding. For  $k = 1$ ,  $v = 0$  with  $z$  in a fixed Lagrangian plane is an invariant Lagrangian strip.

We require attachment maps strict for the selected contact forms, not merely contactomorphisms of their kernels. To obtain them, choose near each attaching isotropic locus a slice transverse to the Reeb field and tangent to the contact hyperplane along that locus. Reeb flow expresses the form as  $da + \beta$  on a product neighborhood of the slice, with  $d\beta$  symplectic. Match the slices by the isotropic symplectic neighborhood theorem with the chosen symplectic normal data. The two resulting primitives differ by a closed one-form restricting to zero on the isotropic locus. A sufficiently small neighborhood retracts onto that locus, so the difference is  $dh$  with  $h = 0$  there. A shift of the Reeb

coordinate by  $h$ , with the appropriate sign, makes the map strict. The same argument works for a Legendrian patch. Extending by Liouville flow then gives collar identifications preserving both the symplectic forms and their primitives.

Choose the strip framings along each source edge so that its two ends give these Legendrian patch identifications. The relative Lagrangian neighborhood theorem extends the strip map to a symplectic map of a thin handle, with the prescribed collar germs. One can see its relative character directly: take tubular coordinates agreeing with the germs, identify their linear symplectic normal data, and apply the relative Moser argument, with its vector field zero on a smaller end collar. Shrinking the finitely many handles makes them embedded and mutually disjoint away from their prescribed attachments. They meet no other sheets, since  $f^{-1}(G) = \Theta$ . Their model primitives glue by the strict collars.

The resulting rounded zero- and one-handlebody  $U_1$  retracts onto  $G$ . Its preimage thickens  $\Theta$ , hence is a disk. The Liouville field is tangent to all its immersed sheets: this holds in the radial balls and in the strip model, and the attachment maps preserve it. Equivalently  $f^*\lambda = 0$  there. It follows that the boundary of the disk is Legendrian and has a positive cylindrical collar. Retain a smaller radial ball at the single-sheet vertex.

*Exterior filling disks with prescribed area.* Represent a free basis of  $\pi_1(U_1)$  by disjoint isotropic circles  $\gamma_j \subset \partial U_1$  avoiding the Legendrian boundary of the source disk. These exist by the subcritical contact embedding theorem [EM02, Theorem 12.4.1]. For circles the formal condition is just a nonzero tangent line in the contact bundle; there is no extension obstruction, and the dimensions permit disjointness and avoidance. The dual handle decomposition shows that  $\partial U_1 \rightarrow U_1$  is an isomorphism on  $\pi_1$ , so these circles may be chosen in the required free homotopy classes.

Join each  $\gamma_j$  by an annulus in  $U_1$  to the corresponding graph cycle. Both boundary integrals of  $\lambda$  vanish: one curve is contact-isotropic and the other lies on sheets with zero primitive. Stokes' theorem gives zero area for the annulus. Glue it to the chosen zero-area filling of that cycle. This specifies a relative homotopy class of disks in  $M$  with boundary  $\gamma_j$  and area zero. Prescribe near the boundary the outward Liouville-flow collar of  $\gamma_j$ ; this collar is isotropic and does not change that area.

Put  $M_1 = M \setminus \text{int } U_1$ . Restoring  $U_1$  from its boundary adds only handles of indices  $2n - 1$  and  $2n$ . Thus the inclusion  $M_1 \rightarrow M$  is an isomorphism on the low relative homotopy problems in dimensions two and three. In particular the specified disk classes, relative to their boundary collars, have representatives in  $M_1$ . Their areas remain zero because the representatives are homotopic relative to the boundary in  $M$ . This is where retaining the filling class is essential:  $\pi_2(M)$  is not zero, and an arbitrary exterior filling would not have the same area. General position relative to the collars makes these smooth disks embedded, pairwise disjoint, and disjoint from  $f(L_0)$  off the prescribed domain attachments. The relevant inequalities are  $2 + 2 < 2n$  and  $2 + n < 2n$ .

*Making the disks isotropic.* Take the prequantization circle bundle  $Q \rightarrow M$  for the integral form  $\omega$ , with connection form  $\vartheta$  normalized by  $d\vartheta = \pi_Q^*\omega$  and circle parameter in  $\mathbb{R}/\mathbb{Z}$ . It is a contact manifold. Each prescribed isotropic disk collar has a horizontal lift. Over its smooth filling disk trivialize the circle bundle and write the connection as  $d\theta + \beta$ . The horizontal boundary lift has winding number

$$-\int_{\partial D^2} \beta = -\int_{D^2} \omega = 0. \quad (4.6)$$

It therefore extends to a smooth lift of the whole disk, relative to a smaller collar. Exact zero area, rather than area merely integral, is what ensures this relative extension. The lifts are embedded and pairwise disjoint because the underlying smooth disks are.

We check the complete formal isotropic condition. Trivialize the pulled-back contact bundle over a lifted disk and fix a frame of its tangent bundle. Ordered isotropic two-frames retract to

$U(n)/U(n-2)$ , whose fundamental group vanishes. Thus the prescribed isotropic derivative on the collar extends to an isotropic monomorphism over the disk. Its inclusion in the ambient tangent bundle and the actual disk derivative are homotopic relative to the collar: the real two-frame space  $V_2(\mathbb{R}^{2n+1})$  has vanishing  $\pi_2$  (indeed it is  $(2n-2)$ -connected). This gives formal isotropic embeddings of the disjoint union of disks with the fixed collar germs. Since  $2 < n$ , the relative  $C^0$ -dense subcritical contact embedding theorem [EM02, Theorem 12.4.1] realizes them by genuine isotropic embeddings.

Their projections to  $M$  are isotropic immersions, because  $d\pi_Q$  identifies the contact hyperplane symplectically with  $TM$ . These projections may be made embedded and pairwise disjoint while preserving all avoidance conditions. To justify the needed transversality within contact-isotropic maps, a contact Hamiltonian  $H$  gives a vector field satisfying

$$\vartheta(X_H) = H, \quad \iota_{X_H} d\vartheta = (R_\vartheta H)\vartheta - dH.$$

Prescribing  $dH$  on the contact hyperplane at a point prescribes an arbitrary projected velocity. At distinct points of the embedded lifted union these choices are independent by disjointly supported Hamiltonians. Relative transversality therefore removes projected double points and intersections between disks, of expected dimension  $4 - 2n < 0$ , and intersections with the immersed  $L_0$ , of expected dimension  $2 - n < 0$ . Take all perturbations zero on a narrower collar. Compactness, the immersion property near the diagonal, and sufficiently small perturbations away from that collar preserve exterior placement and the prescribed attachments. We have obtained disjoint isotropic core disks in  $M_1$ .

*Two-handles and contractibility.* Attach thin index-two handles of (4.5) along these core disks. Choose the attaching parametrization and the symplectic normal framing by restriction from the actual core disk. This removes any framing extension problem: the tangent and normal data of the collar identification are the restrictions of data already defined over the core. The relative isotropic neighborhood theorem and relative Moser argument embed the handles with those collar germs. Their primitives glue by the strict collar construction above. They miss the immersion, and sufficiently thin handles are mutually disjoint. Rounding the positive corners gives a Weinstein domain  $U$  with all the asserted primitive and collar properties.

The domain retracts onto  $G$  with one two-cell attached along each free generator. After contracting a maximal tree, this is a bouquet of circles with its free basis killed by those cells; its cellular complex is acyclic and its fundamental group is trivial, so it is contractible. No two-handle meets the immersion, so the source preimage remains the disk constructed in  $U_1$ . The single-sheet radial ball also remains unchanged.  $\square$

#### 4.4 Cancellation and the parity of double-point degrees

**Lemma 4.8.** *The immersion may be replaced on the source disk of Lemma 4.7 so that every double point has oriented intersection sign  $(-1)^{n/2-1}$ . The replacement preserves the integral grading, spin structure, pushed-forward fundamental class, and the cohomology and homology maps in (4.3).*

*Proof.* Complete  $U$  at its positive boundary and extend the immersed source disk by its positive cylinder. Remove the center of the single-sheet radial Darboux ball. This creates a negative Liouville end, whose Lagrangian asymptote is the standard Legendrian link of a real plane. The source is  $S^{n-1} \times \mathbb{R}$ . The punctured completion is simply connected, since  $U$  is contractible and  $2n > 2$ .

Set  $l = n/2$  and  $m = |I| > 0$ . We use the stabilization traces of [EEMS13, Lemma 3.2 and proof of Lemma 3.4]. The trace from a Legendrian link to its loose stabilization has one double point of sign  $(-1)^{l-1}$ ; the reversed trace has sign  $(-1)^l$ . Insert  $m$  reverse traces in separated collars near

the negative end, running outward from the  $m$ -fold stabilization to the original standard link. The negative link is now loose and the total signed count is

$$I + m(-1)^l = 0.$$

The exact cancellation theorem of Eliashberg–Murphy [EM13, Theorem 2.3] applies with the following verified hypotheses. The target is a simply connected Liouville manifold with a negative end. The immersion is proper, self-transverse, cylindrical at its negative end, and has finitely many double points. Its source is connected. Its primitive is exact because  $H^1(S^{n-1} \times \mathbb{R}; \mathbb{R}) = 0$ . Its signed intersection count is zero, and its single negative component is loose. The additional infinite-width hypothesis in that theorem is required only in dimension  $n = 3$ , whereas here  $n = 514$ . The conclusion is a compactly supported Hamiltonian regular homotopy to an embedding, leaving both tails unchanged.

In the unchanged negative end, insert  $m$  forward traces from the original standard link to the  $m$ -fold stabilization, and fill the center by the real plane. All double points of the filled immersion are now the  $m$  double points of these traces, with sign  $(-1)^{l-1}$ . Its noncylindrical part is compact. A sufficiently long negative Liouville flow in the filled positive completion moves this compact part into  $U$ . The backward flow exists for all time: the compact domain is backward invariant, and on the positive end the flow translates the cylindrical coordinate. Although it rescales the symplectic form, it preserves Lagrangianity. The positive cylinder is unchanged as a subset. Trim at its original boundary collar and glue to the unmodified immersion outside the source disk.

The new source piece is a disk with the same collared boundary. A boundary identification extends to a homeomorphism of the disks; under this comparison, the two maps of the disk into  $U$  are homotopic relative to the boundary because  $U$  is contractible. Thus the closed maps to  $M$  have the same fundamental class and induce the same maps on homology and cohomology. This argument uses contractibility of  $U$ , not asphericity of  $M$ . Orient the replacement compatibly. Its spin structure extends the old boundary spin structure, since the sphere  $S^{n-1}$  has a unique spin structure. Every first homology class of the source can be represented outside the replaced disk, so its Maslov class remains zero. The grading agrees on the common complement after a constant shift, and then extends over the replacement disk.  $\square$

For completeness, with the grading and orientation convention of Section 2, transverse oriented graded Lagrangian planes in complex dimension  $2l$  satisfy

$$(-1)^{\deg(p_-, p_+)} = (-1)^l \text{sign}(p). \quad (4.7)$$

It follows either by rotating the first plane to the second in diagonal unitary coordinates and counting the phase crossings, or by the usual graded Euler-intersection formula. Reversal of the ordered pair changes its degree to  $n - \deg(p_-, p_+)$ , so both ordered degrees have the same parity. For the sign in Lemma 4.8, the right side of (4.7) is  $-1$ . Thus both degrees are odd.

## 4.5 Rational areas and the fixed product labels

**Lemma 4.9.** *The immersion of Lemma 4.8 can be perturbed arbitrarily slightly so that it has the incidence properties and the common area denominator in Proposition 4.1(iv), without changing any of its other conclusions.*

*Proof.* A generic small Hamiltonian perturbation gives transversality to each of the finitely many product labels, keeps its double values off those labels, and avoids their pairwise intersections. Indeed those intersections have dimension strictly less than  $n$ , by the product-label construction,

so their intersection with a generic  $n$ -dimensional immersed submanifold of  $M^{2n}$  is empty. There are now finitely many intersection points involving the immersion, with distinct source preimages except for the prescribed paired double preimages.

Restrict the prequantization bundle and connection to  $L$ . The connection is flat because  $i^*\omega = 0$ . Its holonomy on the torsion subgroup of  $H_1(L; \mathbb{Z})$  already has finite image. On the free part, an arbitrarily small closed one-form graph perturbation changes the holonomy character by its period character. Rational characters are dense in this finite-dimensional character torus, so choose such a perturbation to make the whole holonomy finite. A finite tensor power of the prequantization bundle then has a parallel unit section on  $L$ . Choose a common power also supporting the parallel sections on all the fixed labels.

For each double point, compare the values of the parallel section at its two preimages. For each intersection with a label, compare the section on  $L$  with the chosen section on that label. Exact graph perturbations by functions constant near the relevant source points change these ratios by their endpoint primitive differences while fixing the local sheets. Because all the source patches can be chosen disjoint, we can make the finitely many ratios roots of unity by arbitrarily small such perturbations. These exact perturbations do not change the already finite holonomy. They preserve transversality and create no new intersections. The comparisons between the fixed labels are already roots of unity on every clean component by Lemma 2.2.

Take a further common power killing all these roots of unity. The resulting parallel sections match at every allowed boundary switch, including switches between the two branches of the immersion. For a disk with such lifted boundary segments, parallel transport therefore has holonomy one in that power. The curvature-holonomy formula gives

$$\exp\left(2\pi\sqrt{-1}N_0\int_{D^2}\omega\right)=1$$

for one integer  $N_0 > 0$  independent of the disk and its number of corners. Hence all these areas lie in  $N_0^{-1}\mathbb{Z}$ . Every perturbation used above is a Lagrangian regular homotopy. It preserves the spin structure and grading, the parity of transverse double points, the fundamental class, and the maps in (4.3).  $\square$

*Proof of Proposition 4.1.* Start with Lemma 4.4, choosing the Euler characteristic as in Lemma 4.5. Apply successively Lemmas 4.6, 4.7, and 4.8. Equation (4.7) gives the required odd corner degrees. Finally apply Lemma 4.9. The fundamental class, the degree-two quotient, and the homology injection were established in (4.3) and are preserved at every subsequent step.  $\square$

## 5 An immersed object detected by homology

The construction of Section 4 controls the second Betti number of a Lagrangian immersion. To use that control on the mirror, we need an uncurved object whose degree-two endomorphisms have no additional generators at the double points. This section proves the required statement. Its geometric ingredient is the interior-mark obstruction argument of Fukaya–Oh–Ohta–Ono. We extend that argument to an immersion whose double-point degrees are odd, retaining all switching-node faces. We then pass from solutions at finite energy cuts to an ordinary  $A_\infty$  category over an enlarged field.

## 5.1 Coefficients, gradings, and finite constructions

We use the Novikov field

$$\Lambda = \left\{ \sum_{j \geq 0} a_j T^{\lambda_j} : a_j \in \mathbb{C}, \lambda_j \in \mathbb{R}, \#\{j : \lambda_j \leq E\} < \infty \text{ for every } E \in \mathbb{R} \right\}.$$

Terms with equal exponents are combined. Write  $v$  for the least exponent of a nonzero series,  $v(0) = +\infty$ , and put  $\Lambda_0 = \{v \geq 0\}$  and  $\Lambda_+ = \{v > 0\}$ . The exponent records symplectic area and has degree zero. Algebraic closure and the size of this field are recalled below.

Let  $(X, \omega)$  be a closed symplectic manifold of real dimension  $2n$ , equipped with a homotopy class of trivializations of its canonical complex line bundle. All Lagrangian branes considered here are compact, integrally graded, and spin; the background class is zero. A finite collection is called clean when every pairwise fiber product of its immersions, including a self-fiber product, is clean. On a connected component  $R$  of such a fiber product, Floer cochains are differential forms with the determinant orientation local system and the Maslov degree shift. Thus the underlying complex at energy zero is a direct sum of shifted de Rham complexes. The coefficients are extended from  $\mathbb{R}$  to  $\mathbb{C}$  separately at each energy.

We use suspended  $A_\infty$  signs. If  $|x|' = |x| - 1$ , the identities read

$$\sum_{s=0}^k \sum_{i=1}^{k-s+1} (-1)^{\sum_{j < i} |x_j|'} m_{k-s+1}(x_1, \dots, x_{i-1}, m_s(x_i, \dots, x_{i+s-1}), x_{i+s}, \dots, x_k) = 0. \quad (5.1)$$

Here  $m_k$  has ordinary degree  $2 - k$ , and  $m_{1,0} = d$ . The order of inputs follows the boundary order. Composition in the cohomology category includes the corresponding suspension sign.

We specify the foundational results used in the argument, and then explain the finite-cut convention needed when constant polygons switch branches. The disk and strip construction of [FOOO09, Chapters 7–8], its immersed extension [AJ10, Sections 4–12], and the clean de Rham construction [Fuk25, Section 3] provide the following ingredients:

- (i) Fredholm charts obtained by adding finite-dimensional obstruction spaces supported in component cores, with auxiliary markings and transverse incidence conditions to stabilize the domains;
- (ii) gluing and coordinate changes compatible with normalization of boundary corners, where the obstruction spaces on a broken curve are sums of the spaces on its components;
- (iii) orientations from the spin structures and the determinant lines at the ends;
- (iv) extension, after thickening and outer collaring, of compatible CF-perturbations prescribed on normalized corners, with strong submersivity at the outgoing boundary evaluation;
- (v) signed integration along the fiber, its Stokes formula and its composition formula, and relative versions over a parameter interval;
- (vi) the filtered construction and comparison theorem, including canonical models, passage from finite approximations to a filtered structure up to homotopy, and comparison of choices.

For the clean disk charts and perturbations, precise statements are [Fuk25, Theorem 3.24 and Propositions 3.29–3.30]. The abstract extension and integration statements are those of [FOOO20, Chapters 9–10, 15, 17, and 21]. The construction by core obstruction data, including its relative

and componentwise forms, is developed in [FOOO18a, FOOO18b]; smooth gluing coordinates are understood with the correction in [FOOO24]. We invoke these statements for the ordinary rooted disk problems. The interior-rooted problems used below are constructed separately in Lemma 5.3.

**Lemma 5.1** (Finite clean models). *For a finite clean collection of branes, the disk construction gives a unital curved filtered  $A_\infty$  category over  $\Lambda_0$ , with finite-rank canonical morphism models. The energy-zero differential on the de Rham model is the ordinary differential on its sectors, and the canonical model has those sectors' cohomology as its underlying graded space. Choices, restrictions to subcollections, and finite geometric constructions compare by filtered homotopy equivalences through any prescribed finite energy and arity cut.*

*All these structures and comparisons respect the integral grading. For a self-transverse immersion  $i : L \rightarrow X$  with only double points, its canonical self space is*

$$C^*(L, L) = H^*(L; \mathbb{C}) \otimes \Lambda_0 \oplus \bigoplus_{(p,q) \in R_L} \ell_{pq} \otimes \Lambda_0, \quad (5.2)$$

where  $R_L = \{(p, q) : p \neq q, i(p) = i(q)\}$ , and  $\ell_{pq}$  is a one-dimensional line placed in the intersection degree  $d(p, q)$ . For an embedded brane transverse to  $i$  and avoiding its double values, the mixed canonical spaces are the usual graded intersection lines.

*Proof.* The existence and comparison assertions are the filtered construction theorems just listed. We check the cutoff and grading issues relevant to their use here.

At a fixed finite cut, let  $\delta > 0$  be a lower bound for positive component energies. Such a bound follows from disk compactness and the energy gap for the finite list. Choose  $0 < \varepsilon < \delta$  and assign to an ordinary vertex of energy  $E$  and arity  $k$  the cost

$$c(E, k) = E + \varepsilon(k - 1). \quad (5.3)$$

Costs add under tree substitution. Every positive-energy zero-input vertex has positive cost, as does every zero-energy stable vertex of arity at least two. The zero-energy unary operation is the fixed differential and is handled algebraically. A nontrivial breaking therefore has lower-cost components. An energy and arity bound gives a finite collection of necessary moduli problems and their boundary types: energy bounds control nonconstant components, stability and the arity bound control constant components, and compactness gives finitely many represented classes and charts. In particular, we never bound the number of switches in a constant polygon by its energy alone.

This is the geometric version of the finite  $A_{n,K}$ , or  $A_{N,0}$ , construction. One applies the relative extension theorem at successive costs, including the necessary faces and homotopies. Forgetful-map conditions giving units reduce arity and fit the same induction. Interval versions supply comparisons on common cuts. The filtered lifting theorem then produces a fixed model equivalent to each finite construction through its cut; see [AJ10, Sections 7–11]. This conclusion does not require one transverse perturbation to work simultaneously at all energies. If the relevant energies lie in  $\delta\mathbb{Z}$ , one may choose  $\varepsilon = \delta/4$ ; the cost induction is then literally a discrete induction. More generally one uses the gapped energy set and the equivalent finite energy–arity ordering.

For the grading, cap each transverse end by a path lifted according to the brane grading. The determinant trivialization and the grading make the closed capped Maslov contribution zero. The sphere contribution is also zero because the first Chern class vanishes. Fredholm indices therefore give the integer degree  $2 - k$  for  $m_k$ , with  $T^E$  of degree zero. At a clean component the same index formula includes the ordinary form degree and the constant Maslov shift. Reversing a clean component of dimension  $r$  changes its shift to the complementary shift whose sum with the original

is  $n - r$ . Thus evaluation pushforwards and matching at nodes preserve the integer degree, not just its parity. A Hamiltonian isotopy transports the lifted capping paths.

At energy zero a unary stable nonconstant map cannot occur, and a constant strip with transverse ends contributes no differential on a point sector. The remaining unary operator is  $d$ . Transfer with respect to this differential gives the displayed finite-dimensional cohomology spaces. The unit can be retained in this transfer by choosing the contraction to fix the constant function. On transverse mixed sectors there is nothing to transfer. These observations prove (5.2) and the assertion about mixed spaces.  $\square$

A cochain  $b$  of ordinary degree one and positive valuation twists the operations by inserting arbitrarily many copies of  $b$  in all gaps. The sums converge: an insertion contributes a uniformly positive amount to the valuation for each fixed cochain. Write

$$\Delta(b) = \sum_{k \geq 0} m_k(b, \dots, b).$$

The equation  $\Delta(b) = 0$  is the zero-curvature Maurer–Cartan equation. We use this equation, rather than the weaker equation permitting a multiple of the unit.

## 5.2 The statement for an immersed brane

**Proposition 5.2** (An immersed object with homological detection). *Let  $(X^{2n}, \omega)$  and a finite clean collection be as above, with  $n$  even. Fix a compatible almost complex structure  $J$  admitting no nonconstant  $J$ -holomorphic sphere. Suppose a label  $i : L \rightarrow X$  satisfies:*

- (a) *its self-intersections are transverse double points, and both ordered intersection degrees at each double point are odd;*
- (b) *the map  $i_* : H_{n-2}(L; \mathbb{Q}) \rightarrow H_{n-2}(X; \mathbb{Q})$  is injective;*
- (c) *areas of the self-disk problems for  $L$ , including polygons with switches, belong to a locally finite additive submonoid  $G \subset \mathbb{R}_{\geq 0}$ .*

*Fix positive-valuation zero-curvature bounding cochains on any selected other labels. There is an algebraically closed field  $\kappa$  containing  $\Lambda$ , of cardinality at most continuum, and a unital uncurved  $\kappa$ -linear  $A_\infty$  category containing those bounded labels and a new object, denoted  $\mathbf{L}$ , with the following properties.*

- (i) *The full category on the other bounded labels is the scalar extension of their original category.*
- (ii)  $\dim_\kappa H^2 \text{Hom}(\mathbf{L}, \mathbf{L}) \leq b_2(L)$ .
- (iii) *If  $D$  is one of the selected bounded labels, is embedded, and is transverse to  $i$  and avoids its double values, the underlying mixed spaces involving  $D$  and  $\mathbf{L}$  are the graded intersection lines for  $D$  and  $i$ .*
- (iv) *If, in addition,  $D$  is connected and its bounded self-Floer cohomology has the full graded dimensions of  $H^*(D)$ , composition followed by its top trace gives perfect pairings*

$$H^p \text{Hom}(D, \mathbf{L}) \otimes H^{n-p} \text{Hom}(\mathbf{L}, D) \rightarrow \kappa \quad (p \in \mathbb{Z}).$$

*The trace in (iv) is nonzero on the one-dimensional top self-Hom of  $D$ . The proposition asserts an ordinary object over  $\kappa$ ; it does not assert that  $L$  has a bounding cochain in the original Novikov field.*

Here locally finite means that  $G \cap [0, E]$  is finite for every  $E$ . Fix  $\delta > 0$  with  $G \cap (0, \delta) = \emptyset$ ; if  $G = \{0\}$ , any positive  $\delta$  will do. All comparisons used for the self-label keep  $J$  and the immersion fixed, so their energy indices belong to  $G$ . The discrete rational-area condition supplied by Proposition 4.1 is a special case.

The proof has three steps. We first construct the interior-mark operations needed to detect the first surviving curvature coefficient. We then remove curvature at any finite energy cut in the geometric model. Finally we transport these approximate solutions to one fixed finite model and pass to a residue field.

### 5.3 An interior mark and its boundary strata

For the remainder of the geometric argument, use the single immersion  $i : L \rightarrow X$ . Its diagonal sector is  $\Omega^*(L; \mathbb{C})$ ; a switch sector is the shifted point line  $\ell_{pq}$ . Ordinary disk operations have a distinguished boundary output. We also consider stable disks with one interior marked point,  $k$  cyclically ordered boundary inputs, and no distinguished boundary output. The boundary lift to  $L$  is continuous between designated switches. The word “cyclically” specifies the action relabeling the  $k$  boundary inputs; integrations take place on the labeled spaces, without dividing by this action.

For  $\eta \in \Omega^*(X; \mathbb{C})$ , pull back  $\eta$  by the interior evaluation and integrate it together with the boundary inputs. The resulting scalar operation is written  $p_k^\eta$ . Its terms carry their area weights. A stable constant disk with one boundary input and the interior mark is included. The constant problem with no boundary input is unstable and is not included.

On a component with continuous boundary lift the Maslov index is zero. The real map index is therefore  $n$ . Subtracting the three disk automorphism parameters and adding the marked-point parameters gives virtual dimension  $n + k - 2$  for  $k$  inputs and a boundary output, and  $n + k - 1$  for  $k$  boundary inputs and the interior mark. These are the dimension formulas used below.

**Lemma 5.3** (Finite interior-mark construction). *Fix the ordinary rooted disk data through a finite energy–arity cut, enlarged to include all factors of the required boundary strata. The preceding interior-mark spaces admit oriented Kuranishi structures and CF-perturbations relative to those data, with the following features:*

- (i) *Every boundary product is rooted at the component carrying the interior mark. Its other vertices carry the ordinary rooted disk data. Both diagonal matching nodes and inverse switching nodes are included.*
- (ii) *The additional interior evaluation is strongly smooth. No submersivity is required of it. At a diagonal matching node the ordinary child evaluation is strongly submersive for the CF-perturbation.*
- (iii) *The interior-rooted data are equivariant under cyclic relabeling of their external boundary inputs, with the determinant orientation signs. The ordinary rooted child data need not be cyclic.*
- (iv) *The integrations obey Stokes and composition on these boundary products. At energy zero,*

$$p_{1,0}^\eta(x) = \pm \int_L i^* \eta \wedge x \tag{5.4}$$

*on closed forms. The sign is fixed and nonzero.*

*Proof.* We construct the interior-rooted spaces with all immersed boundary labels. Their local boundary problems are the ordinary diagonal and transverse-jump problems of [AJ10, Sections 4–5]; the relative construction below specifies their organization when no boundary output is chosen. It uses the mixed parent–child organization of [AFOOO26, Propositions 16.3 and 16.14], retaining reciprocal internal switches even when the external word is empty. The parent is always additional interior-rooted data. Its identification with a boundary-rooted bulk operation is neither required nor imposed.

*Compactification and normalization.* Keep the interior mark in immersed stable-map compactness. At a boundary node the two lifts either agree in  $L$  or end at distinct preimages  $p, q$  of a double value. In the latter case the two normalized flags have the reciprocal labels  $(p, q)$  and  $(q, p)$ . These flags disappear from the external word on gluing; they remain part of the normalized stratum. In particular, an empty external word does not restrict its compactification to the diagonal sector.

There are no nonconstant spheres. A connected tree of constant spheres attached at one interior node and carrying at most the one interior mark cannot be stable: it has fewer than three external special points in total. The degeneration in which the entire boundary collapses would retain a nonconstant sphere, and is also absent. All remaining strata are disk trees. Constant stable vertices are retained, including the disk with the interior mark and one boundary node. The constant disk with no boundary input or node and only the interior mark is unstable and gives no stratum.

Let  $v_*$  be the vertex carrying the interior mark. Orient every edge toward  $v_*$ . A vertex other than  $v_*$  then has exactly one outgoing flag, which is its ordinary disk output; every remaining flag is an input. A normalized corner is the product of the moduli problems on these vertices, with the diagonal matching equations and reciprocal-label matches along its edges. For any collection of edges, cutting them in any order produces this same oriented tree and the same factors. The root of each separated component is either  $v_*$  or its unique edge toward  $v_*$ . This rule also describes intermediate faces, so the identifications on iterated normalized corners agree.

Stabilization respects these identifications. Forgetting auxiliary marks contracts only constant unstable components. Contracting a non-root such component carries the unique path toward  $v_*$  to the unique path toward  $v_*$ ; hence the output flag on every surviving ordinary component is unchanged. A constant component carrying the interior mark and an attaching boundary node is stable and cannot be contracted. If it has no such node, it is the excluded constant zero-input root. An unstable constant bivalent disk has either two diagonal nodes on the same sheet or two reciprocal switch nodes: its two constant boundary arcs rule out exactly one switch. Its contraction preserves the labels of the surviving flags. The two-switch constant strip is the identity under the dual cap-line identification fixed by determinant gluing; it is not a new orientation choice [AJ10, Sections 4.3 and 5]. Thus a contraction performs the same pairing of the ordered cap lines whichever order of normalization is used. For a chain of these components, the neck lengths concatenate as  $T = T_1 + \cdots + T_j$ . This is associative. Forgetting is interpreted in the admissible category of [AFOOO26, Section 14.3]: its pullbacks of admissible obstruction and CF data are admissible and smooth in the gluing profiles used for integration. The forgetting map itself need not be smooth in ordinary corner coordinates. This admissible compatibility gives compatibility with stabilization as well as with cutting a fixed tree.

These constructions are equivariant under isomorphisms of decorated trees. Such an isomorphism fixes the interior mark and preserves the directed paths to it, so it never changes an ordinary child’s output. Relabeling external inputs permutes their positions without rerooting any ordinary child. The same statement holds for an automorphism permuting several isomorphic subtrees. Normalized corners retain their ordering of the cut edges; changing that ordering permutes the outward normal lines with their determinant sign. Their subsequent finite quotients therefore have the required corner identifications and automorphism weights.

*Relative charts and smoothing.* At a smooth nonconstant root choose auxiliary marked points on compact cores, in free orbits of its finite automorphism group, and impose transverse incidence conditions. Choose obstruction sections supported away from the distinguished mark, the nodes, and the strip ends. Their span can make the augmented Cauchy–Riemann operator surjective: a cokernel vector annihilating all such sections would vanish on an open core and then vanish by unique continuation. Taking the span of the stabilizer translates makes this an invariant finite-dimensional obstruction space. If effective charts are used, include vectors  $D_u\xi$  for map variations  $\xi$  supported on disjoint free stabilizer-orbit balls in the core. These variations lie in the augmented kernel and carry a faithful stabilizer representation modulo infinitesimal reparametrizations; they make the thickened chart effective. No submersivity is requested at the interior evaluation, which is used only to pull back  $\eta$ .

At a broken root use the direct sum of the obstruction data on its normalized components. Ordinary vertices carry the fixed ordinary data. At a diagonal edge the child output evaluation is submersive, so component surjectivity remains surjectivity after imposing the matching equation in  $L$ . All matching equations can be solved recursively from the root toward the leaves: having chosen the parent variation, choose the child’s variation with the required output value and continue with its children. Joint submersivity of all boundary evaluations on one component is not needed. At a reciprocal switch the matching space is a point; the operator has precisely the transverse fixed-jump asymptotic condition of the immersed disk charts. The linear and nonlinear gluing problems are therefore those of the ordinary immersed construction, with the interior-mark coordinate added on the root component. A constant root stabilized by that mark is kept as a marked component in this description.

Transport obstruction sections from the compact cores into the plumbed domains. Exponential gluing estimates give charts through the smoothing parameters, with the corrected smooth gluing profiles cited above. Coordinate changes are obtained from componentwise obstruction data. We specify their relative choice, so that compatibility is more than a choice of unrelated local thickenings. Use the quasi-component construction of [FOOO18b, Conditions 8.12–8.15, Lemmas 8.19–8.20, and Definition 8.21]. A quasi-component records a transported compact-core obstruction choice and its auxiliary stabilization data. On a boundary tree take the componentwise union of the already chosen indices: those of its interior root and the fixed indices of its ordinary vertices. The directed-tree identifications show that this prescription agrees on every iterated boundary and stabilizer quotient.

Extend the corresponding open family of indices inside a proper family over a sufficiently small boundary neighborhood, as in [FOOO18b, Proposition 8.18]. Complete its finite cover by new compact-core choices on the root away from a smaller neighborhood, and saturate the choices under the finite automorphism groups. Choose the new compact-core spaces generically transverse to the finitely many inherited spaces and their stabilizer translates, as in the quasi-component construction. Surjectivity and these direct-sum conditions are then open; shrinking the neighborhoods preserves them. Properness gives the semicontinuity inclusion of obstruction spaces under changes of the central map, as in [FOOO18b, Proposition 8.17]. These inclusions and forgetting the auxiliary marks define the coordinate changes; they satisfy the cocycle conditions because the index inclusions compose and the admissible forgetting operations commute with normalization. The original boundary families are unchanged on the smaller collar. This is the componentwise proof of [FOOO18b, Theorems 5.3 and 5.4], with its additional root data specified relatively. The local diagonal and fixed-jump operators and the augmented matching surjectivity verified above are exactly what that construction uses. The fixed ordinary factors are retained.

*Finite extension of perturbations.* Let  $\delta > 0$  be a lower bound for positive component energies, and choose  $0 < \varepsilon < \delta$ . Assign cost  $E + \varepsilon(k - 1)$  to an ordinary vertex of energy  $E$  with  $k$  inputs and cost  $E + \varepsilon k$  to an interior-rooted vertex. These costs add under boundary splitting. Every

ordinary stable vertex has positive cost: the zero-energy unary strip is unstable and represents the separately fixed de Rham differential, and a zero-energy stable ordinary vertex has at least two inputs. The constant one-input interior root has cost  $\varepsilon$ ; the constant zero-input root is excluded. Each nontrivial breaking therefore has lower-cost factors. The locally finite energy set, the arity cutoff, and stability give a finite collection of problems and normalized corner types, after adjoining all required faces. This includes constant switching polygons; their number of inputs is controlled by arity, not by their zero energy.

Induct on this finite order. On every normalized boundary prescribe the product of the already constructed interior-rooted parent perturbation and the fixed ordinary child perturbation. The prescriptions agree on higher corners by the directed-tree and stabilization identifications. Diagonal fiber products are defined because the child output is strongly submersive for its fixed CF-perturbation; reciprocal switches are products over points. First extend these compatible normalized-boundary products to a corner neighborhood by outer collaring, as in the collar step of [AFOOO26, Proposition 14.8 and Remark 14.7]. The required agreement on iterated normalized boundaries is precisely the agreement verified above, including their stabilizer quotients. Apply [AFOOO26, Proposition 14.5] with the map to a point and with the closed smaller corner neighborhood as its prescribed set. It supplies a compatible thickening and extends the CF-perturbation over the new root space while retaining it near that set. Composition on the prescribed collar still uses the original ordinary child perturbations. There is no requirement to forget a boundary input on the new root space and no new submersion condition at its interior mark.

For cyclic relabeling and finite stabilizers, take invariant obstruction spaces and include the finite-group representations in the auxiliary parameter spaces of the CF-perturbations. The prescribed corner products are equivariant, so the same relative extension is equivariant. This uses equivariant families, not single-valued equivariant transverse sections. Finite quotients are integrated with their ordinary stabilizer weights. Stokes and composition now apply to every prescribed product. The identical construction over a parameter interval, relative to its endpoints, supplies finite-cut comparisons whenever the ordinary choices vary.

*Orientations and the zero-input identity.* Use the spin determinant orientations and the graded cap lines at all transverse flags, with the local normalization specified below. At a reciprocal node the two cap lines are paired in their ordered duality; at a diagonal node they are replaced by the matching determinant of  $TL$ . Determinant gluing is associative. Permuting normalized smoothing normals or cap lines gives exactly their Koszul signs, including at an automorphism quotient. Stabilizing marks together with their incidence constraints contribute canceling orientation factors. These are the orientation conventions for ordinary immersed disk composition; adding the even-dimensional interior-mark coordinate does not alter its cap-line rule.

For  $k = 0$ , pull back the closed  $(n-2)$ -form  $\eta$  to the virtual  $(n-1)$ -space. Each boundary product has a single ordinary curvature child and an interior-rooted parent with one input. Composition first identifies its contribution with  $p_1^\eta(m_0)$ , with the common Stokes sign. This includes every reciprocal-switch product. Only after composition do we use the grading: a curvature output has degree two, while each branch generator has odd degree, so those child outputs vanish. There is no sphere or boundary-collapse term. Stokes therefore gives  $p_1^\eta(m_0) = 0$  for closed  $\eta$  of degree  $n-2$ .

At energy zero the one-input diagonal root is the regular constant disk space  $L$ . Its one boundary mark and one interior mark fix the domain; its evaluations are the identity and  $i$ . Integrating a boundary form gives (5.4), with the fixed orientation sign. The boundary-rooted operations were never changed in this construction.  $\square$

## 5.4 The signed obstruction identity

We record the signs because the obstruction argument uses the sum over all cyclic positions, including its two extreme cases. On the diagonal sector the spin cappings are the ordinary embedded-disk cappings. The local orientation calculation at a diagonal node is therefore the embedded calculation, even if the moduli space has switching-node strata elsewhere.

Here is an explicit normalization of that calculation. Put the output test form first, followed by the boundary input forms, in the geometric integral. Write  $X_0$  for the degree of the output test form; in the interior-rooted case this form is  $\eta$ . At a smooth boundary slot of degree  $X_j$ , the signed boundary-rooted and interior-rooted operations have exponent

$$\sum_{j=1}^k jX_j + \binom{k-1}{2} + \epsilon_0, \quad (5.5)$$

where  $\epsilon_0$  is the one fixed bare smooth-node gluing sign. The binomial coefficient has its polynomial meaning also at  $k = 0$ . This is a choice of the signed  $m, p$  conventions together; the unstable unary operation is  $m_{1,0} = d$ . Reversing all boundary conventions replaces both operations by their  $A_\infty$  opposites and gives the same conclusions.

For clarity, one can verify the normalization as follows. In the determinant line order put the capped disk operator and inverse disk automorphism orientation first, then the marked-point tangent factors in boundary order. The former combined factor has odd parity when  $n$  is even, each smooth boundary mark has odd parity, and the interior mark has even parity. Consider a child containing  $s$  consecutive inputs starting at position  $i$ , with  $r = k - i - s + 1$  trailing inputs. Put parent before child in the product orientation. Moving the child block through the trailing factors, with outward normal first, gives the bare boundary exponents

$$i + sr + \epsilon_0 \quad \text{for a boundary-rooted parent,} \quad i - 1 + sr + \epsilon_0 \quad \text{for an interior-rooted parent.} \quad (5.6)$$

The tangent of the diagonal matching copy of  $L$  has even dimension and contributes no further parity. The difference of one in these two formulas is precisely the change from the odd boundary-output tangent factor to the even interior-mark factor.

Let  $X_z$  be the ordinary degree of the child output form. Its degree formula gives  $X_z = \sum_{\text{child}} X_j + 2 - s$ . In the Fubini comparison, moving child inputs past trailing inputs and then moving their output back into the node slot contributes  $F = (\sum_{\text{child}} X_j + X_z) \sum_{\text{trailing}} X_j$ . Let  $A_k$  be the exponent in (5.5) for the original input word, and let  $A_P, A_C$  be the corresponding parent and child exponents, with  $X_z$  in the parent node slot. Put  $I = \sum_{j < i} (X_j - 1)$ , and denote the appropriate bare exponent in (5.6) by  $B$ . Direct substitution gives

$$A_k + A_P + A_C + B + F \equiv \begin{cases} I + \sum_j X_j + k + 1, & \text{boundary output,} \\ I + \sum_j X_j + k, & \text{interior output} \end{cases} \pmod{2}.$$

After the input degrees have canceled, the constant part of this calculation is the identity

$$k + s(k - s + 1) + \binom{k-1}{2} + \binom{k-s}{2} + \binom{s-1}{2} = 0 \pmod{2}.$$

This polynomial identity also covers  $s = 0$  and  $s = k$ . For Stokes with a boundary output, the parity condition for the integrated form is  $X_0 + \sum X_j = k + 1$  modulo two; with an interior output it is  $X_0 + \sum X_j = k$  modulo two. Thus in both cases the displayed exponent is  $I + X_0$ .

Write the raw Stokes equation as  $\int_{\partial\mathcal{M}} \Omega - \int_{\mathcal{M}} d\Omega = 0$ , where  $\Omega$  is the product of the test form and input forms. Multiply this whole equation by  $(-1)^{A_k+X_0}$ . After rewriting a boundary product in the signed operations, its exponent is  $A_k + X_0 + A_P + A_C + B + F \equiv I$ , as required. For the differential of the  $i$ th input, the raw exterior-derivative sign is  $X_0 + \sum_{j<i} X_j$ , and the minus sign in the Stokes equation contributes one. The operation with that differentiated input has exponent  $A_k + i$ . Its normalized exponent is therefore

$$(A_k + X_0) + 1 + X_0 + \sum_{j<i} X_j + (A_k + i) \equiv \sum_{j<i} (X_j - 1) \pmod{2}.$$

This gives the same insertion sign and accounts for  $m_{1,0} = d$ . Thus the interior Stokes formula has the suspended insertion signs of (5.1), and no output differential when  $\eta$  is closed. This is the  $\mathfrak{p}$  convention of [FOOO09, Section 3.8]; the calculation is also the pure-sector normalization in [Opea, Section 6].

A cyclic move of the first boundary input to the last has sign

$$(-1)^{(X_1-1)\sum_{j>1}(X_j-1)}. \quad (5.7)$$

Indeed the boundary tangents contribute  $k-1$ . Permuting the input forms contributes  $X_1 \sum_{j>1} X_j$ , and changing (5.5) contributes  $(k-1)X_1 + \sum_{j>1} X_j$ . Their sum is the exponent displayed above. These determinant computations also hold on the thickened charts: obstruction directions and their duals are inserted compatibly with gluing, and adding stabilization marks together with incidence constraints inserts canceling orientation factors.

**Lemma 5.4** (Interior obstruction identity). *Assume the hypotheses on  $L$  in Proposition 5.2. Let  $b$  be a positive-valuation degree-one cochain supported on the diagonal de Rham sector. At any finite cut at which the expressions are defined, and for every closed  $\eta \in \Omega^{n-2}(X; \mathbb{C})$ ,*

$$\sum_{r \geq 0} p_{r+1}^\eta(\Delta(b), b, \dots, b) = 0 \quad (5.8)$$

*through that cut. The curvature  $\Delta(b)$  is supported on the diagonal sector and has ordinary degree two.*

*Proof.* Every  $m_k(b, \dots, b)$  has degree two. A switch sector is a point line of odd degree and has no degree-two cochain. Hence the curvature has no switch component. In the boundary formula for an interior-rooted disk with diagonal inputs, a child whose output is a switch similarly pushes forward to zero: its required output degree would be two on that shifted point. We include these faces in the geometric construction and use this degree calculation only after taking their child pushforwards. Composition of correspondences then makes their entire contributions zero.

All remaining faces have a diagonal child output. Their signs are the ones calculated above. Apply Stokes to the interior-rooted spaces with  $k > 0$  copies of  $b$ , and divide the resulting equation by  $k$ . A child carrying a block of  $s$  external inputs has  $k$  cyclic placements. For  $0 < s < k$  these are the starts of the block. For  $s = 0$  they are the  $k$  cyclic gaps. For  $s = k$  the child carries all inputs, and its output node occupies one of the  $k$  gaps, giving the  $k$  possible rooted linear orders on that child. Input differentials likewise occur in  $k$  positions. Since  $b$  has suspended degree zero, moving the curvature input back to the first parent slot gives no sign by (5.7). The normalized contribution is therefore

$$\sum_{r+s=k} p_{r+1}^\eta(m_s(b, \dots, b), b, \dots, b),$$

with the differential included in  $m_1$ .

For  $k = 0$  the boundary is a single curvature child attached to the interior-rooted parent, including the stable constant one-input parent. Its contribution is  $p_1^\eta(m_0)$ . There is no unstable constant  $p_0$  contribution, and no collapsed-boundary sphere term by Lemma 5.3. Summing over  $k$  proves (5.8). The sum is finite through the chosen cut because  $b$  has positive valuation. All divisions take place in characteristic zero.  $\square$

## 5.5 Removing curvature through a finite cut

**Lemma 5.5** (Approximate bounding cochains). *Under the hypotheses of Proposition 5.2, for every  $E > 0$  there is a finite geometric construction and a diagonal degree-one cochain  $b_E$  with  $v(b_E) \geq \delta$  such that  $v(\Delta(b_E)) > E$ . These solutions can be transported to a fixed finite-rank curved model, giving degree-one cochains  $\tilde{b}_E$  with a common positive valuation lower bound and with curvature valuations tending to infinity.*

*Proof.* Work first with an energy cut high enough for all operations about to be used. Start with  $b = 0$ . Suppose the coefficients of  $\Delta(b)$  below  $\lambda \in G \setminus \{0\}$  vanish, and let  $o_\lambda$  be its coefficient at  $T^\lambda$ . The curved  $A_\infty$  equation, evaluated on  $b$ , says

$$\sum_{r,t \geq 0} m_{r+1+t}(b^{\otimes r}, \Delta(b), b^{\otimes t}) = 0.$$

At energy  $\lambda$  only the zero-energy unary term acts on  $o_\lambda$ : every other displayed term contains either positive operation energy or an additional  $b$  input. Consequently  $do_\lambda = 0$ .

Apply Lemma 5.4. At the same energy, the only surviving term is the zero-energy  $p_1$  term on  $o_\lambda$ . Thus

$$\int_L i^* \eta \wedge o_\lambda = 0 \quad \text{for every closed } \eta \in \Omega^{n-2}(X; \mathbb{C}). \quad (5.9)$$

By Poincaré duality this says that the image of  $\text{PD}_L[o_\lambda] \in H_{n-2}(L; \mathbb{C})$  under  $i_*$  is zero. The assumed rational injection remains injective over  $\mathbb{C}$ , so  $[o_\lambda] = 0$ . Choose a diagonal one-form  $c_\lambda$  with  $dc_\lambda = -o_\lambda$  and replace  $b$  by  $b + T^\lambda c_\lambda$ . The only change in the curvature at energy  $\lambda$  is  $T^\lambda dc_\lambda$ , which removes that coefficient. All other new terms have greater energy.

There are only finitely many elements of  $G$  below  $E$ , and every new exponent is a sum of operation and correction exponents, hence still belongs to  $G$ . This induction therefore produces the first assertion, with all correction terms of valuation at least  $\delta$ . Every required arity is bounded: evaluation through energy  $E$  on such cochains uses at most  $E/\delta$  inserted inputs. Closing this finite list under the boundary products gives exactly a finite cost construction of the preceding lemmas. No simultaneous perturbation over all energies was used.

Now choose one finite-rank curved model from Lemma 5.1. For each  $E$ , construct the geometric operations far enough to compare with this model at the cut needed to evaluate the comparison on  $b_E$ . A filtered  $A_\infty$  comparison sends a cochain to  $\sum_{k \geq 0} f_k(b_E, \dots, b_E)$ , and its homomorphism equation sends vanishing curvature through the cut to vanishing curvature through that cut. Include the zero-input term  $f_0$  in this bookkeeping. It has strictly positive valuation.

Uniform positivity can be arranged before extending the single-label construction to the full list. For the single label, the geometric operations and finite comparisons use the fixed locally finite monoid  $G$ ; their zero-input terms thus start at the same positive gap. Transfer to the canonical model uses zero-energy contraction data and sums of these energies. Restriction from the full clean category may be chosen with this single-label model, or compared afterward by one fixed filtered equivalence. In the latter case take the smaller of the preceding gap and the positive valuation of that fixed equivalence's zero-input term. This is still a positive lower bound independent of  $E$ .

Lift any truncated coefficients to Novikov coefficients. The transported cochains need not be diagonal, which is harmless: the construction of the approximate solutions has already been completed in the geometric model. Taking the comparison cuts to infinity yields the required sequence in the fixed model.  $\square$

## 5.6 From finite solutions to a category over a field

**Lemma 5.6.** *The valued field  $\Lambda$  is complete and algebraically closed. Its value group is  $\mathbb{R}$ , its residue field is  $\mathbb{C}$ , and its cardinality is  $2^{\aleph_0}$ .*

*Proof.* Every nonempty support satisfying the displayed condition has a least element. The condition also makes addition and convolution multiplication well defined: the supports are bounded below, and only finitely many pairs of their exponents can contribute below any given bound. A nonzero series has the form  $aT^\lambda(1+u)$ , where  $a \in \mathbb{C}^\times$  and either  $u = 0$  or  $v(u) > 0$ . The geometric series  $\sum_{j \geq 0} (-u)^j$  satisfies the support condition and inverts  $1+u$ . Thus  $\Lambda$  is a field. Its monomials and constant terms give the assertions about the value group and residue field.

A Cauchy sequence for the absolute value  $|x| = \exp(-v(x))$  has eventually constant coefficients below every fixed real bound. These stabilized coefficients define a series in  $\Lambda$ : below each bound its support agrees with that of one term of the Cauchy sequence. This series is the limit, proving completeness.

We use the standard extension theorem for complete nonarchimedean fields: a nonarchimedean absolute value on a complete field extends uniquely to every finite extension [Con, Theorem 6.4]. Let  $F/\Lambda$  be a finite Galois extension of degree  $d$ , equipped with that extension, and write  $w$  for its additive valuation. Uniqueness makes  $w$  invariant under  $\text{Gal}(F/\Lambda)$ . Consequently

$$dw(x) = v(N_{F/\Lambda}(x)) \quad (x \in F^\times).$$

Since  $\mathbb{R}$  is divisible and an ordered abelian group is torsion-free, this equality shows that the value group of  $F$  is still  $\mathbb{R}$ .

Its residue field has degree at most  $d$  over  $\mathbb{C}$ . Indeed, a finite linearly independent family of residue classes lifts to a  $\Lambda$ -linearly independent family: rescale any putative relation so that its coefficients have nonnegative valuation and at least one has valuation zero, and then reduce the relation. The residue field is therefore  $\mathbb{C}$ , since  $\mathbb{C}$  is algebraically closed. Every Galois automorphism consequently acts trivially on this residue field.

For  $x \in F^\times$ , divide by a monomial of valuation  $w(x)$  to obtain a unit  $y$ . All conjugates of  $y$  have the same nonzero residue, so

$$\overline{\text{Tr}_{F/\Lambda}(y)} = d\bar{y} \neq 0.$$

Here residue characteristic zero is essential. Thus  $\text{Tr}_{F/\Lambda}(x) \neq 0$  for every nonzero  $x$ . The  $\Lambda$ -linear trace map  $F \rightarrow \Lambda$  is injective, forcing  $d = 1$ . Every finite algebraic extension in characteristic zero is contained in a finite Galois extension. Hence  $\Lambda$  is algebraically closed.

Every support is countable, being the union over integral bounds of finite sets. There are at most  $(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$  countable sequences of real exponents and complex coefficients. The opposite cardinality bound follows from the inclusion of the constant series  $\mathbb{C}$ .  $\square$

Choose a nonprincipal ultrafilter  $\mathfrak{u}$  on  $\mathbb{N}$  and form the countable ultrapower  ${}^*\Lambda = \Lambda^{\mathbb{N}}/\mathfrak{u}$ . Thus two sequences represent the same element precisely when they agree on a set belonging to  $\mathfrak{u}$ . The valuation takes values in the ordered group  ${}^*\mathbb{R} = \mathbb{R}^{\mathbb{N}}/\mathfrak{u}$ . Every comparison of such values means a

comparison on a set belonging to  $\mathbf{u}$ . Set

$$\begin{aligned} O_b &= \{x \in {}^*\Lambda : v(x) \geq -c \text{ for some } c \in \mathbb{R}\}, \\ \mathfrak{m}_\infty &= \{x \in {}^*\Lambda : v(x) > c \text{ for every } c \in \mathbb{R}\}, \\ \kappa &= O_b / \mathfrak{m}_\infty. \end{aligned} \tag{5.10}$$

Zero belongs to both subsets. We call a sequence of scalars bounded below if its valuations have a common real lower bound; this condition in particular puts its ultrapower class in  $O_b$ .

**Lemma 5.7.** *The ring  $O_b$  is a valuation ring with maximal ideal  $\mathfrak{m}_\infty$ . Its residue field  $\kappa$  is algebraically closed, and constant sequences induce an embedding  $\Lambda \hookrightarrow \kappa$ . Its cardinality is  $2^{\aleph_0}$ .*

*If the entries of a sequence of finite matrices are bounded below, and a fixed polynomial identity in these entries holds with an error whose valuation tends to  $+\infty$ , the reduced matrices over  $\kappa$  satisfy the identity exactly.*

*Proof.* The elements of  ${}^*\mathbb{R}$  bounded above and below by real constants form a convex subgroup. Quotienting the value group by this subgroup gives a valuation whose ring is  $O_b$  and whose positive-value ideal is  $\mathfrak{m}_\infty$ . Explicitly,  $O_b$  is a subring, and any nonzero  $x \in O_b \setminus \mathfrak{m}_\infty$  has valuation bounded on both sides, so  $x^{-1} \in O_b$ . An element outside  $O_b$  has inverse in  $\mathfrak{m}_\infty$ . These observations also verify directly the valuation-ring and maximal-ideal assertions.

Each nonzero constant sequence has finite real valuation and is a unit of  $O_b$ . Its image in  $\kappa$  is nonzero, proving the claimed embedding; no quotient by  $\Lambda_+$  is involved here.

The ultrapower is algebraically closed. To find a root of a monic polynomial with coefficients in  ${}^*\Lambda$ , represent its finitely many coefficients by sequences and choose a root over  $\Lambda$  at each index. The sequence of roots solves the polynomial in the ultrapower. A monic polynomial over  $\kappa$  lifts to a monic polynomial over  $O_b$ . Every root of this lift belongs to  $O_b$ : if a root had negative coarsened valuation, its highest power would have strictly smaller valuation than every other term of the polynomial, which is incompatible with their sum being zero. Reducing its factorization over  ${}^*\Lambda$  proves algebraic closure of  $\kappa$ .

The countable ultrapower has cardinality at most  $(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$ ; the embedded copy of  $\mathbb{C}$  gives the reverse inequality for  $\kappa$ . Finally, reduction  $O_b \rightarrow \kappa$  respects finite sums and products. Any sequence with valuation tending to  $+\infty$  belongs to  $\mathfrak{m}_\infty$ , which proves the polynomial assertion.  $\square$

We apply this construction to matrices of operations after inserting each approximate bounding cochain. All infinite sums involved in an individual insertion have already converged over  $\Lambda$  before passing to sequences. The following statement therefore involves only finite algebraic identities.

**Proposition 5.8.** *Let  $\mathcal{I}$  be a finite set of objects and, for every ordered pair, let  $V(X, Y)$  be a finite-dimensional graded complex vector space with a chosen homogeneous basis. For every  $j \geq 1$ , suppose that curved strictly unital  $A_\infty$  operations  $m_{k,j}$  are defined on the spaces  $V(X, Y) \otimes_{\mathbb{C}} \Lambda$ , with common units in  $V(X, X)^0$ . Assume that:*

- (i) *for every fixed object word and every matrix entry of its operation, the sequence of entries is bounded below in valuation;*
- (ii) *every coefficient of  $m_{0,j}(X)$  has valuation tending to  $+\infty$ , for every  $X \in \mathcal{I}$ .*

*Reduction of the operation matrices defines a strictly unital uncurved  $A_\infty$  category  $\mathcal{C}_\kappa$  on the spaces  $V(X, Y) \otimes_{\mathbb{C}} \kappa$ . If the operations on a subset  $\mathcal{J} \subseteq \mathcal{I}$  are independent of  $j$  and define an uncurved category  $\mathcal{B}$  over  $\Lambda$ , then the full subcategory on  $\mathcal{J}$  is exactly  $\mathcal{B} \otimes_\Lambda \kappa$ . Moreover,*

$$\dim_\kappa H^2 \mathcal{C}_\kappa(X, X) \leq \dim_{\mathbb{C}} V(X, X)^2.$$

*Proof.* For a fixed external word, the curved  $A_\infty$  equation is a finite sum of compositions of operation matrices, including the finitely many insertions of  $m_0$ . It is therefore a finite polynomial identity in bounded-below entries. Lemma 5.7 preserves that identity, and hypothesis (ii) makes every curvature insertion zero. The resulting identities are precisely the uncurved  $A_\infty$  equations. The unit equations are also finite polynomial identities and retain their stated common units. On  $\mathcal{J}$ , constant matrices reduce along the embedding  $\Lambda \hookrightarrow \kappa$ , proving the scalar-extension assertion. The final inequality is the dimension bound for cohomology of a finite-dimensional complex.  $\square$

**Lemma 5.9.** *Let  $A_j$  be square matrices over  $\Lambda$  of a fixed size whose entries are bounded below in valuation. If  $v(\det A_j) = 0$  for all sufficiently large  $j$ , their reduced matrix over  $\kappa$  is invertible. In particular, this applies when  $A_j = A^{(0)} + E_j$ , where  $A^{(0)}$  is a fixed invertible complex matrix and every entry of  $E_j$  has strictly positive valuation. No common positive lower bound on those valuations is required.*

*If such matrices in every degree define, after reduction, pairings  $C^p \otimes D^{n-p} \rightarrow \kappa$  between finite complexes, compatible with their differentials, then the induced pairings  $H^p(C) \otimes H^{n-p}(D) \rightarrow \kappa$  are perfect.*

*Proof.* The determinant commutes with reduction. The class of  $(\det A_j)_j$  has valuation zero and therefore does not belong to  $\mathfrak{m}_\infty$ . Its reduction is nonzero. In the indicated special case,  $\det(A^{(0)} + E_j) - \det A^{(0)}$  has positive valuation for each  $j$ , so  $v(\det A_j) = 0$ .

A nondegenerate chain pairing identifies each  $C^p$  with the dual of  $D^{n-p}$  and intertwines their differentials with the dual differentials, with the chain-pairing signs. Duality of finite vector spaces is exact, so passing to cohomology gives the asserted perfect pairings.  $\square$

*Remark 5.10.* Every algebraically closed field containing  $\Lambda$  and of cardinality  $2^{\aleph_0}$  is abstractly isomorphic to  $\mathbb{C}$ : it is algebraically closed of characteristic zero and has transcendence degree  $2^{\aleph_0}$  over  $\mathbb{Q}$ , since it contains  $\mathbb{C}$  and has cardinality  $2^{\aleph_0}$ . Such an identification is an abstract field isomorphism; it is not required to fix the distinguished field of complex constants.

## 5.7 The trace pairing and the proof of the proposition

*Proof of Proposition 5.2.* Take a fixed finite-rank curved category as in Lemma 5.1, restricted to  $L$  and the selected bounded labels. On  $L$  use a sequence of cochains from Lemma 5.5, with residual curvatures tending to infinite valuation. On the other labels use their fixed bounding cochains. Twist separately for every member of this sequence. Each coefficient of every fixed operation has valuation bounded below by zero. Proposition 5.8 therefore gives an uncurved category over  $\kappa$ . Operations involving only the other labels were constant in the sequence, so its full subcategory on them is their scalar extension. Strict units also survive.

The underlying graded self space of  $\mathbf{L}$  is still (5.2), now over  $\kappa$ . Its degree-two dimension is  $b_2(L)$ , since no switch degree is two. Cohomology has dimension at most its chain space in that degree, proving (ii). The mixed-space assertion follows from the same unchanged underlying spaces.

It remains to prove the pairing assertion. For an embedded connected label  $D$ , its canonical self space is  $H^*(D) \otimes \Lambda$ . Its bounded differential is zero: in a finite-dimensional graded complex, equality of the cohomology dimensions and the underlying dimensions in every degree forces every differential to have rank zero. Let  $\mathrm{tr}_D : H^n(D) \otimes \Lambda \rightarrow \Lambda$  be integration over the oriented source. Extend it by zero in other degrees.

Before taking the residue field, form the matrix, in complementary degrees, of the composition pairing on the two mixed chain spaces. With the suspended convention above this can be written

$$\langle x, y \rangle = (-1)^{|x|} \mathrm{tr}_D m_2(x, y), \quad x \in C^*(D, L), \quad y \in C^*(L, D),$$

using boundary input order. At energy zero an intersection generator pairs only with its reciprocal generator at the same point. The corresponding configuration is a constant triangle whose smooth output lies on  $D$ . Removing that smooth marking gives the constant transverse strip; the marking fixes translation. The transverse strip operator is invertible, so the local count is  $\pm 1$ . Its evaluation on  $D$  represents the class of that point, and transfer to the canonical self model preserves its top integral. If a submersive thickening is used for the pushforward, the resulting Thom representative has the same integral. There is no zero-area configuration pairing different intersection points.

Reversing the intersection exchanges degrees  $p$  and  $n - p$ . Consequently the leading pairing matrix in each such pair of degrees is a signed permutation matrix. Every bounding-cochain insertion has strictly positive valuation. Each determinant thus has valuation exactly zero at every stage of the approximation. By Lemma 5.9 it remains nonzero over  $\kappa$ , giving a perfect pairing of chain spaces.

The resulting category is uncurved, and the self differential of  $D$  is zero. The  $m_2$  identity therefore gives

$$\langle m_1 x, y \rangle + (-1)^{|x|} \langle x, m_1 y \rangle = 0.$$

The perfect chain pairing identifies one finite complex with the graded dual of the other shifted by  $n$ , and hence gives perfect cohomology pairings. The trace is nonzero on the top self-Hom of  $D$  by its definition and the full-rank hypothesis. This proves (iv) and the proposition.  $\square$

*Remark 5.11.* The field enlargement is compatible with earlier enlargements of the coefficient field. Two algebraically closed extensions of  $\Lambda$  of cardinality at most continuum embed over  $\Lambda$  into a common algebraically closed field of the same cardinal bound: choose an algebraically closed overfield of a compositum of copies, or use transcendence bases over  $\Lambda$ . All ordinary finite-dimensional Hom calculations and perfect pairings persist under this extension.

## 6 Comparison of the finite Fukaya categories

The mirror theorems used in Section 2 employ ordinary transverse Floer constructions. The immersion of Section 4 belongs to the clean, potentially curved theory of Section 5. This section supplies the bridge between these constructions on the embedded labels. We first compare operations on strictly ordered collections, where repeated labels do not occur. We then recover the full categories from infinitely many ordered Hamiltonian copies. Finally we explain the grading and ordinary tensor product in the Künneth theorem. The ordered-copy strategy is also used in [Opea, Section 10.2]; we prove the needed reconstruction explicitly. No mirror theorem for a bounding-cochain deformation of the additional immersion is used.

**Proposition 6.1** (Comparison on the ordinary labels). *Let  $\mathcal{L}$  be either a finite collection of embedded integrally graded spin Lagrangian spheres in the projective symplectic quartic, or a finite collection of integral symmetric graph tori and their Hamiltonian copies in a standard symplectic torus. Use zero background class and complex Novikov coefficients. Put the labels in mutually transverse position without triple intersections, with their gradings transported by the perturbations.*

*The clean category on these labels, with zero bounding cochains for a common disk-free almost complex structure, and their ordinary transverse Fukaya category are equivalent up to quasi-equivalence after an algebraically closed coefficient extension of cardinality at most continuum. The equivalence preserves the labeled objects and their integral gradings. Its restriction to a smaller finite collection has the same conclusion. In particular these comparisons may be used with the undeformed enhanced mirror theorems of Seidel and Abouzaid.*

The coefficient extensions in the statement serve to assemble finite comparisons; they do not prescribe a new Novikov completion of a perfect hull. The proof below makes this distinction explicit.

## 6.1 A common structure and a regular directed calculation

Put down countably many slightly displaced copies of each original label, in layers indexed by the nonnegative integers. Choose them pairwise transverse and without triples and transport their gradings. Within each layer fix an order, and order first by the layer. The common almost complex structure will be chosen for this entire countable collection, including every copy.

For a graded sphere in a symplectic K3, the expected dimensions of unparametrized simple spheres and unmarked simple disks are respectively  $-2$  and  $-1$ . Generic compatible almost complex structures exclude them. Multiple spheres reduce to simple spheres, and the disk decomposition theorem reduces existence of a disk to existence of a simple disk; see [Sei15, Lemma 8.4]. A countable intersection of the corresponding residual sets therefore gives one compatible  $J$  without spheres or individual-label disks for any countable collection of the spheres under consideration. On the torus,  $\pi_2(Y) = \pi_2(Y, D_A) = 0$  and Hamiltonian invariance of relative area give the stronger exclusion for every compatible structure. We fix such a  $J$ .

For any finite set of these copies, restrict the clean operations to words  $L_0 < \dots < L_k$ . All corners then have transverse point sectors. An individual-label bubble is excluded, and a constant stable polygon is impossible because it would give a common point of at least three labels. Thus their moduli problems are transverse-ended strip and polygon problems, possibly requiring virtual perturbation, with the same fixed  $J$ .

We compare them with ordinary regular polygons having zero Hamiltonian term. It is useful to restrict all varying almost complex structures to pullbacks of  $J$  by Hamiltonian diffeomorphisms: this preserves the absence of spheres, including during comparison families.

**Lemma 6.2** (Regularity within Hamiltonian pullbacks). *On the strictly ordered words of a finite subcollection, consistent domain-dependent regular Floer data can be chosen with zero Hamiltonian term and with almost complex structure a Hamiltonian pullback of  $J$  at each domain point. The boundary value is  $J$ , and the structures on the strip ends are independent of the strip length coordinate and equal to  $J$  near both boundary heights. On domain-boundary strata the data may be prescribed by mutually compatible, already regular component data of this type; the remaining choices can be made relative to those prescriptions.*

*Proof.* Describe a pullback by a Hamiltonian generating function in an auxiliary time interval  $[0, 1]$ . These generating functions, rather than the endpoint diffeomorphisms, are the extension variables. Choose them zero on boundary arcs and use strip data equal to zero near heights 0 and 1. Functions prescribed on components of a broken polygon extend by splicing with the common strip functions on necks. The prescriptions agree at multiple breaks because every component and every pairwise end has already been assigned its data. Smooth extension from collars of the compactified domain spaces, followed by extension over their interiors, gives consistent data. Additional variations can vanish to infinite order at every face while retaining arbitrary freedom on interior core patches.

We verify that this restricted family suffices for transversality. For a nonconstant strip  $u(s, t)$ , put  $v = \partial_s u$ . The strip-injective-point theorem gives a dense open set where  $v \neq 0$ , where  $u(s, t)$  is distinct from the limiting intersection values, and where it is attained at no other length coordinate with the same height. This is the strip form of the injective-point argument in [FHS95, Sections 3–5]. The Lagrangian-strip version is spelled out in [KS02, proof of Lemma 5.12, the set  $R_1(u)$ ]: the injectivity argument uses the equation on interior strip patches, finite energy, and the transverse

asymptotic limits. On a small such patch, vary a Hamiltonian that vanishes to first order along the image curve at each height. In the compatible metric, its Hessian is an arbitrary symmetric endomorphism  $A$  satisfying  $Av = 0$ . The derivative of its Hamiltonian vector field is  $J'A$ , where  $J'$  is the current structure. Differentiating the pullback structure in the strip equation shows, up to the fixed convention sign, that this variation contributes

$$AJ'v.$$

As  $A$  ranges over symmetric endomorphisms annihilating  $v$ , these vectors fill  $v^\perp$ . To see surjectivity, work on  $v^\perp$ , where  $J'v \neq 0$ , and choose a symmetric matrix with any prescribed value on  $J'v$ . Localize this Hessian along the injective curve and in the height parameter. Such endpoint variations are realized by varying the auxiliary generating function, by composing its endpoint flow with the chosen small Hamiltonian variation.

If  $\mu$  is a universal cokernel vector, it is therefore parallel to  $v$  on this dense open set. In particular  $\langle \mu, J'v \rangle = 0$ , and both assertions extend by continuity where needed. Translation invariance of the strip equation gives  $D_u v = 0$ . For a compactly supported scalar function  $f$ , the Leibniz rule gives

$$D_u(fv) = (\partial_s f)v + (\partial_t f)J'v.$$

Pairing with the cokernel vector and integrating yields

$$\partial_s \langle \mu, v \rangle = 0$$

in the interior height band. Decay at the transverse ends, or integrability in the unweighted strip spaces, forces this constant to be zero. Hence  $\mu$  vanishes on that band, and unique continuation for the adjoint operator gives  $\mu = 0$ . Constant strips have the nondegenerate transverse-end operator and do not occur as nonconstant trajectories.

For a polygon with at least three sides, one may localize the variation at a domain point as well as in the target. At a point where  $du \neq 0$ , the Hessian has no condition of annihilating a curve tangent, so its variations give the required surjectivity directly. A nonconstant solution has such a point in any free interior core region by unique continuation; constant solutions were excluded by the position of the labels. Relative Sard–Smale therefore supplies regular data, retaining the prescribed faces, ends, and boundary values. Apply its countably many conditions on the countably many relative homotopy classes and words. Variations may decay to the prescribed face data arbitrarily rapidly, so the smooth consistency and infinite-order agreement at gluing faces are retained. No uniform regular gluing length over unbounded energies is required.  $\square$

## 6.2 The finite-cut comparison and its analytic hypotheses

We next connect the constant structure to the data in Lemma 6.2. Multiply each auxiliary generating function by a parameter varying from 0 to 1, chosen constant near the parameter endpoints. This is a collared path of domain data. It retains the boundary value  $J$  and remains within Hamiltonian pullbacks, hence has neither sphere bubbles nor individual-label disk bubbles.

**Lemma 6.3** (Comparison on strictly ordered words). *For a finite ordered subcollection as above and every fixed energy and arity cut, the clean operations with constant  $J$  and the regular operations of Lemma 6.2 are related by a filtered  $A_\infty$  pseudo-isotopy through that cut. The comparison and its inverse have no energy loss, preserve the integral grading, and are linear isomorphisms modulo positive valuation on every strict arrow. Zero bounding cochains suffice on both ends and throughout all constant- $J$  changes of clean models used in this assertion.*

*Proof.* We apply the disk and strip Kuranishi construction, gluing, relative CF-perturbation, and integration results summarized in Lemma 5.1. The equations here have domain-dependent coefficients, so we verify the hypotheses of those constructions for this parameter problem.

*Compactification.* Rescaling an interior bubble freezes a compatible pullback of  $J$  and would produce a sphere, which is excluded. Rescaling at a smooth boundary point freezes  $J$  and would produce a disk on a single label, also excluded. At strip ends and necks the same conclusion uses the fixed strip coefficient there. A tree attached to a smooth point of a boundary arc cannot consist entirely of constant disks: with no added inputs, it has an unstable leaf. The only remaining degenerations are transverse-end strip breakings and polygon domain breakings, with strictly ordered components. Gromov compactness applies to the marked problem with area bound and to its parameter version. Areas add on these trees. For a finite ordered list the positive energy gap follows from compactness and transverse strip asymptotics: a limit of nonconstant curves with area tending to zero would be a constant stable polygon or a nonconstant transverse strip of zero energy. Neither is possible. Through a fixed cut there are finitely many areas, since near a limiting tree relative homotopy fixes the area as the sum of the component areas. Alternatively the rational-area convention of Lemma 2.2 gives this discreteness directly. All nonconstant operations in the comparison therefore have positive filtration.

*Local obstruction problems.* The linearized equation is the ordinary Cauchy–Riemann Fredholm problem with transverse asymptotics, supplemented by smooth domain and interval parameters. On polygon cores its coefficients are smooth in those parameters; on strip components they are height-dependent and translation invariant. Choose stabilization and incidence constraints on cores, and choose obstruction sections supported away from nodes, ends, and auxiliary marked points. Elliptic cokernel detection by such test sections gives surjectivity of each thickened component problem, including surjectivity on the fibers over the interval. Consequently the evaluation to the interval is weakly submersive. There is no moving asymptotic intersection value. Auxiliary markings only provide slices: all coefficients pull back from the original punctured polygon, so neither stabilization nor a change of slice changes the equation itself.

*Gluing estimates.* Matched ends carry the identical height-dependent strip operator. Its transverse spectral gap is uniformly positive in a sufficiently small neighborhood of any fixed parameter value. Choose weights strictly within that gap. Preglue with the component data spliced to the common strip data on the necks. The errors are the usual cutoff-derivative terms of size  $O(1/R)$  for neck length  $R$  and the exponentially decaying map errors. The extra coefficient errors have the same exponential estimates: the coefficients are smooth in the plumbing parameters and agree to the prescribed jets with the component and strip coefficients, while a plumbing parameter is exponentially small in the neck length. These estimates hold for every fixed finite number of derivatives. The parametrix from the surjective thickened component problems therefore has exactly the small-error estimate needed for the Neumann-series correction. On cores, parameter derivatives are bounded. On ends they multiply decaying map derivatives, because the intersection limits remain fixed, so they are bounded in the chosen weighted norms as well.

The differentiable gluing argument now applies to component, interval, and length derivatives. Derivatives of the additional core coefficients remain bounded; length derivatives of their plumbing errors, in the fixed component/end charts, remain exponentially small. Decreasing the exponential weights when taking derivatives and using the usual smooth gluing profile gives the collared corner charts. Strip-trajectory breaking uses coefficient data independent of the trajectory length and hence adds no coefficient error. Thus these domain-dependent data introduce neither a new Fredholm matching condition nor an uncontrolled gluing term in the disk gluing theorem. In particular domain-independence of  $J$  is unnecessary for this application of its local estimates.

*Corner compatibility and relative perturbation.* For each tree take the direct sum of the component

core obstruction spaces, extending them off the stratum by the just-described gluing and stabilization charts. Components carry the same interval parameter; hence a normalized corner is a product of component spaces over that interval. Evaluations at the transverse polygon corners are to fixed point sectors. Transporting incidence slices and core obstructions gives the coordinate changes, and iterated breakings give the same sums of component data in either order. This is precisely the disk-component-wise corner compatibility required by the construction theorem. The added interval parameter is matched at every vertex, and fiberwise thickening makes all required projections submersive. The relative version prescribes the endpoint structures, or compares them by their collared pseudo-isotopies, and then extends the compatible normalized-corner CF-perturbations after thickening and outer collaring.

At the constant endpoint the moduli compactifications are those of the clean construction on these strict words. At the regular endpoint one takes the ordinary counts, using zero obstruction spaces on the regular, automorphism-free problems. No forgetting-mark compatibility is needed on strictly ordered words. The construction is made through an energy and arity cut, not at all arities for a fixed energy. The additive tree cut of Lemma 5.1 makes the boundary induction finite and includes every operation and homotopy needed in the pseudo-isotopy identities. The signed Stokes and fiber-product composition formulas then give those identities.

*Orientations and gradings.* Use the same capping paths and spin trivializations in both constructions. Gluing their determinant lines transports an intersection orientation line and its reciprocal contraction together. The zero background class and the vanishing capped Maslov contributions leave no disk-class-dependent sign twist. Unitary frame paths with the same grading lift are compared in  $SU$ , which is simply connected; spin framings may be stabilized in low rank. The remaining determinant permutation conventions depend only on indices and input order. In suspended degrees, let their difference be  $r_k(a_1, \dots, a_k)$ . Determinant gluing identifies the sums of these exponents under insertion in any slot. Rescaling the generator lines by degree signs first makes  $r_1 = 0$ . Inserting a unary vertex then shows that each  $r_k$  is independent of each  $a_i$ ; inserting a binary vertex gives  $r_k = (k - 1)r_2$ . One overall convention sign removes this final difference. These are identities of determinant lines for formal gluing data, so arbitrary indices can be used without requiring a holomorphic solution for every test. The comparison therefore identifies the signed operations, with the same suspension and variance conventions. The integer dimension formulas give homogeneous maps and homotopies.

All equations in this path are area equations, with no Hamiltonian term. Hence the comparisons have no energy loss. On a strict point sector the zero-energy differential is zero; the comparison has invertible zero-energy linear term. Single-label curvature stays zero because all such disks are excluded. This proves every assertion of the lemma.  $\square$

The regular data just used can be included in the ordinary full category. On the remaining pair strips choose nondegenerate Hamiltonian chords and generic Hamiltonians, retaining boundary structure  $J$ . One may still take interior-height structures to be pullbacks of  $J$ : Hamiltonian transport changes the strip problem to a boundary problem to which the variation proof of Lemma 6.2 applies. On interiors of the other polygon domain spaces, unrestricted domain-point Hamiltonian variations give regularity. Extend the Hamiltonian terms and generating functions consistently from the previously prescribed faces. Sphere exclusion and the boundary disk exclusion persist for this choice. Alternatively, ordinary comparison data on the quartic may use Seidel's sphere-bubble dimension argument in moduli spaces of dimension at most one [Sei15, Section 8c]. Interior variations flat at faces have the same compactness and gluing limits as the prescribed data; their errors and derivatives are exponentially small on plumbing necks, and neck length times such an error remains bounded. Thus the ordinary invariance construction also works relative to our prescribed sector.

The area coefficients agree. Seidel uses mod-rational action lines determined by flat prequantum multisections. After extension to  $\Lambda$ , choosing action offsets identifies these with the area-weighted intersection lines by rescaling their generators. This agrees on every strict word; changing offsets rescales input and output lines compatibly. On the torus the ordinary weakly exact construction uses the same area weights from the outset.

### 6.3 Passing finite comparisons to a field

The following criterion records exactly what a coarsened residue preserves in a sequence of finite-cut comparisons.

**Lemma 6.4.** *Suppose complexes and comparison maps over  $\Lambda$  are given at successive cutoffs on fixed finite-dimensional graded spaces with fixed homogeneous bases, with all their matrix entries bounded below. Assume that their chain-map equations have errors tending to infinite valuation. Their reductions are chain maps over  $\kappa$ . They are quasi-isomorphisms under either of the following sufficient conditions:*

- (a) *homotopy inverse maps and the two inverse homotopies have bounded-below matrices and satisfy their equations with errors tending to infinite valuation;*
- (b) *the source and target have equal dimensions in every degree, and in every degree the comparison determinants have valuation zero for all sufficiently large cutoffs.*

*The same reduction applies to each fixed-word equation for  $A_\infty$  comparison functors whose zero-input components vanish, provided the component matrices are bounded below and the equation errors tend to infinite valuation.*

*Proof.* All stated equations involve finite sums and products of entries. Lemma 5.7 therefore gives exact chain-map and, in case (a), homotopy-inverse equations. In case (b), Lemma 5.9 makes the reduced chain map invertible in every degree, hence a chain isomorphism. A fixed-word  $A_\infty$  functor equation with zero-input functor component zero is again a finite polynomial identity. Its curvature terms vanish upon reduction under the hypotheses of Proposition 5.8.  $\square$

For example, a filtered comparison whose linear term is invertible modulo  $\Lambda_+$  satisfies condition (b). This includes filtered homotopy equivalences on point-sector models whose differentials vanish modulo  $\Lambda_+$ : reducing the homotopy-inverse equations then makes the linear terms mutually inverse. Quasi-isomorphism at each cutoff alone would not give this conclusion under coarsened reduction, since a nonzero entry of valuation tending to infinity reduces to zero.

We have now compared every finite strictly ordered calculation. To obtain an actual functor from comparisons through arbitrarily large cuts, take the fixed finite clean models supplied by Lemma 5.1. Their finite-cut comparisons can be chosen with zero bounding cochains throughout because the common constant  $J$  has no single-label disks. Transfer on strict point-sector words does not require additional objects. The resulting comparison matrices and their inverse matrices have valuation bounded below, since there is no energy loss. Apply Lemma 6.4 to these sequences of matrices. Every fixed word then satisfies the exact functor equations over its residue field  $\kappa$ . The zero-energy invertible linear terms give quasi-isomorphisms on strict arrows, either directly or by reducing the inverse comparison and the homotopies at the same time. This procedure uses fixed models before passing to the field; it does not evaluate a nonconvergent limiting bounding cochain.

Repeated copies are isomorphic in both full categories. In the ordinary torus category this is weakly exact continuation; for the quartic spheres it is Seidel's sphere invariance, including

independence of the regular boundary structure [Sei15, Lemma 8.14]. In clean theory use Hamiltonian invariance with transported bounding cochains [Fuk25, Section 15]. It gives isomorphisms after inverting the Novikov parameter; finite Hofer energy causes no restriction over the field. Zero transports to zero on the torus by the disk exclusion. On a sphere the canonical model has no degree-one term, so its bounding-cochain class is unique. The gradings are transported throughout. Each individual label has the full ordinary self-cohomology ranks, and consequently the degree-zero Hom between two copies is one-dimensional in either full category.

The algebraic lemmas below assemble these facts into the full comparison and are also useful for keeping track of the fields.

## 6.4 Recovering a full category from ordered copies

The comparisons on strictly ordered polygons omit the endomorphisms of a label and all arrows pointing backwards in the order. Repeated copies of each label recover this missing information. The precise requirement is that an arrow from one copy to the next become invertible in each full category. We first explain how comparisons on finite lists can be made simultaneously, and then prove the reconstruction statement.

We use ordinary  $A_\infty$  module calculus as in [Kel01, Sei08]. All categories in this subsection are uncurved,  $\mathbb{Z}$ -graded, strictly unital  $A_\infty$  categories over a field. All functors are ordinary  $A_\infty$  functors, with zeroth component  $F^0 = 0$ . A finite Hom model means that every morphism space is finite-dimensional as a total graded vector space; the bound may depend on the pair of objects. The linear term of an  $A_\infty$  functor is therefore a map of bounded finite-dimensional complexes. A quasi-equivalence means a functor inducing quasi-isomorphisms on these complexes and an essentially surjective functor on  $H^0$ ; equivalence up to quasi-equivalence allows a zigzag of such functors.

**Lemma 6.5.** *Let  $K$  be an algebraically closed field of cardinality  $2^{\aleph_0}$  containing  $\Lambda$ , and let  $\mathfrak{u}$  be a nonprincipal ultrafilter on  $\mathbb{N}$ . Its countable ultrapower  $K^{\mathbb{N}}/\mathfrak{u}$  is an algebraically closed extension of the same cardinality. For complexes with fixed finite-dimensional graded spaces, an ultrafilter-large sequence of quasi-isomorphisms remains a quasi-isomorphism in this ordinary ultrapower.*

*More generally, suppose countably many unknown scalars are subject to countably many polynomial equations and nonvanishing conditions over  $K$ , and every finite subsystem has a solution in  $K$ . The whole system has a solution in this ultrapower.*

*Proof.* The algebraic-closure and cardinality proofs are the coordinatewise root argument and cardinality calculation of Lemma 5.7. Unlike coarsened reduction, the ordinary ultrapower retains every sequence that is nonzero on a set belonging to  $\mathfrak{u}$ .

For a matrix of fixed size, its rank in the ultrapower is the unique integer equal to its coordinate ranks on a set belonging to  $\mathfrak{u}$ . This follows directly from the minors: higher-order minors vanish on that set, and one of the finitely many minors of the claimed rank is nonzero on a set belonging to  $\mathfrak{u}$ . Apply this observation to the differentials of the mapping cones. The rank equalities expressing acyclicity in each degree are preserved, so the induced maps remain quasi-isomorphisms.

For the final assertion, enumerate the conditions, choose at index  $j$  a solution of the first  $j$  conditions, and assign arbitrary values to unknowns not yet involved. Each condition holds for all sufficiently large indices and therefore holds in the ultrapower.  $\square$

In particular, finite-sublist comparison functors into fixed finite models can be assembled over one further ordinary ultrapower when their equations form a countable system. Their linear quasi-isomorphisms survive by the preceding lemma.

**Lemma 6.6** (Simultaneous finite-list comparisons). *Let  $k$  be a field, let  $\mathcal{D}$  be a strictly unital uncurved  $A_\infty$  category with countably many objects and finite Hom models, and let*

$$\mathcal{D}_1 \subset \mathcal{D}_2 \subset \cdots \subset \mathcal{D}$$

*be full subcategories on finite sets of objects whose union contains every object. Let  $\mathcal{C}_1, \dots, \mathcal{C}_s$  be strictly unital uncurved  $A_\infty$  categories with finite sets of objects and finite Hom models, and fix object maps  $f_a : \text{Ob}(\mathcal{D}) \rightarrow \text{Ob}(\mathcal{C}_a)$ . Fix also a set  $\mathcal{P}$  of ordered pairs of objects of  $\mathcal{D}$ . Suppose that for every positive integer  $N$  and every  $a$  there is a strictly unital functor*

$$F_{a,N} : \mathcal{D}_N \longrightarrow \mathcal{C}_a$$

*with object map  $f_a$ , whose linear term is a quasi-isomorphism on each pair in  $\mathcal{P}$  belonging to  $\mathcal{D}_N$ . There is a field extension  $K/k$  and strictly unital functors*

$$F_a : \mathcal{D} \otimes_k K \longrightarrow \mathcal{C}_a \otimes_k K$$

*with those object maps and quasi-isomorphisms on every pair in  $\mathcal{P}$ . If  $k$  is algebraically closed and has cardinality at most  $\mathfrak{c} = |\mathbb{R}|$ , the same is true of  $K$ . If, in addition,  $k$  has characteristic zero and contains a copy of  $\mathbb{C}$ , then  $K$  is abstractly isomorphic to  $\mathbb{C}$ .*

*Proof.* Choose a nonprincipal ultrafilter  $\mathcal{U}$  on the positive integers and set

$$K = k^{\mathbb{N}}/\mathcal{U}.$$

Here two sequences are identified when they agree on a set belonging to  $\mathcal{U}$ . This quotient is a field: a nonzero class has a representative  $(a_N)$  with  $a_N \neq 0$  on a set in  $\mathcal{U}$ , and its inverse is represented by  $a_N^{-1}$  on that set, with arbitrary values elsewhere. Constant sequences embed  $k$  into  $K$ .

Fix homogeneous bases in all source and target morphism spaces. For a fixed composable word of source objects, the coefficients of each corresponding functor component  $F_{a,N}^j$  are defined for all sufficiently large  $N$ . Fill in the finitely many missing values arbitrarily and take their classes in  $K$ . These classes define  $F_a^j$ . For each fixed word, the  $A_\infty$  functor equation is a finite sum of products of the structure coefficients and the functor coefficients. It holds for all sufficiently large  $N$ , so it holds in  $K$ . The same argument proves the strict unit identities. Thus all components together define the stated functors. No compatibility between the separately chosen  $F_{a,N}$  is required.

We check quasi-isomorphisms explicitly. Fix a pair in  $\mathcal{P}$  and a target  $\mathcal{C}_a$ . For sufficiently large  $N$ , the mapping cone of  $F_{a,N}^1$  is an acyclic bounded complex of finite-dimensional vector spaces. Such a complex is contractible: split its cycles and boundaries degree by degree. Choose a contraction  $h_N$  with

$$d_N h_N + h_N d_N = \text{id}.$$

The cone has one fixed finite graded underlying vector space, since the object map and the Hom models are fixed. Consequently this equation is a finite matrix identity. Taking classes of the coefficients gives a contraction of the cone of  $F_a^1$ . Hence  $F_a^1$  is a quasi-isomorphism. This argument applies to each pair separately and requires no uniform dimension bound over all pairs.

The coordinatewise root and cardinality arguments in Lemma 6.5 apply to any algebraically closed  $k$  with  $|k| \leq \mathfrak{c}$ : they give algebraic closure of  $K$  and  $|K| \leq |k|^{\aleph_0} \leq \mathfrak{c}$ . If  $k$  contains  $\mathbb{C}$ , both  $K$  and  $\mathbb{C}$  have transcendence degree  $\mathfrak{c}$  over  $\mathbb{Q}$ . The field-identification argument of Remark 5.10 therefore gives  $K \cong \mathbb{C}$ .  $\square$

*Remark 6.7.* The isomorphism  $K \cong \mathbb{C}$  is an isomorphism of abstract fields. It need not fix the copy of  $\mathbb{C}$  originally embedded in  $K$ . When it is used to regard the resulting objects as complex objects, all defining coefficients and all maps must be transported by the same isomorphism. In particular, the categorical comparison is established over  $K$  before making this identification.

We now specify the ordered category used in the reconstruction. Let  $I = \{1, \dots, \ell\}$ , with  $\ell \geq 1$ , be a finite set of labels. Its copies are objects  $x_{\alpha, m}$ , where  $\alpha \in I$  and  $m \geq 0$ , ordered first by  $m$  and then by  $\alpha$ . Thus every fixed object precedes  $x_{\alpha, m}$  for all sufficiently large  $m$ , for each label  $\alpha$ . An ordered category on these copies has

$$\mathrm{Hom}_{\mathcal{D}}(x, x) = ke_x, \quad \mathrm{Hom}_{\mathcal{D}}(y, x) = 0 \quad \text{if } x < y,$$

where  $e_x$  is its strict unit; its other morphisms go from earlier to later objects. We call these other pairs strict arrows. If the polygon calculation initially provides a nonunital category with only strict arrows, adjoining the  $e_x$  gives this strictly unital category. A functor on the original category extends by  $F^1(e_x) = e_{F(x)}$  and by setting every higher component with a unit input equal to zero. The strict unit identities then give all equations involving a unit.

**Lemma 6.8** (Reconstruction from ordered copies). *Let  $k$  be a field, and let  $\mathcal{D}$  be an ordered strictly unital uncurved  $A_\infty$  category on the objects  $x_{\alpha, m}$  just described, with finite Hom models. Let  $\mathcal{C}_1$  and  $\mathcal{C}_2$  be strictly unital uncurved  $A_\infty$  categories with finite Hom models and object sets  $\mathrm{Ob}(\mathcal{C}_a) = \{X_{a, \alpha} : \alpha \in I\}$  for  $a = 1, 2$ . Suppose that there are strictly unital functors*

$$F_a : \mathcal{D} \longrightarrow \mathcal{C}_a, \quad F_a(x_{\alpha, m}) = X_{a, \alpha}, \quad a = 1, 2,$$

*whose linear terms are quasi-isomorphisms on every strict arrow. Suppose further that there are classes*

$$t_{\alpha, m} \in H^0 \mathrm{Hom}_{\mathcal{D}}(x_{\alpha, m}, x_{\alpha, m+1})$$

*such that  $H^0(F_a)(t_{\alpha, m})$  is invertible for both  $a = 1, 2$ . Then  $\mathcal{C}_1$  and  $\mathcal{C}_2$  are equivalent up to quasi-equivalence, matching the objects with the same label.*

*Proof.* We use the dg category of strictly unital right  $A_\infty$  modules with all higher module morphisms, normalized to vanish when an algebra input is a unit. A module quasi-isomorphism means a quasi-isomorphism at every object. Over a field every such quasi-isomorphism is a homotopy equivalence in this dg category. Indeed, an objectwise acyclic module  $M$  admits a contraction at each object. Extend these contractions successively in the number of algebra inputs. At each stage the remaining contraction equation has an obstruction in a product of ordinary Hom complexes with contractible target  $M(x)$ . The obstruction is closed because the morphism differential squares to zero and the lower-arity equations have been solved. It is therefore exact, and the next component can be chosen. The strict-unit normalization is preserved by the differential, so the induction stays within normalized morphisms. There is one component for each finite input word, and the induction requires no convergence of infinite sums. Thus  $M$  is contractible as an  $A_\infty$  module. Applying this to the cone proves the assertion about quasi-isomorphisms. Consequently this dg category enhances the derived category  $D(\mathcal{D})$ , and its morphism complexes compute  $\mathrm{RHom}_{\mathcal{D}}$ . Restriction and Yoneda are enhanced functors into this particular module model.

Write

$$h_x = \mathrm{Hom}_{\mathcal{D}}(-, x)$$

for the right representable module. The module Yoneda lemma gives, naturally in a module  $N$ ,

$$\mathrm{RHom}_{\mathcal{D}}(h_x, N) \simeq N(x). \quad (6.1)$$

Under this identification, a morphism induced by  $x \rightarrow y$  acts by the module's contravariant action  $N(y) \rightarrow N(x)$ . Direct sums and mapping cones exist in this module category and are computed objectwise.

For the moment fix one of the functors  $F : \mathcal{D} \rightarrow \mathcal{C}$  and omit its index. The restriction of the target representable at  $X_\beta$  is the module

$$N_\beta = F^* \operatorname{Hom}_{\mathcal{C}}(-, X_\beta), \quad N_\beta(x) = \operatorname{Hom}_{\mathcal{C}}(F(x), X_\beta).$$

The restricted Yoneda functor sends  $X_\beta$  to  $N_\beta$ . We will prove that it is cohomologically fully faithful and that each  $N_\alpha$  is determined, up to isomorphism, by  $\mathcal{D}$  and the chosen successor classes  $t_{\alpha,m}$ .

Fix  $\alpha$  and abbreviate  $x_{\alpha,m}$  to  $x_m$  and  $t_{\alpha,m}$  to  $t_m$ . Choose closed representatives for the module maps  $h_{x_m} \rightarrow h_{x_{m+1}}$  induced by  $t_m$ . Let  $s$  be their direct sum shift, taking the  $m$ th summand to the  $(m+1)$ st. Define the telescope  $T_\alpha$  by the triangle

$$\bigoplus_{m \geq 0} h_{x_m} \xrightarrow{1-s} \bigoplus_{m \geq 0} h_{x_m} \longrightarrow T_\alpha \longrightarrow \left( \bigoplus_{m \geq 0} h_{x_m} \right) [1]. \quad (6.2)$$

Thus  $T_\alpha$  is the mapping cone of  $1 - s$ . Its isomorphism class in  $D(\mathcal{D})$  depends only on the sequence of classes  $t_m$ .

Put  $v_m = H^0(F)(t_m)$ , an invertible endomorphism of  $X_\alpha$  in  $H^0(\mathcal{C})$ . Define invertible classes  $u_m$  by

$$u_0 = \operatorname{id}_{X_\alpha}, \quad u_{m+1} = u_m v_m^{-1}.$$

By (6.1),  $u_m \in H^0 N_\alpha(x_m)$  defines a morphism  $f_m : h_{x_m} \rightarrow N_\alpha$  in  $D(\mathcal{D})$ . The identity  $u_{m+1} v_m = u_m$  says precisely that these morphisms are compatible with the successor maps. Applying  $\operatorname{Hom}_{D(\mathcal{D})}(-, N_\alpha)$  to (6.2) therefore gives a morphism

$$\theta_\alpha : T_\alpha \longrightarrow N_\alpha$$

whose composite with the map from each summand  $h_{x_m}$  is  $f_m$ . Compatibility in the derived category suffices: exactness of the Hom sequence of this triangle supplies the required morphism, without an additional choice of higher compatibility data.

This morphism is a quasi-isomorphism. To see this, evaluate at an arbitrary object  $y$  of  $\mathcal{D}$ . Evaluation preserves direct sums and mapping cones. On a direct sum of any sequence of vector spaces,  $1 - s$  is injective: the first component of an element in its kernel is zero, and induction gives the same for every later component. The cohomology sequence of the evaluated triangle consequently gives

$$H^j T_\alpha(y) = \varinjlim_m H^j \operatorname{Hom}_{\mathcal{D}}(y, x_m). \quad (6.3)$$

For all sufficiently large  $m$ , the arrow  $y \rightarrow x_m$  is strict. On cohomology the evaluated map  $f_m$  is

$$H^j \operatorname{Hom}_{\mathcal{D}}(y, x_m) \xrightarrow{H^j(F^1)} H^j \operatorname{Hom}_{\mathcal{C}}(F(y), X_\alpha) \xrightarrow{u_m \circ (-)} H^j \operatorname{Hom}_{\mathcal{C}}(F(y), X_\alpha).$$

Both arrows are isomorphisms. Their composites are compatible with the direct system. They therefore identify the limit in (6.3) with  $H^j N_\alpha(y)$ , and this identification is induced by  $\theta_\alpha$ . Since  $y$  and  $j$  were arbitrary,  $T_\alpha \simeq N_\alpha$  in  $D(\mathcal{D})$ .

It remains to check morphisms between these modules; this also verifies that the comparison just obtained is the one induced by restricted Yoneda. Apply  $\operatorname{RHom}_{\mathcal{D}}(-, N_\beta)$  to (6.2). By (6.1), the

two sums become products of the complexes  $N_\beta(x_m)$ . For each  $j$  the resulting cohomology sequence is the short exact sequence

$$\begin{aligned} 0 \longrightarrow \varprojlim_m^1 H^{j-1} N_\beta(x_m) &\longrightarrow H^j \operatorname{RHom}_{\mathcal{D}}(T_\alpha, N_\beta) \\ &\longrightarrow \varprojlim_m H^j N_\beta(x_m) \longrightarrow 0. \end{aligned} \quad (6.4)$$

For completeness, this sequence is simply the kernel–cokernel sequence for  $1 - \tau$  on products of cohomology groups. If  $\tau_m : H^j N_\beta(x_{m+1}) \rightarrow H^j N_\beta(x_m)$  denotes the transition map, then

$$(1 - \tau)((a_m)_m) = (a_m - \tau_m(a_{m+1}))_m.$$

Its kernel is  $\varprojlim$ , and its cokernel is the sequential  $\varprojlim^1$ . Products of complexes of vector spaces commute with cohomology, which justifies this description directly from the triangle.

Each  $\tau_m$  is precomposition by  $v_m$  and is therefore an isomorphism. The displayed product map is surjective: given  $(b_m)$ , set  $a_0 = 0$  and recursively choose  $a_{m+1} = \tau_m^{-1}(a_m - b_m)$ . Thus the left term of (6.4) vanishes. Evaluation at  $m = 0$  identifies its right term with  $H^j \operatorname{Hom}_{\mathcal{C}}(X_\alpha, X_\beta)$ . More explicitly, a class  $w$  in this last group corresponds to the compatible sequence  $(wu_m)_m$ . The restricted Yoneda image of  $w$ , composed with  $f_m$ , is exactly this class under module Yoneda. Hence the isomorphism calculated above is induced by the actual map

$$H^j \operatorname{Hom}_{\mathcal{C}}(X_\alpha, X_\beta) \longrightarrow H^j \operatorname{RHom}_{\mathcal{D}}(N_\alpha, N_\beta).$$

Restricted Yoneda is cohomologically fully faithful.

Apply this argument to both  $F_1$  and  $F_2$ . In the dg module category fixed above, their restricted representables with label  $\alpha$  are both isomorphic in  $H^0$  to the same object  $T_\alpha$ . Take the full dg subcategory on all these restricted representables and on the  $T_\alpha$ . Each restricted Yoneda functor into this category is cohomologically fully faithful. It is also essentially surjective: every object of this full dg subcategory is isomorphic in  $H^0$  to one of its images with the same label. This gives the asserted zigzag of quasi-equivalences.  $\square$

*Remark 6.9.* In the application, the successor classes require no special normalization. The two repeated copies are isomorphic in each full category, and their degree-zero Hom space is one-dimensional. A nonzero class in that space is consequently a nonzero scalar multiple of an isomorphism, and hence is itself invertible. The quasi-isomorphism on strict arrows implies that any nonzero choice of  $t_{\alpha,m}$  has this property simultaneously for both target categories.

*Proof of Proposition 6.1.* Take as ordered source the fixed ordinary calculation on the countable system of copies, with formal strict units adjoined. Its restriction to each finite number of layers maps to the two full categories of the original labels: for the ordinary target use invariance and the copy isomorphisms; for the clean target first use Lemma 6.3 and the coarsened reduction, and then use clean Hamiltonian invariance. All finite reductions can use the same coarsened ultrapower of  $\Lambda$ : choose the finite-cut sequences for each finite list separately in that one field construction. Thus the fixed source and both fixed finite targets are over a common field  $\kappa$ . Take finite Hom models in the targets, and compose with quasi-inverses over this field when needed. These are uncurved functors with zero nullary component. They induce quasi-isomorphisms on all strict arrows.

Lemma 6.6 assembles the finite-layer functors, after one ordinary field ultrapower, into two functors on the full ordered source. The source has one-dimensional degree-zero cohomology between successive copies of a label. Choose any nonzero class there. Its image in each target is nonzero, and hence an isomorphism, because those copies are isomorphic and their degree-zero Hom space is

a line. Lemma 6.8 now identifies the two full categories, matching the labels. All structure maps, inverse maps, and contractions were homogeneous, so this identification respects the integral grading. The field statements follow from Lemma 5.7 and Lemma 6.6.  $\square$

## 6.5 Products, integral gradings, and ordinary perfect hulls

The remaining categorical input constructs the product collection. We state separately the published filtered theorem and the interpretation over a field that will be used in the proof.

**Theorem 6.10** (Filtered Künneth theorem). *Let  $(X_i, \omega_i)$  be compact symplectic manifolds and let  $\mathcal{L}_i$  be finite clean collections of relatively spin immersed Lagrangians with specified background data. Denote by  $\mathrm{Fuk}^{\mathrm{st}}(X_i; \mathcal{L}_i)$  the uncurved categories of these labels with zero-curvature positive-valuation bounding cochains. There is a strict unital filtered  $A_\infty$  functor*

$$\mathrm{Fuk}^{\mathrm{st}}(X_1; \mathcal{L}_1) \otimes \mathrm{Fuk}^{\mathrm{st}}(X_2; \mathcal{L}_2) \longrightarrow \mathrm{Fuk}^{\mathrm{st}}(X_1 \times X_2; \mathcal{L}_1 \times \mathcal{L}_2)$$

*which is a homotopy equivalence onto its image. On objects it sends  $((L_1, b_1), (L_2, b_2))$  to the product label equipped with an induced bounding cochain  $b_1 \times b_2$ , canonical up to gauge equivalence.*

This is [Fuk25, Situation 16.8, Theorem 16.9, and Corollary 16.10]; the algebra-level Künneth theorem was obtained independently by Amorim [Amo17]. We use this theorem only for embedded factors and with zero background. Products of the finite clean collections are clean, since their fiber products are the products of the factor fiber products.

**Proposition 6.11** (Ordinary graded product category). *Suppose the manifolds and branes in Theorem 6.10 have the Calabi–Yau volume gradings used in this paper. Choose finite subcollections of bounded objects and finite-rank canonical Hom models. After including their bounding cochains and extending to the Novikov field, the full category of the induced product objects is quasi-equivalent to the ordinary enhanced tensor product of the two uncurved factor categories. This comparison respects the integral grading. The same conclusion holds after further coefficient extension. Consequently, if the two ordinary perfect hulls are  $\mathrm{Perf}(X)$  and  $\mathrm{Perf}(Z)$  for smooth projective varieties, the product perfect hull is  $\mathrm{Perf}(X \times Z)$ .*

*Proof.* We first verify the integer grading, because a parity-graded tensor statement would not suffice. Cap all boundary conditions using the lifts prescribed by the brane gradings. The volume trivializations make the capped disk and sphere Maslov contributions zero. At a product corner take the direct sum of the two caps; at a corner on only one factor take the identity cap on the other. The Fredholm symbols with boundary conditions are direct sums, so the map-index contributions add. For matching strips, equivalently folded diagonal boundaries, choose the diagonal lift whose identity strip has degree zero. Its two seam contributions cancel.

Apply this convention to the tri-module construction of [Fuk25, Sections 5 and 16]. Cap the two disk problems separately. At a product corner the shift is the sum of the two shifts, while de Rham input and pushforward use the ordinary degree on the unsplit product of intersection components. The matched domain parameters are counted once. For example, a disk with  $k + 1$  corners has domain dimension  $k - 2$ ; together with its capped map index this yields output degree equal to the sum of input degrees plus  $2 - k$ . Replacing its boundary conditions by product conditions adds their map indices without adding a second copy of  $k - 2$ . The extra module domains and comparison homotopies obey the same rule: marked-point, homotopy, and matching dimensions are counted once. On a clean intersection component  $C$  in a factor of complex dimension  $n_i$ , the two reversed corner shifts sum to  $n_i - \dim C$ . This is exactly the dimension correction in the fiber-product or

pushforward at that node. Thus all index equalities used by the tensor and tri-module constructions are equalities in  $\mathbb{Z}$ . Use homogeneous contractions in their algebraic steps; in particular the induced bounding cochains have degree one. The orientations reduced modulo two agree with the signs in the quoted theorem, using spin framings stabilized along the caps.

We next justify forgetting the filtration. First twist all operations by the chosen bounding cochains. These insertions are convergent in the filtered models, and the resulting categories have  $m_0 = 0$ , and the strict functor in Theorem 6.10 has nullary component  $F^0 = 0$ . Subsequent formulas for an  $A_\infty$  functor or module on any fixed word are therefore finite arity formulas; no new infinite bounding-cochain insertion is made after forgetting the filtration.

Use the tensor model of [Fuk25, Section 16.1], whose objects are exterior bi-Yoneda representables. Restrict to the selected bounded factor objects. The restricted bi-Yoneda objects are the representables for these full subcategories, and Yoneda evaluation in each variable still computes their Homs. Hence this restriction gives the same full product comparison. Tensor is invariant under filtered homotopy equivalence, so the factor categories may now be replaced by their finite-rank Hom models.

On finite-rank Homs, completed tensor followed by extension to the field is the ordinary tensor. Compare the filtered bi-Yoneda model over the field with the ordinary bi-Yoneda model. Objectwise their values coincide, since they are just tensors of finite-rank spaces. A filtered premorphism maps to an ordinary premorphism component by component at each word length. This is a map of complexes and respects composition, because the categories are now uncurved and these formulas are finite at each word. It is a quasi-isomorphism on Homs between the selected objects: in both models, Yoneda evaluation identifies that Hom complex up to homotopy with the same finite tensor complex over the field. Explicitly, the unit contraction on a representable bar resolution applies in each variable; it is a finite formula in each word and preserves the filtration in the filtered model. Thus the two Yoneda evaluations and their unit homotopies commute with the comparison just described.

This argument does not require the entire premorphism space, an infinite product of components, to commute with scalar extension. Only its Yoneda contraction and the resulting finite Hom complex are used. The ordinary bi-Yoneda tensor computes the ordinary enhanced tensor product: replacing the factors by dg models gives the usual dg tensor product and its representable modules. This proves the asserted quasi-equivalence and explains why no completed perfect hull appears. Further field extension preserves these finite Hom quasi-isomorphisms and the split-closed hull.

Finally, external products of perfect generators of smooth projective  $X$  and  $Z$  split-generate  $\text{Perf}(X \times Z)$ , as follows from exterior products of finite perfect diagonal resolutions. Their Hom complexes and compositions are the ordinary tensor products. This proves the final claim.  $\square$

## 7 A perfect complex with a controlled second extension group

We now pass from the immersion to algebraic geometry. The role of Floer theory in this step is precise: it supplies an object with finite-dimensional morphism cohomology containing a perfect complex as a direct summand, and it computes the Euler pairings of that complex. We do not require a mirror-symmetry theorem for the immersion itself.

Let  $\mathcal{D}$  be the finite collection of embedded product labels in  $M = P \times Y$  supplied by Proposition 2.3. Let  $i : L \rightarrow M$  be the immersion of Proposition 4.1. The incidence conclusions of Proposition 4.1 make the intersections with  $L$  transverse and the entire finite list clean. Choose the split almost complex structure  $J_P \times J_Y$  used in Proposition 2.3:  $J_P$  has no nonconstant holomorphic spheres, and  $Y$  has none because  $\pi_2(Y) = 0$ . A holomorphic sphere in the product projects to

holomorphic spheres in the two factors, so it too is constant. The common area denominator and the homology injection required below are part of Proposition 4.1. Use Proposition 5.2 to adjoin  $L$  with zero curvature. The comparisons of Proposition 6.1, including their compatibility with the chosen bounding cochains, identify the full subcategory generated by  $\mathcal{D}$  after perfect hull with

$$\mathcal{B} \simeq \text{Perf}(S \times B).$$

All statements are over one common algebraically closed coefficient field obtained by the field extensions in those propositions.

## 7.1 Representability and a splitting

Here is the categorical argument that makes the topological bound useful. The categories in the next lemma are enhanced triangulated categories: one may use pretriangulated differential graded categories or strictly unital  $A_\infty$  categories and their perfect hulls. Proper means that the total cohomology of each morphism complex is finite-dimensional; smooth means that the diagonal bimodule is perfect.

**Lemma 7.1** (A splitting detected by relative traces). *Let  $\mathcal{A}$  be a proper enhanced category over a field  $k$ , and let  $\mathcal{B} \subset \mathcal{A}$  be a full smooth proper subcategory closed under shifts, cones, and direct summands. Suppose that its Serre functor is the shift  $[n]$ . Let  $D_1, \dots, D_m$  split-generate  $\mathcal{B}$ , with  $\text{Hom}^0(D_j, D_j) = k$  for every  $j$ . Let  $L \in \mathcal{A}$ . Assume that there are nonzero linear functionals*

$$t_j : \text{Hom}^n(D_j, D_j) \longrightarrow k$$

for which composition induces perfect pairings

$$\text{Hom}^r(D_j, L) \otimes \text{Hom}^{n-r}(L, D_j) \longrightarrow k, \quad (u, v) \longmapsto t_j(vu)$$

for every  $j$  and  $r$ . Then there is an object  $\mathcal{E} \in \mathcal{B}$  which is a direct summand of  $L$  in  $\mathcal{A}$ . Its restricted right Yoneda module is isomorphic to that of  $L$ . In particular,

$$\dim_k \text{Ext}^2(\mathcal{E}, \mathcal{E}) \leq \dim_k \text{Hom}^2(L, L).$$

*Proof.* Restrict the right Yoneda module of  $L$  to  $\mathcal{B}$ . It is a proper module. Smoothness makes it perfect [TV07, Lemma 2.8 and Corollary 2.9]: express the diagonal bimodule as a summand of a finite iterated extension of exterior products of representable modules, and tensor this expression with the restricted module. Its values on the finitely many left factors are perfect complexes of  $k$ -vector spaces by properness. The resulting expression is a finite iterated extension of representables, with a possible direct summand. Thus the module is represented by some  $\mathcal{E} \in \mathcal{B}$ .

The representing module isomorphism corresponds, by Yoneda, to a map  $f : \mathcal{E} \rightarrow L$ . In the triangle

$$\mathcal{E} \xrightarrow{f} L \longrightarrow C \longrightarrow \mathcal{E}[1],$$

the object  $C$  is right orthogonal to  $\mathcal{B}$ . To obtain the other orthogonality, fix  $D = D_j$ . Serre duality and  $\text{Hom}^0(D, D) = k$  give  $\text{Hom}^n(D, D) = k$ . Consequently the nonzero functional  $t_j$  is a nonzero multiple of its Serre trace. Composition followed by this trace gives perfect pairings between  $\text{Hom}^r(D, \mathcal{E})$  and  $\text{Hom}^{n-r}(\mathcal{E}, D)$ , without any simplicity assumption on  $\mathcal{E}$ .

Cohomological associativity identifies

$$\text{Hom}^{n-r}(L, D) \xrightarrow{\circ f} \text{Hom}^{n-r}(\mathcal{E}, D)$$

with the dual of

$$\mathrm{Hom}^r(D, \mathcal{E}) \xrightarrow{f^\circ} \mathrm{Hom}^r(D, L).$$

The latter is an isomorphism by representability. Hence the former is also an isomorphism. The triangle gives  $\mathrm{Hom}^*(C, D_j) = 0$  for every  $j$ . Since the  $D_j$  split-generate,  $\mathrm{Hom}^*(C, \mathcal{B}) = 0$ . In particular the connecting morphism  $C \rightarrow \mathcal{E}[1]$  vanishes, and the triangle splits. The inclusion and retraction of this direct summand split  $\mathrm{Hom}^*(\mathcal{E}, \mathcal{E})$  off  $\mathrm{Hom}^*(L, L)$ .  $\square$

The hypotheses of Lemma 7.1 hold here. The variety  $S \times B$  is smooth projective with trivial canonical bundle and complex dimension  $n = g + 2$ , so its Serre functor is  $[n]$ . Each product label is connected. The no-disk self-Floer computations on its two factors and the enhanced product comparison give its full graded self-cohomology, in particular one-dimensional degree-zero and degree- $n$  spaces. Proposition 5.2 supplies the nonzero top traces and the required pairings with  $L$ . Finally, the point sectors at the double points of  $L$  have odd degree, so none contributes to the degree-two chain space. The finite-rank model has degree-two dimension  $b_2(L)$ . We conclude that

$$\dim \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) \leq \dim \mathrm{Hom}^2(L, L) \leq b_2(L) = D_{\mathrm{tot}} - 19. \quad (7.1)$$

## 7.2 Recovering the algebraic K3 component of the Mukai vector

The coefficient field is algebraically closed of characteristic zero, contains  $\mathbb{C}$ , and has cardinality at most that of  $\mathbb{C}$ . It is therefore abstractly isomorphic to  $\mathbb{C}$ : both fields have transcendence degree continuum over  $\overline{\mathbb{Q}}$ . Transport  $S$ ,  $B$ , and  $\mathcal{E}$  by such an isomorphism. This operation preserves algebraic cycles, dimensions of Ext groups, and the algebraic Mukai lattice and Euler pairings used in their construction. We henceforth use the same notation for the resulting complex varieties and complex.

Recall the algebraic Mukai space

$$\Lambda_S^{\mathrm{alg}} = H^0(S, \mathbb{Q}) \oplus \mathrm{NS}(S)_{\mathbb{Q}} \oplus H^4(S, \mathbb{Q}),$$

the orthogonal decomposition  $\Lambda_S^{\mathrm{alg}} = \Delta \oplus \mathcal{N}$ , and the bases  $d_1, d_2, d_3$  and  $n_1, \dots, n_{18}$  introduced in Section 2. Put

$$v(\mathcal{E}) = \mathrm{ch}(\mathcal{E}) \sqrt{\mathrm{td}(S \times B)}, \quad \beta_l^\epsilon = \sum_{j \in J_l} e^{\epsilon j \theta}.$$

Here  $\epsilon \in \{1, -1\}$  is the common slope sign in the torus mirror correspondence. Since  $B$  has trivial tangent bundle, its Todd class is 1. This is the Mukai-vector convention of [Muk87, Definition 2.1 and Proposition 2.2].

**Proposition 7.2** (The two properties of the perfect complex). *There is a perfect complex  $\mathcal{E}$  on  $S \times B$  satisfying (7.1). Moreover, for some  $\rho \in \mathbb{Q}^\times$ , the projection of  $v(\mathcal{E})$  to  $\Lambda_S^{\mathrm{alg}} \otimes H^*(B, \mathbb{Q})$  is  $\rho p$ , where*

$$p = \sum_{l=1}^3 d_l \beta_l^\epsilon + \epsilon \sum_{i=1}^{18} n_i B(G_i). \quad (7.2)$$

*Proof.* Only the assertion about the Mukai vector remains to be proved. The representing map identifies all morphism cohomology from an embedded product label to  $\mathcal{E}$  with that to  $L$ . The latter has the ordinary transverse-intersection chain generators, with their integral degrees. Its Euler characteristic is therefore the oriented intersection number, with the uniform sign fixed by the grading conventions. It depends only on  $i_*[L] = \mu \mathrm{PD}(\alpha)$ , where  $\mu \neq 0$  is the realization multiple in Proposition 4.1.

The K3 Euler pairing identifies the primitive quartic classes of the sphere labels with the algebraic Mukai classes of their images. For the torus factor, intersections of the graph classes  $F(e^{B(A)})$  and  $F(e^{B(A')})$  are the determinant of the slope difference, up to a fixed orientation sign. This is also the Euler pairing of the associated line bundles, by Riemann–Roch on  $B$ . The common slope convention replaces  $e^{B(A)}$  by  $e^{\epsilon B(A)}$ . By linearity, the same comparison sends  $F(\beta_l)$  to  $\beta_l^\epsilon$  and  $F(B(G_i))$  to  $\epsilon B(G_i)$ . The latter assertion can equally be read off by comparing the linear coefficient in the polynomial  $e^{tB(G_i)}$ ; the exponential classes span the space in question.

It remains to check that the three other summands in  $\alpha$  do not affect these Euler tests. The summands with quartic degree zero and four have zero pairing with a sphere class of degree two, by degree. The remaining summand has quartic factor  $a$ , which is perpendicular to every primitive sphere class. Thus the Euler pairings of  $v(\mathcal{E})$  with every product test are those of a single nonzero rational multiple of the right side of (7.2). The factor includes  $\mu$  and the uniform orientation convention.

These pairings determine the asserted projection. Indeed, its torus coefficients are Hodge classes, since the K3 algebraic factors are Tate classes and  $v(\mathcal{E})$  is a rational Hodge class in every cohomological degree. The chosen graph exponentials span all the Hodge classes of  $B$ , and the sphere classes span  $\Lambda_S^{\text{alg}}$ . The Euler pairing is nondegenerate on this tensor product. For the K3 factor this is nondegeneracy of the algebraic Mukai pairing. For  $B$ , its Hodge group is the reductive diagonal  $\text{SL}_2$ ; Poincaré duality restricts to a nondegenerate pairing on invariants. To see the last assertion, apply an equivariant projection onto invariants to the second factor of a nonzero duality pairing with an invariant first factor. The Euler involution on cohomology preserves the invariant space and is invertible, so it preserves nondegeneracy. This proves the claimed formula.  $\square$

The next section uses only the two properties in Proposition 7.2. The upper bound leaves at least nineteen directions in the kernel of the Hochschild action on  $v(\mathcal{E})$ . The explicit component (7.2) will force exactly those directions to be ordinary polarized deformations, and will make the complex semiregular.

## 8 A semiregular complex and polarized deformations

Section 7 supplies a perfect complex on one product  $S \times B$ . We now show that this complex deforms while the polarized K3 surface moves in all nineteen directions. Its second Chern character then supplies a nonzero correspondence for very general polarized K3 surfaces. The conditional reduction of [Opeb] will recover the prescribed Kuga–Satake correspondence and specialize it to every fiber.

All varieties in this section are complex. Thus the coefficient field constructed earlier has already been identified abstractly with  $\mathbb{C}$ . Retain the notation

$$X = S \times B, \quad B = E^g, \quad g = 512, \quad D_{\text{tot}} = 22 + \binom{2g}{2}.$$

The starting surface has Picard rank 19. Its rational ample class  $h$  satisfies  $h^2 = 2d$ , and its primitive integral ray generator will be denoted by  $h_0$ . Recall the orthogonal decomposition

$$\Lambda_S^{\text{alg}} = \Delta \perp \mathcal{N}, \quad \Delta = H^0(S, \mathbb{Q}) \oplus \mathbb{Q}h \oplus H^4(S, \mathbb{Q}), \quad \mathcal{N} = h^\perp \cap \text{NS}(S)_\mathbb{Q}.$$

The bases are  $d_1, d_2, d_3$  of  $\Delta$  and  $n_1, \dots, n_{18}$  of  $\mathcal{N}$ , with  $(n_i, n_j) = -\lambda_i \delta_{ij}$ .

For the complex  $\mathcal{E} \in \text{Perf}(X)$  of Proposition 7.2, there are  $\rho \in \mathbb{Q}^\times$  and  $\epsilon \in \{1, -1\}$  such that

$$\dim_{\mathbb{C}} \text{Ext}_X^2(\mathcal{E}, \mathcal{E}) \leq D_{\text{tot}} - 19, \quad \rho^{-1}v(\mathcal{E}) = p + \tau, \quad (8.1)$$

where  $v(\mathcal{E}) = \text{ch}(\mathcal{E})\sqrt{\text{td}(X)}$  and

$$p = \sum_{l=1}^3 d_l \otimes \beta_l^\epsilon + \epsilon \sum_{i=1}^{18} n_i \otimes B(G_i), \quad \beta_l^\epsilon = \sum_{j \in J_l} e^{\epsilon j \theta}. \quad (8.2)$$

Here  $B(A) = z^t A w$ ,  $\theta = B(I)$ , and the three sets  $J_l$  contain nine distinct positive integers in total. The generators  $z_i, w_i$  are now the normalized complex Hodge forms on  $B$  from Section 2. For rational symmetric  $A$ , the form  $B(A)$  represents a rational divisor class; the individual generators need not be rational Betti classes. The matrices  $G_i$  are those of Proposition 3.1. The K3 factors of  $\tau$  belong to the rank-three transcendental space  $T_0 = T(S)$ ; each homogeneous part of  $p + \tau$  is a rational Hodge class. The scalar  $\rho^{-1}$  only normalizes the displayed class and does not change  $\mathcal{E}$ .

## 8.1 The Hochschild action and its trace

We use homological Hochschild indices:

$$\text{HH}_j(X) \simeq \bigoplus_{u-v=j} H^v(X, \Omega_X^u), \quad \text{HH}^2(X) \simeq \bigoplus_{u+v=2} H^v(X, \wedge^u T_X).$$

For the action, these identifications are the Todd-corrected Hochschild–Kostant–Rosenberg identifications. The compatibility theorem of Calaque–Rossi–Van den Bergh identifies cap product with the action of polyvectors on forms [CRVdB12, Theorem 1.4]. On homology the correction is left multiplication by  $\sqrt{\text{td}(X)}$ . Consequently the Hochschild Chern character becomes  $v(\mathcal{E})$ .

Fix the total-complex convention in which antiholomorphic cochain factors precede holomorphic form factors. A polyvector-valued cochain acts by contraction and exterior multiplication, with the Koszul signs from moving contractions past cochain factors. On  $H^1(T_X)$  this is the usual infinitesimal period action. In the present product the correction on this summand is the identity:  $\text{td}(B) = 1$  and  $\sqrt{\text{td}(S)} = 1 + [\text{pt}_S]$ , whose positive-degree term cannot contract a single vector. These observations also fix the interpretation of ordinary deformations inside  $\text{HH}^2(X)$ . Betti and de Rham normalizations of Chern classes are identified throughout; changing the convention for  $2\pi i$  rescales the corresponding polyvector coordinates and leaves the calculation unchanged.

For a perfect complex  $F$  on a smooth projective variety, write

$$\text{sr}_F : \text{Ext}^2(F, F) \longrightarrow \bigoplus_{j \geq 0} H^{j+2}(\Omega^j), \quad u \longmapsto \left( \frac{1}{j!} \text{tr}(\text{At}(F)^j \circ u) \right)_{j \geq 0}. \quad (8.3)$$

Changing the conventional sign of the Atiyah class changes component signs only. We call  $F$  *semiregular* if this map is injective. This is the Buchweitz–Flenner semiregularity map for perfect complexes [BF03].

**Lemma 8.1.** *For a perfect complex  $F$  on a smooth projective complex variety, cap product with its Hochschild Chern character factors as*

$$\text{HH}^2(X) \xrightarrow{\text{ev}_F} \text{Ext}_X^2(F, F) \xrightarrow{\text{tr}_F^{\text{HH}}} \text{HH}_{-2}(X).$$

*Under uncorrected HKR the second arrow is (8.3). Under corrected HKR it is that map followed by multiplication by  $\sqrt{\text{td}(X)}$ . In particular, the two trace maps have the same kernel.*

*Proof.* The factorization is also recorded in [BF08, Corollary 6.3.2 and Theorem 6.4.2]. We describe it to specify the map and its normalization. Let  $\delta : X \rightarrow X \times X$  be the diagonal and put  $\mathcal{H} = L\delta^* \mathcal{O}_\delta$ . The adjunction unit

$$\eta : \mathcal{O}_\delta \longrightarrow \delta_* \mathcal{H}$$

evaluated as a morphism of kernels on  $F$  gives  $F \rightarrow F \otimes^{\mathbf{L}} \mathcal{H}$ . Its partial trace is the Hochschild Chern character. If  $u : F \rightarrow F[2]$  is inserted before this map, the same partial trace gives  $\mathrm{tr}_F^{\mathrm{HH}}(u)$ .

An element  $b \in \mathrm{HH}^2(X)$  is a degree-two endomorphism of the diagonal kernel. Naturality of the adjunction unit gives the commuting square between  $b$ ,  $L\delta^*b$ , and  $\eta$ . Evaluating this square on  $F$  and taking the partial trace shows that applying  $L\delta^*b$  to the character equals inserting  $\mathrm{ev}_F(b)$  before the trace. This is precisely the asserted factorization.

The normalized HKR image of  $\eta$  is the exponential of the universal Atiyah class; see [C05, Proposition 4.4 and Theorem 4.5]. After evaluation on  $F$ , insertion of  $u$  therefore gives the components in (8.3). Finally,  $\sqrt{\mathrm{td}(X)}$  has constant term one, so multiplication by it is an automorphism of the finite-dimensional Hodge-cohomology target.  $\square$

Let

$$\Phi : \mathrm{HH}^2(X) \longrightarrow \mathrm{HH}_{-2}(X), \quad b \longmapsto b \lrcorner (p + \tau), \quad \mathcal{K} = \ker \Phi.$$

There is no mixed degree-one summand in  $\mathrm{HH}^2(X)$ , because  $\mathrm{HH}^1(S) = 0$ . Hence

$$\mathrm{HH}^2(X) = \mathrm{HH}^2(S) \oplus \mathrm{HH}^2(B), \quad \dim \mathrm{HH}^2(X) = 22 + \binom{g}{2} + g^2 + \binom{g}{2} = D_{\mathrm{tot}}. \quad (8.4)$$

Lemma 8.1 and (8.1) imply

$$\dim \mathcal{K} \geq 19. \quad (8.5)$$

The remaining task is to bound this kernel from above using the prescribed component  $p$ . The transcendental component  $\tau$  has not been prescribed. We shall show that its absence would leave at most two kernel directions, whereas its presence allows at most nineteen. Comparison with (8.5) will force  $\tau \neq 0$  and equality, yielding both semiregularity and ordinary polarized deformations.

## 8.2 Identifying the nineteen directions

Choose  $0 \neq \sigma \in H^{2,0}(S)$ . Contraction with  $\sigma$  gives an isomorphism

$$\mathrm{HH}^2(S) \longrightarrow \mathrm{HH}_0(S), \quad b \longmapsto s = b \lrcorner \sigma. \quad (8.6)$$

Indeed the three summands  $H^0(\wedge^2 T_S)$ ,  $H^1(T_S)$ , and  $H^2(\mathcal{O}_S)$  map respectively to  $H^0(\mathcal{O}_S)$ ,  $H^1(\Omega_S^1)$ , and  $H^2(\Omega_S^2)$ . In these coordinates, the action of  $b$  on  $\mathrm{HH}_0(S)$  takes values in the one-dimensional space  $\mathrm{HH}_{-2}(S) = H^2(\mathcal{O}_S)$  and is a nondegenerate bilinear pairing. Its two outer summands pair with each other, and its middle summand is cup product up to a nonzero normalization and sign. In particular it is nondegenerate on  $\Lambda_S^{\mathrm{alg}} \otimes \mathbb{C}$  and makes that subspace orthogonal to  $T_0^{1,1}$ .

Under (8.6), the tangent space to polarized K3 deformations is

$$\{s \in H^{1,1}(S) : (s, h) = 0\} = \mathcal{N}_{\mathbb{C}} \oplus T_0^{1,1}. \quad (8.7)$$

It has dimension 19. Its complement in  $\mathrm{HH}_0(S)$  is  $\Delta_{\mathbb{C}}$ . Thus vanishing of the three  $\Delta$  coordinates of  $s$  means exactly that the Hochschild direction is an ordinary polarized K3 direction.

On  $B$ , identify Hodge cohomology with the exterior algebra on the  $z_i, w_i$ , where  $z_i$  has type  $(1, 0)$  and  $w_i$  has type  $(0, 1)$ . An arbitrary element of  $\mathrm{HH}^2(B)$  acts as

$$A_2 = W(C) + \frac{1}{2} \sum_{i,j} R_{ij} \partial_{z_i} \partial_{z_j} + e_D, \quad e_D = \sum_{i,j} D_{ij} w_i \partial_{z_j}, \quad (8.8)$$

where  $C, R$  are skew matrices and  $D$  is arbitrary. Harmless choices of the order of two contractions are included in  $R$ . The three terms represent  $H^2(\mathcal{O}_B)$ ,  $H^0(\wedge^2 T_B)$ , and  $H^1(T_B)$ , respectively. The last direction preserves the principal polarization precisely when

$$e_D \theta = 0 \iff D = D^t.$$

The polarization class is fixed here; the deformed abelian variety need not retain a product decomposition.

**Proposition 8.2.** *For the complex  $\mathcal{E}$  satisfying (8.1)–(8.2), the following hold.*

1. *The transcendental component  $\tau$  is nonzero.*
2. *The space  $\mathcal{K}$  consists of ordinary polarized directions on both factors. Its projection to the tangent space of polarized K3 deformations is an isomorphism. In particular  $\dim \mathcal{K} = 19$ .*
3. *The complex  $\mathcal{E}$  is semiregular, and  $\dim \operatorname{Ext}_X^2(\mathcal{E}, \mathcal{E}) = D_{\text{tot}} - 19$ .*

*Proof.* We first establish two alternatives using only the form of the class  $p + \tau$ . If  $\tau = 0$ , then  $\dim \mathcal{K} \leq 2$ . If  $\tau \neq 0$ , then  $\mathcal{K}$  consists of polarized directions and injects into the nineteen-dimensional K3 tangent space. We will use the geometric lower bound only after this calculation.

Write the coefficient of  $\sigma$  in  $\tau$  as  $\tau^- \in \operatorname{HH}_{-2}(B)$ . We first record

$$\tau^- = 0 \iff \tau = 0. \quad (8.9)$$

The transcendental Hodge structure  $T_0$  is simple. To see this, split any rational Hodge substructure orthogonally using a polarization. Since  $H^{2,0}(T_0)$  is one-dimensional, at least one of the two summands has no  $(2, 0)$  part. That summand also has no  $(0, 2)$  part and is therefore of type  $(1, 1)$ , whereas  $T_0$  has no nonzero rational  $(1, 1)$  class. The summand is zero. Each homogeneous Künneth component of  $\tau$ , contracted on the K3 factor with its nondegenerate pairing, is a rational Hodge map from  $T_0$  to an appropriate Tate twist of  $H^*(B)$ . A nonzero such map is injective. It cannot vanish on  $H^{0,2}(T_0)$ , which pairs with  $\sigma$ . This proves (8.9).

Let  $(s, A_2)$  belong to  $\mathcal{K}$ . Project its action on  $p + \tau$  to the three  $\Delta$  rows of  $\operatorname{HH}_0(S) \otimes \operatorname{HH}_{-2}(B)$ . If  $s_l$  are the coordinates of the  $\Delta$  projection of  $s$ , the equations are

$$A_2 \beta_l^\epsilon + s_l \tau^- = 0 \quad (1 \leq l \leq 3). \quad (8.10)$$

Only the action on the  $\sigma$  coefficient of  $\tau$  can contribute to these rows. Product cap signs are even, and a uniform convention sign can be absorbed into the  $s_l$ .

Apply Lemma 3.2, replacing the nine exponents by  $\epsilon j$ . It gives

$$C = R = 0, \quad D = D^t, \quad s_l \tau^- = 0. \quad (8.11)$$

The lemma applies to an arbitrary common  $\tau^-$ . Its point is that the nine spaces  $e^{\epsilon j \theta} \wedge^2 \langle w_1, \dots, w_g \rangle$  are independent, whereas all three rows have the same additional vector. One three-exponent row must vanish separately, forcing the two skew parameters and the skew part of  $D$  to vanish.

Suppose first that  $\tau = 0$ . The  $\mathcal{N}$  rows now give

$$e_D B(G_i) = 0 \quad (1 \leq i \leq 18).$$

Since  $D$  and  $G_i$  are symmetric, this is equivalent to  $[D, G_i] = 0$ . The scalar-commutant assertion of Proposition 3.1 makes  $D$  scalar. The  $\operatorname{HH}_{-2}(S) \otimes \operatorname{HH}_0(B)$  row says that the linear functional

determined by  $s$  annihilates every  $d_l$  and  $n_i$ . Indeed the  $\beta_l^\epsilon$  are independent modulo degree-two classes by Lemma 3.2, and the  $B(G_i)$  are linearly independent. Therefore  $s$  lies in the one-dimensional space  $T_0^{1,1}$ . There is at most one scalar parameter in  $D$  and one parameter in  $s$ , giving  $\dim \mathcal{K} \leq 2$ . This proves the first alternative. Here  $(s, D) = (0, I)$  is indeed a kernel direction, since  $e_I$  kills  $\theta$  and every  $B(G_i)$ .

Assume now that  $\tau \neq 0$ , so  $\tau^- \neq 0$  by (8.9). Equations (8.11) now force all  $s_l = 0$ . By (8.7) and (8.8), every element of  $\mathcal{K}$  is an ordinary polarized direction on both factors.

It remains to prove that projection of  $\mathcal{K}$  to the K3 tangent space is injective. If  $s = 0$ , the  $\mathcal{N}$  rows again give  $[D, G_i] = 0$ , so  $D = tI$ . The component  $\tau^+$  opposite  $\tau^-$ , namely the coefficient of  $\bar{\sigma}$ , belongs to  $\mathrm{HH}_2(B)$  and is nonzero: the total class is rational, hence real, and complex conjugation exchanges these two coefficients. Put

$$e = \sum_i w_i \partial_{z_i}, \quad f = \sum_i z_i \partial_{w_i}, \quad H = [f, e].$$

These operators form an  $\mathfrak{sl}_2$ -action on the exterior algebra, with  $H$  acting on Hodge degree  $(u, v)$  by  $u - v$  and  $e$  lowering that weight by two. In every finite-dimensional  $\mathfrak{sl}_2$  representation the lowering operator is injective on positive-weight spaces: its kernel consists of lowest-weight vectors, whose weights are nonpositive. Hence  $e\tau^+ \neq 0$ . The  $\bar{\sigma}$  row of  $te(p + \tau) = 0$  forces  $t = 0$ .

The projection is thus injective into a nineteen-dimensional space, proving the second alternative. In either case the algebraic calculation gives  $\dim \mathcal{K} \leq 19$ . Now (8.5) excludes the first alternative and forces equality in the second. Hence  $\tau \neq 0$ , the projection is an isomorphism, and  $\dim \mathcal{K} = 19$ . Therefore  $\mathrm{rank} \Phi = D_{\mathrm{tot}} - 19$ . The factorization in Lemma 8.1 passes through an Ext space of dimension at most this number. Equality follows, and its trace arrow must be injective. This proves semiregularity and the asserted Ext dimension.  $\square$

### 8.3 Integrating the directions while preserving the class

The preceding argument determines actual polarized tangent directions. We next integrate them while retaining the whole Chern character. The integration uses the action of the period-domain groups on cohomology, rather than an inference from the dimension of a Hodge locus's tangent space.

**Lemma 8.3.** *There is a nineteen-dimensional analytic germ of polarized deformations of  $(S, B)$  through the starting point such that its projection to the polarized K3 deformation germ is an isomorphism and every homogeneous component of  $v(\mathcal{E})$  remains a Hodge class under flat transport. The same assertion holds for  $\mathrm{ch}(\mathcal{E})$ .*

*Proof.* Use marked local period domains for  $h_0$ -polarized K3 surfaces and principally polarized abelian varieties. Their compact duals are homogeneous spaces for the complex orthogonal group on  $h^\perp \subset H^2(S, \mathbb{C})$  and the complex symplectic group on  $H^1(B, \mathbb{C})$ , respectively. Their holomorphic tangent spaces at the starting filtration identify with the  $(-1, 1)$  pieces of the corresponding Lie algebras. For the K3 domain there is no additional  $(-2, 2)$  piece: an orthogonal infinitesimal transformation from the one-dimensional  $(2, 0)$  line to the  $(0, 2)$  line must vanish.

These Lie algebras act on the cohomology used in  $p + \tau$ . On the K3 they act on primitive  $H^2$  and fix  $h, H^0, H^4$ ; on  $B$  they act on the exterior powers of  $H^1(B)$ . Their  $(-1, 1)$  actions are exactly the ordinary period actions used in the cap calculation. For example, the K3 action sends  $\sigma$  to  $s$  and a class  $x \in H^{1,1}$  to the multiple of  $\bar{\sigma}$  prescribed by preservation of cup product. On the torus it is  $e_D$ .

By Proposition 8.2, each element  $k \in \mathcal{K}$  therefore gives an element of this product Lie algebra which annihilates  $v(\mathcal{E})$ . Choose a basis  $k_1, \dots, k_{19}$  and consider

$$t \longmapsto \exp \left( \sum_{i=1}^{19} t_i k_i \right) F_0$$

in the product compact dual. Every linear combination in the exponent annihilates the class, so its exponential fixes it. For sufficiently small complex  $t$ , the resulting filtrations are in the polarized period domains, since the positivity condition is open. Each fixed component in  $H^{2j}(X, \mathbb{Q})$  still belongs to  $F^j$ . As that component is real, it also belongs to  $\bar{F}^j$ , and hence has type  $(j, j)$ .

The derivative of the K3 projection is the isomorphism in Proposition 8.2. The holomorphic inverse function theorem therefore identifies the image germ with the K3 period germ. Local Torelli for K3 surfaces and the local period description of polarized abelian varieties realize it by polarized families. These are the usual marked period constructions; see [Huy16, Chapter 6] for the K3 factor.

Finally  $\mathrm{td}(B) = 1$  and  $\sqrt{\mathrm{td}(S)} = 1 + [\mathrm{pt}_S]$  throughout the family. This Todd class is flat and remains Hodge, as does its inverse. The assertion for  $\mathrm{ch}(\mathcal{E})$  follows.  $\square$

#### 8.4 Lifting and spreading the perfect complex

We use the following consequence of semiregularity for perfect complexes. Suppose  $X_A$  is smooth and proper over a local Artinian  $\mathbb{C}$ -algebra  $A$ , and  $A \rightarrow A'$  is a square-zero quotient. The obstruction to lifting a perfect complex from  $X_{A'}$  to  $X_A$  lies in its degree-two Ext group with the square-zero ideal as coefficients. The images of this obstruction under the Buchweitz–Flenner maps measure the failure of the horizontal Chern characters to belong to the requisite Hodge filtration on  $X_A$ . When the Hodge-to-de Rham spectral sequence degenerates, vanishing of this failure makes the semiregularity images themselves zero. This is the smooth proper case of [Pri24, Corollary 2.25 and Remark 2.27]. It applies to perfect complexes without any simplicity hypothesis.

**Proposition 8.4.** *Let  $2e = h_0^2$  be the primitive ray degree of the starting ample class. There is a finite-type irreducible complex base  $Q$ , with a dominant map to a chart of the moduli space of primitively amply polarized K3 surfaces of degree  $2e$ , carrying smooth projective families  $\mathcal{S}_Q, \mathcal{B}_Q$  and a perfect complex  $\mathcal{E}_Q$  on their product. At every complex point  $q \in Q$  the cycle*

$$\gamma_q = \mathrm{ch}_2(\mathcal{E}_Q|_{\mathcal{S}_q \times \mathcal{B}_q}) \in \mathrm{CH}^2(\mathcal{S}_q \times \mathcal{B}_q)_{\mathbb{Q}}$$

*induces a nonzero map  $h_q^\perp \rightarrow H^2(\mathcal{B}_q, \mathbb{Q})$ .*

*Proof.* Choose smooth algebraic local moduli charts  $U_S, U_B$  through  $(S, h_0)$  and the principally polarized  $B$ , with universal smooth projective families. Level structures and, if necessary, an étale chart provide these families and their polarizing line bundles. Shrink to affine charts. The analytic germ in Lemma 8.3 gives a morphism of formal germs

$$\mathrm{Spec} A \longrightarrow U_S \times U_B, \quad A = \mathbb{C}[[t_1, \dots, t_{19}]].$$

Let  $X_A$  be the pulled-back product family and put  $A_m = A/(t_1, \dots, t_{19})^{m+1}$  and  $X_m = X_A \times_A A_m$ . Formal comparison with the analytic germ says that the horizontal extensions of all Chern-character components of  $\mathcal{E}$  belong to  $F^j H_{\mathrm{dR}}^{2j}(X_A/A)$ .

We construct compatible perfect complexes  $\mathcal{E}_m$  on  $X_m$  by induction, starting with  $\mathcal{E}_0 = \mathcal{E}$ . The kernel

$$I_m = \ker(A_{m+1} \rightarrow A_m)$$

is annihilated by the maximal ideal. The obstruction to extending  $\mathcal{E}_m$  therefore belongs to

$$\mathrm{Ext}_X^2(\mathcal{E}, \mathcal{E}) \otimes_{\mathbb{C}} I_m. \quad (8.12)$$

The de Rham Chern characters of  $\mathcal{E}_m$  are the restrictions of the horizontal starting classes. This follows from compatibility of the Chern character with the Gauss–Manin connection and its nilpotent, or crystalline, identification. Thus the Hodge-filtration condition required for the extension is precisely the one already established on the formal germ.

Smooth projective families in characteristic zero have locally free Hodge cohomology, formation compatible with base change, and degeneration of the relative Hodge-to-de Rham spectral sequence. The preceding semiregularity theorem consequently annihilates the image of the obstruction in (8.12) under  $\mathrm{sr}_{\mathcal{E}} \otimes I_m$ . That map is injective by Proposition 8.2, so the obstruction is zero. Choose a lift together with its identification on  $X_m$ . Induction gives the required compatible system.

Derived Grothendieck existence for the proper, flat, finitely presented family over the complete noetherian ring  $A$  gives a pseudo-coherent complex  $\mathcal{E}_A$  restricting compatibly to the  $\mathcal{E}_m$ ; see [The26, Tag 0DIG]. We verify that this algebraization is perfect. Choose a uniform Tor-amplitude interval  $[u, v]$  for  $\mathcal{E}$  on the quasi-compact closed fiber. For every  $x \in X_0$ , derived restriction gives

$$(\mathcal{E}_A)_x \otimes_{\mathcal{O}_{X_A, x}}^{\mathbf{L}} k(x) \simeq \mathcal{E}_x \otimes_{\mathcal{O}_{X_0, x}}^{\mathbf{L}} k(x).$$

The residue-field criterion for Tor-amplitude of a pseudo-coherent complex makes  $(\mathcal{E}_A)_x$  have amplitude  $[u, v]$  [The26, Tag 068V]. For  $j \notin [u, v]$ , the coherent sheaf  $H^j(\mathcal{E}_A)$  therefore has support disjoint from  $X_0$ . Properness maps this support to a closed subset of  $\mathrm{Spec} A$ . Every nonempty closed subset of this local scheme contains its closed point, so the support must be empty. Hence  $\mathcal{E}_A$  is bounded coherent. The ring  $A$  is regular and  $X_A$  is smooth over it, so  $X_A$  is regular; bounded coherent complexes on a regular scheme are perfect [The26, Tags 036D and 0FDC].

Both the map to the affine finite-type charts and the perfect complex descend to a finitely generated  $\mathbb{C}$ -subalgebra  $A_0$  of  $A$ . For the complex this is continuity of perfect complexes under filtered limits, applied to the base-changed finite-presentation projective family; see [The26, Tag 09RF]. Enlarge  $A_0$  to include the finitely many functions defining the maps to both moduli charts. Set  $Q = \mathrm{Spec} A_0$  and let  $\mathcal{E}_Q$  denote the descended complex. Since  $A_0 \subset A$  is a domain,  $Q$  is irreducible. Evaluation at  $t = 0$  gives a complex point  $q_0$  of  $Q$  with fiber  $\mathcal{E}$ .

The map  $Q \rightarrow U_S$  is dominant. Indeed the map of completed local rings furnished by the K3 projection of the analytic germ is an isomorphism: its derivative is invertible and both rings are power-series rings in nineteen variables. In particular the coordinate-ring map from the chosen irreducible chart into  $A$  is injective. It factors through  $A_0$ , so that map is injective as well, which proves dominance.

We check nonvanishing before spreading it. The  $H^2(S) \otimes H^2(B)$  component of  $v(\mathcal{E})$  is the same as that of  $\mathrm{ch}_2(\mathcal{E})$ , since the positive-degree Todd term is supported in  $H^4(S)$ . For every  $j$  the action on the primitive class  $n_j$  is consequently

$$(\mathrm{ch}_2(\mathcal{E}))_*(n_j) = -\rho \epsilon \lambda_j B(G_j) \neq 0. \quad (8.13)$$

The  $\Delta$  terms vanish on  $n_j$  by orthogonality, and the transcendental terms do also. Each  $G_j$  is nonzero and the map  $A \mapsto B(A)$  is injective, proving the last inequality.

The family over  $Q$  is pulled back from smooth projective families over the smooth charts. Its analytic cohomology therefore forms local systems even if  $Q$  is singular. The Chern character of the perfect complex restricts to a locally constant section of these local systems: locally represent it by a bounded complex of vector bundles and use the topological Chern character. The polarizing class

and the Künneth projections are flat as well. Thus  $\gamma_q$  gives a morphism from the primitive local system of  $\mathcal{S}_Q$  to  $R^2(\mathcal{B}_Q)_*\mathbb{Q}$ . The analytification of the irreducible complex variety  $Q$  is connected, so the rank of this morphism is constant. Equation (8.13) makes that rank positive everywhere. Finally a perfect complex defines a class in algebraic  $K$ -theory and its Chern character on every smooth fiber is a rational algebraic cycle, as asserted.  $\square$

## 8.5 All degrees and the specified Kuga–Satake map

We finish by stating precisely how the family just constructed meets the input of [Opeb]. For a primitively polarized K3 surface  $(S', h')$ , put  $V' = (h')^\perp \subset H^2(S', \mathbb{Q})$ . Theorem 1.1 of [Opeb] says that, if  $\text{Hg}(V') = \text{SO}(V', q)$ , a nonzero algebraic map  $V' \rightarrow H^2(C, \mathbb{Q})$  for any abelian variety  $C$  suffices to realize the prescribed full Kuga–Satake map. The full-group condition holds very generally in each polarized component. Theorem 1.2 of that paper needs this nonzero algebraic input only very generally and concludes algebraicity of the prescribed map at every point of the component, for all the stipulated standard data on its transcendental Hodge structure. Its proof specializes actual cycles representing the entire prescribed flat class.

*Proof of Theorem 1.1.* Fix  $d > 0$  and perform the construction with the rational ample class of square  $2d$  chosen in Section 2. Let  $2e = h_0^2$  be its primitive ray degree. The dominant finite-type map in Proposition 8.4 has constructible image containing a nonempty Zariski open subset of the K3 chart. Every complex point in that image has a complex point above it and hence a nonzero algebraic primitive-to- $H^2$  correspondence. The moduli space of primitively amply polarized K3 surfaces of fixed degree is irreducible; the image therefore supplies the very-general input in its degree- $2e$  component. Applying [Opeb, Theorems 1.1–1.2], the prescribed Kuga–Satake correspondence is algebraic for every surface in that component.

It remains to pass from  $2e$  to the initially chosen degree  $2d$ . The two degrees have the same rational squareclass because  $h_0$  and  $h$  span the same rational ray. Let  $(S', h')$  be any Picard-rank-one surface of degree  $2d$ . In a marking of its K3 lattice  $L$ , let  $u$  be the image of  $h'$  and choose a primitive vector  $v \in L$  of square  $2e$ . There is  $b \in \mathbb{Q}_{>0}$  with  $q(bv) = q(u)$ . Witt extension gives a rational isometry  $\psi$  of  $L_{\mathbb{Q}}$  satisfying  $\psi(u) = bv$ . It identifies the polarized period domains for these two lines.

Write  $\ell$  for the marked period of  $S'$ . Its rational  $(1, 1)$  subspace is exactly  $\mathbb{Q}u$ . Since  $\psi$  is rational, the rational  $(1, 1)$  subspace of the transferred period  $\psi\ell$  is exactly  $\mathbb{Q}v$ . Thus no integral class of square  $-2$  perpendicular to  $v$  can be of type  $(1, 1)$  there: it would have to be a rational multiple of the positive vector  $v$ . The transferred period avoids all the root hyperplanes excluded by the polarized period theorem. That theorem realizes it by a K3 surface with primitive ample polarization of degree  $2e$  and Picard rank one, with the sign of the ample generator chosen consistently.

Consequently, for every Picard-rank-one  $(S', h')$  of degree  $2d$ , there is a projective K3 surface  $S''$  of degree  $2e$  and a rational Hodge isometry

$$H^2(S', \mathbb{Q}) \xrightarrow{\sim} H^2(S'', \mathbb{Q})$$

carrying the polarization line to the polarization line. Buskin’s theorem realizes this isometry by a rational algebraic correspondence [Bus19, Theorem 1.1]. Compose it with the algebraic Kuga–Satake correspondence already obtained for  $S''$ . Since  $T(S') = (h')^\perp$  and likewise for  $S''$ , the resulting algebraic map from the primitive cohomology of  $S'$  to the second cohomology of an abelian variety is nonzero. Composition has codimension two: the intermediate K3 surface has dimension two. The Picard-rank-one locus is the complement of the countable union of proper algebraic Noether–Lefschetz loci in the polarized moduli space. Hence this pointwise construction supplies the very-general input required by [Opeb, Theorem 1.2] in degree  $2d$ .

Since  $d$  was arbitrary, [Opeb, Corollary 1.3] applies. Every projective K3 surface admits a primitive ample integral class, obtained by dividing any ample integral class along its primitive ray, and hence lies in one of the components just covered. The conclusion of [Opeb] retains the full ordered even-Clifford construction, the chosen rational polarizations and standard data, and transport to the stipulated isogenous model. It therefore gives a cycle

$$\Gamma_S \in \mathrm{CH}^2(S \times A_S \times A_S)_{\mathbb{Q}}, \quad (\Gamma_S)_*x = \kappa_S(x) \quad (x \in T(S)),$$

with exactly the map in Theorem 1.1. □

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